

The privilege of HCNA/HCNP/HCIE:

With any Huawei Career Certification, you have the privilege on <http://learning.huawei.com/en> to enjoy:

- 1、e-Learning Courses: Logon <http://learning.huawei.com/en> and enter [Huawei Training/e-Learning](#)
 - If you have the HCNA/HCNP certificate: You can access Huawei Career Certification and Basic Technology e-Learning courses.
 - If you have the HCIE certificate: You can access all the e-Learning courses which marked for HCIE Certification Users.
 - Methods to get the HCIE e-Learning *privilege*: Please associate HCIE certificate information with your Huawei account, and email the account to Learning@huawei.com to apply for HCIE e-Learning privilege.
- 2、Training Material Download
 - Content: Huawei product training material and Huawei career certification training material.
 - Method: Logon <http://learning.huawei.com/en> and enter [Huawei Training/Classroom Training](#), then you can download training material in the specific training introduction page.
- 3、Priority to participate in [Huawei Online Open Class \(LVC\)](#)
 - The Huawei career certification training and product training covering all ICT technical domains like R&S, UC&C, Security, Storage and so on, which are conducted by Huawei professional instructors.
- 4、Learning Tools:
 - [eNSP](#): Simulate single Router&Switch device and large network.
 - [WLAN Planner](#): Network planning tools for WLAN AP products.
- In addition, Huawei has built up [Huawei Technical Forum](#) which allows candidates to discuss technical issues with Huawei experts, share exam experiences with others or be acquainted with Huawei Products.
- Statement:

This material is for personal use only, and can not be used by any individual or organization for any commercial purposes.

Huawei Certification

HCNA-HNTD ENTRY

**Huawei Networking Technology and Device
Lab Guide**



Huawei Technologies Co.,Ltd.

Copyright © Huawei Technologies Co., Ltd. 2016.

All rights reserved.

Huawei owns all copyrights, except for references to other parties. No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.
All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The information in this manual is subject to change without notice. Every effort has been made in the preparation of this manual to ensure accuracy of the contents, but all statements, information, and recommendations in this manual do not constitute the warranty of any kind, express or implied.



Huawei Certification

HCNA-HNTD Huawei Networking Technology and Device

Entry Lab Guide

Version 2.2

Huawei Certification System

Relying on its strong technical and professional training and certification system and in accordance with customers of different ICT technology levels, Huawei certification is committed to providing customers with authentic, professional certification, and addresses the need for the development of quality engineers that are capable of supporting Enterprise networks in the face of an ever changing ICT industry. The Huawei certification portfolio for routing and switching (R&S) is comprised of three levels to support and validate the growth and value of customer skills and knowledge in routing and switching technologies.

The Huawei Certified Network Associate (HCNA) certification level validates the skills and knowledge of IP network engineers to implement and support small to medium-sized enterprise networks. The HCNA certification provides a rich foundation of skills and knowledge for the establishment of such enterprise networks, along with the capability to implement services and features within existing enterprise networks, to effectively support true industry operations.

HCNA certification covers fundamentals skills for TCP/IP, routing, switching and related IP network technologies, together with Huawei data communications products, and skills for versatile routing platform (VRP) operation and management.

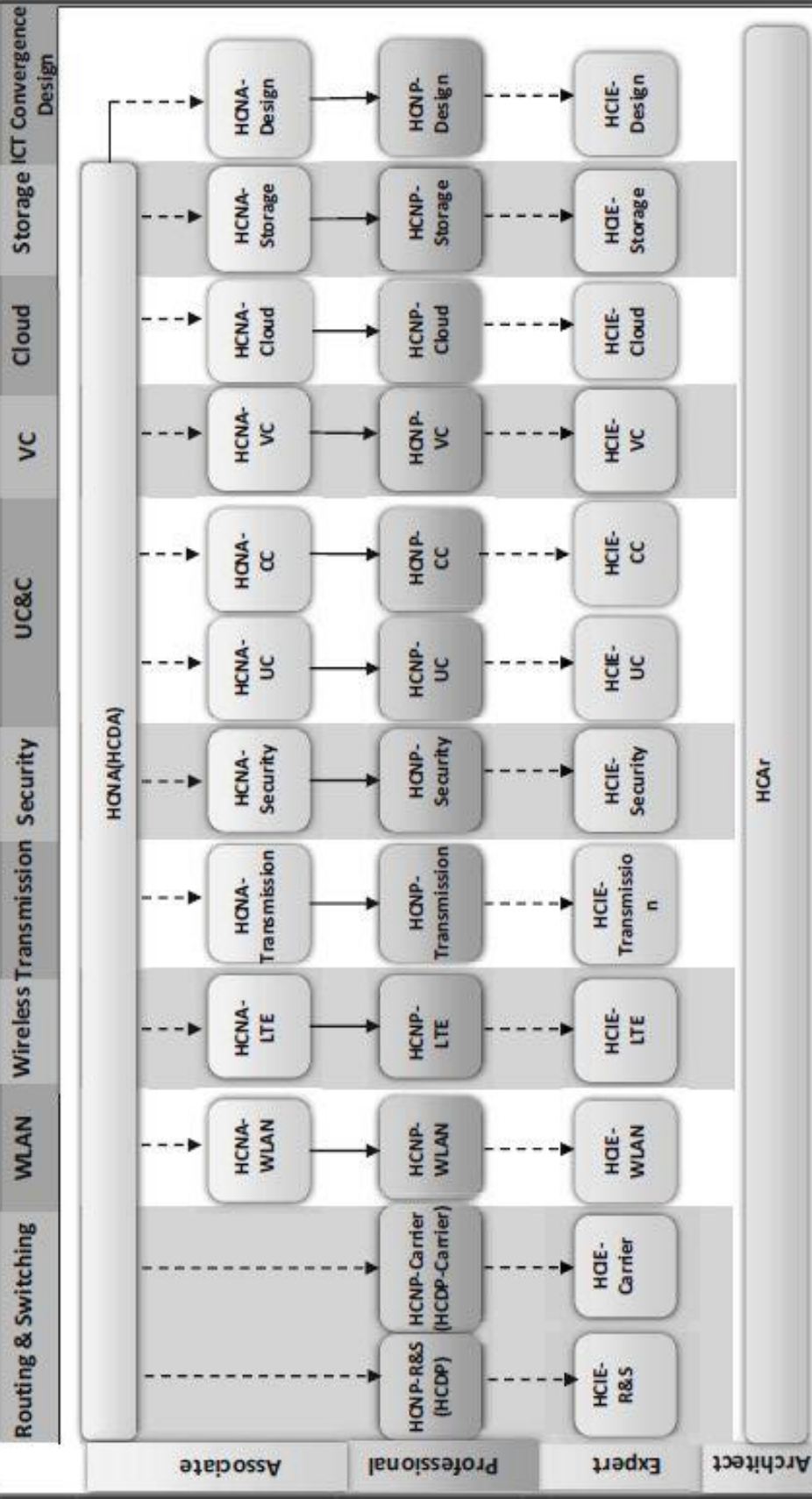
The Huawei Certified Network Professional (HCNP-R&S) certification is aimed at enterprise network engineers involved in design and maintenance, as well as professionals who wish to develop an in depth knowledge of routing, switching, network efficiency and optimization technologies. HCNP-R&S consists of three units including Implementing Enterprise Routing and Switching Network (IERS), Improving Enterprise Network Performance (IENP), and Implementing Enterprise Network Engineering Project (IEEP), which includes advanced IPv4 routing and switching technology principles, network security, high availability and QoS, as well as application of the covered technologies in Huawei products.

The Huawei Certified Internet Expert (HCIE-R&S) certification is designed to imbue engineers with a variety of IP network technologies and proficiency in maintenance, for the diagnosis and troubleshooting of Huawei products, to equip engineers with in-depth competency in the planning, design and optimization of large-scale IP networks.

Huawei Certification Portfolio

ICT Career Certification

Proposed Advanced relationship
 Necessary advanced relationship



Reference Icons



Router



L3 Switch



L2 Switch



Cloud



Ethernet link



Serial link

Lab Environment Specification

In order to ensure that the configuration given in this lab is supported on all devices, it is recommended that the following device models and VRP versions be used:

Identifier	Device Model	VRP version
R1	AR 2220E	Version 5.160 (AR2200 V200R007C00SPC600)
R2	AR 2220E	Version 5.160 (AR2200 V200R007C00SPC600)
R3	AR 2220E	Version 5.160 (AR2200 V200R007C00SPC600)
S1	S5720-36C-EI-AC	Version 5.160 (S5720 V200R008C00SPC500)
S2	S5720-36C-EI-AC	Version 5.160 (S5720 V200R008C00SPC500)
S3	S3700-28TP-EI-AC	Version 5.70 (S3700 V100R006C05)
S4	S3700-28TP-EI-AC	Version 5.70 (S3700 V100R006C05)

CONTENTS

MODULE 1 ESTABLISHING BASIC NETWORKS WITH ENSP	1
LAB 1-1 BUILDING BASIC IP NETWORKS	1
MODULE 2 BASIC DEVICE NAVIGATION AND CONFIGURATION	12
LAB 2-1 BASIC DEVICE NAVIGATION AND CONFIGURATION	12
MODULE 3 STP AND RSTP	23
LAB 3-1 CONFIGURING STP	23
LAB 3-2 CONFIGURING RSTP	40
MODULE 4 ROUTING CONFIGURATION	49
LAB 4-1 CONFIGURING STATIC ROUTES AND DEFAULT ROUTES	49
LAB 4-2 CONFIGURING RIPV1 AND RIPV2	66
LAB 4-3 RIPV2 ROUTE AGGREGATION AND AUTHENTICATION	81
LAB 4-4 OSPF SINGLE-AREA CONFIGURATION	97
MODULE 5 FTP AND DHCP	114
LAB 5-1 CONFIGURING FTP SERVICES	114
LAB 5-2 IMPLEMENTING DHCP	122

Module 1 Establishing Basic Networks with eNSP

Lab 1-1 Building Basic IP Networks

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Set up and navigate the eNSP simulator application.
- Establish a simple peer-to-peer network in eNSP.
- Perform capture of IP packets using Wireshark within eNSP.

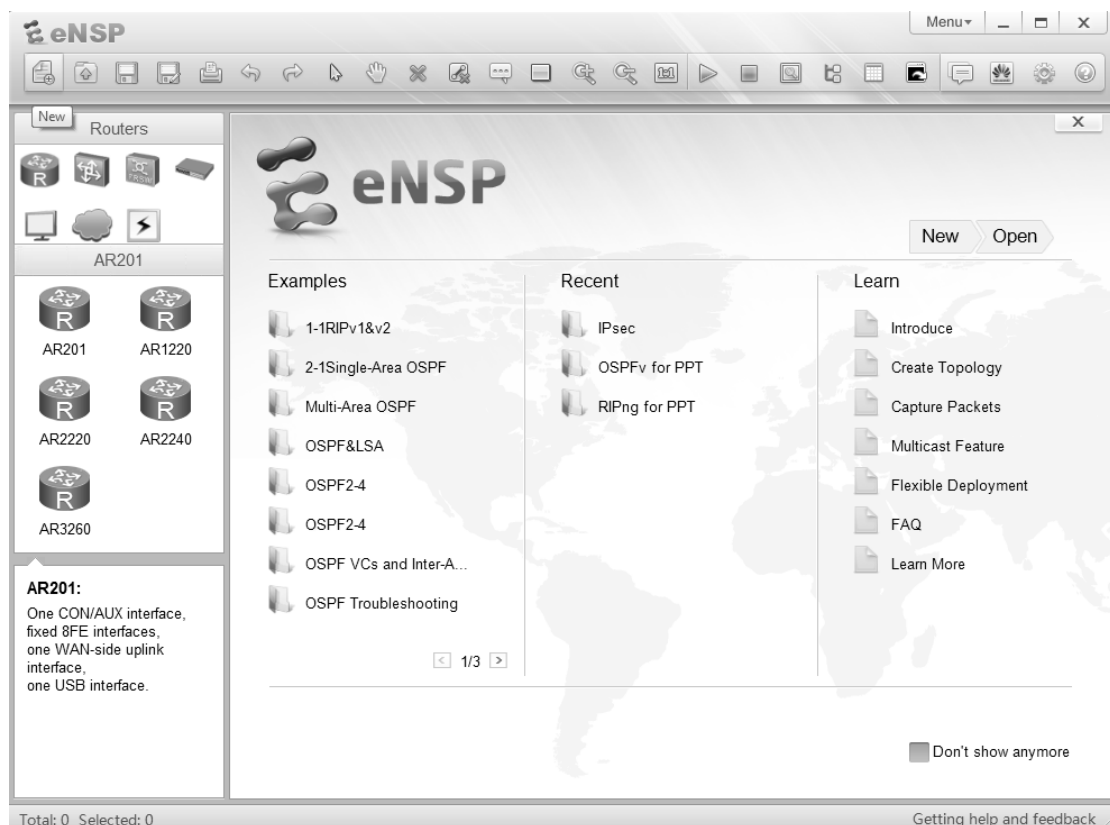
The fundamental network behavior can be understood through the application of packet capture tools to the network. The use of Huawei' s simulator platform eNSP is capable of supporting both the implementation of technologies and the capture of packets within the network to provide a comprehensive knowledge of IP networks.

Tasks

Step 1 **Initiate eNSP**

This step introduces how to start and navigate the eNSP simulator application for rapid development of TCP/IP knowledge and familiarity with network operation. If eNSP is not available, please inform the course instructor

After launching eNSP, the following application user interface will be presented. The left panel houses the icons that represent the various products and devices that are supported within eNSP, while the central panel provides lab examples for practice scenarios.

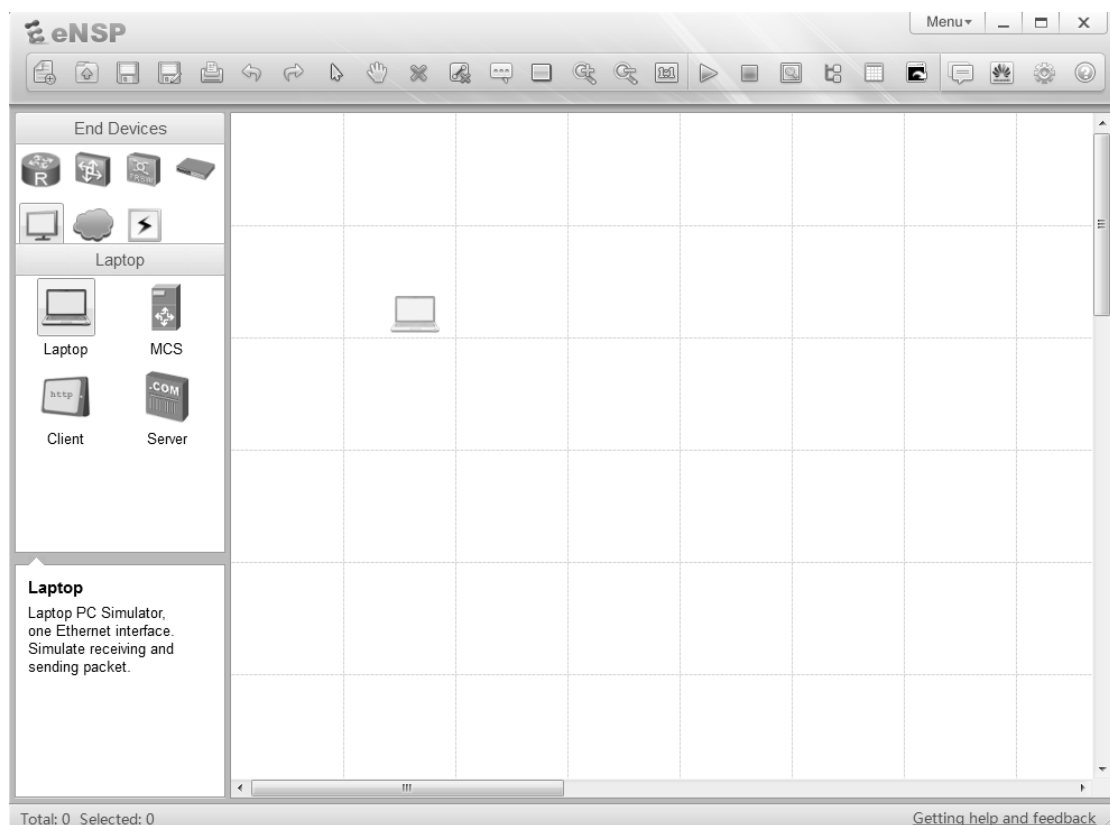


After launching eNSP, users should select the New operator in the top left corner of the application window to begin a new lab session.

The user will be presented with a canvas on which to establish a network topology for practice and analysis of network behavior. In this example a simple peer-to-peer network using two end systems is to be established.

Step 2 **Build a Topology**

Select the End Device icon in the top left panel to reveal a list of end devices that can be applied. Select the Laptop icon and drag it to the canvas, release the icon to place it on the canvas.

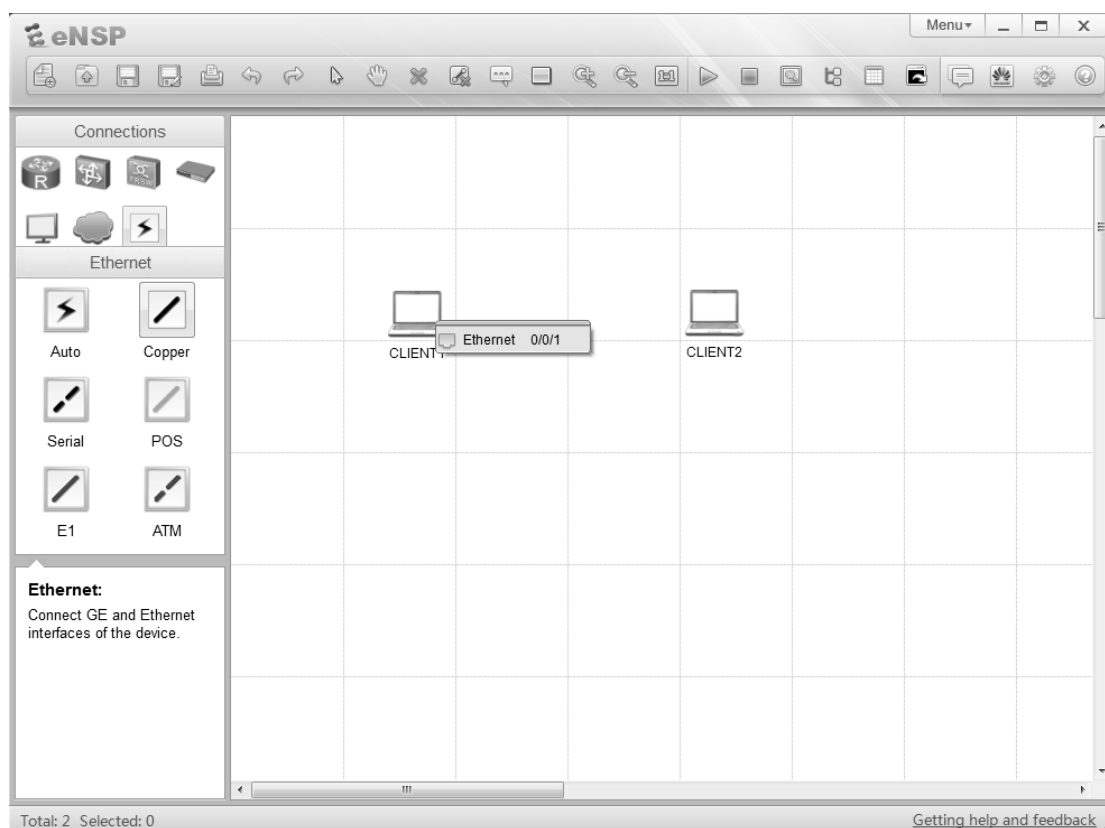


The same action should be taken to position a second laptop on the canvas for establishing the peer-to-peer network topology.

The devices on the canvas represent simulated end systems that can be used to emulate real world operations.

Step 3 **Establish a physical medium**

Select the connections icon from the upper left panel to reveal a list of media that can be applied to the topology. Select the copper (Ethernet) medium from the list. Once the icon has been clicked, the cursor will represent a connector to show the current role of the cursor as a connector. Click on the client device to reveal a list of port interfaces supported by the simulated device. For the client click the option for Ethernet 0/0/1 to apply the connection.

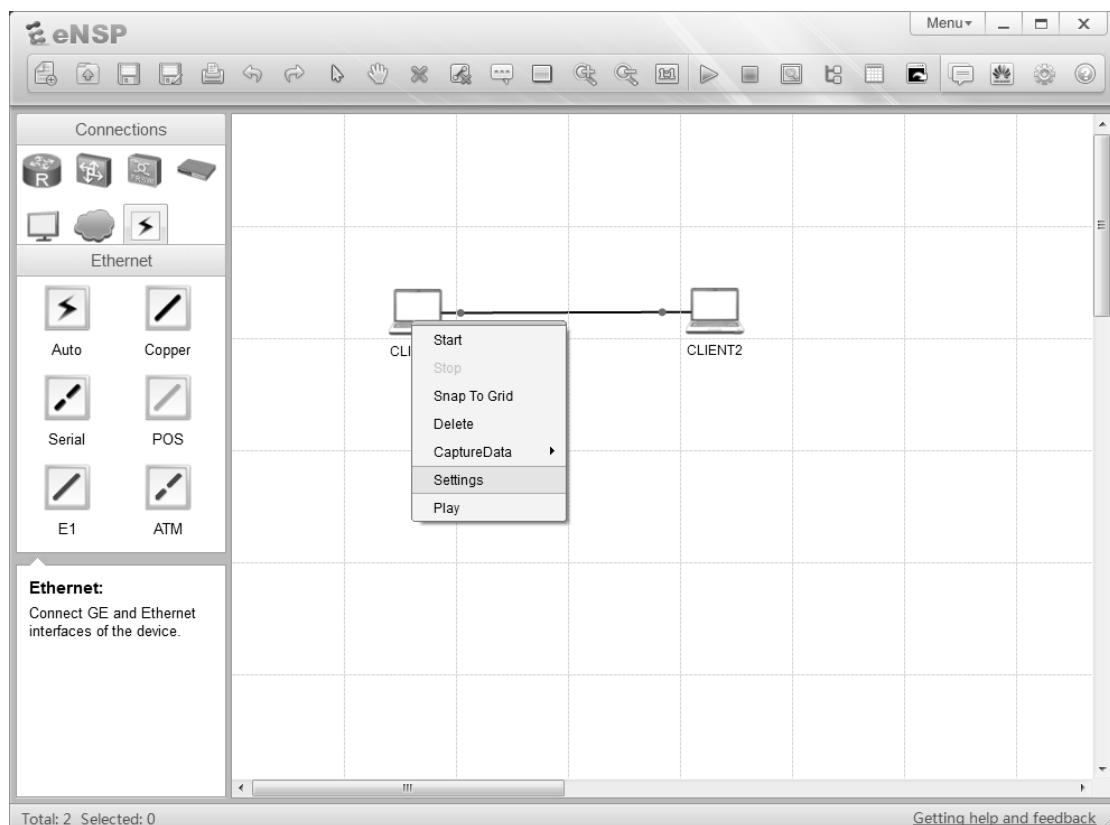


Once this has been achieved, click on the peering device to apply the opposite end of the medium to the end system. Again select the interface Ethernet 0/0/1 to establish the medium between the two devices and complete the construction of a peer-to-peer topology.

The establishment of a point-to-point network reveals a connection with two red dots on the medium that represent the current state of the interfaces to which the medium connects as down.

Step 4 **Access the end system settings**

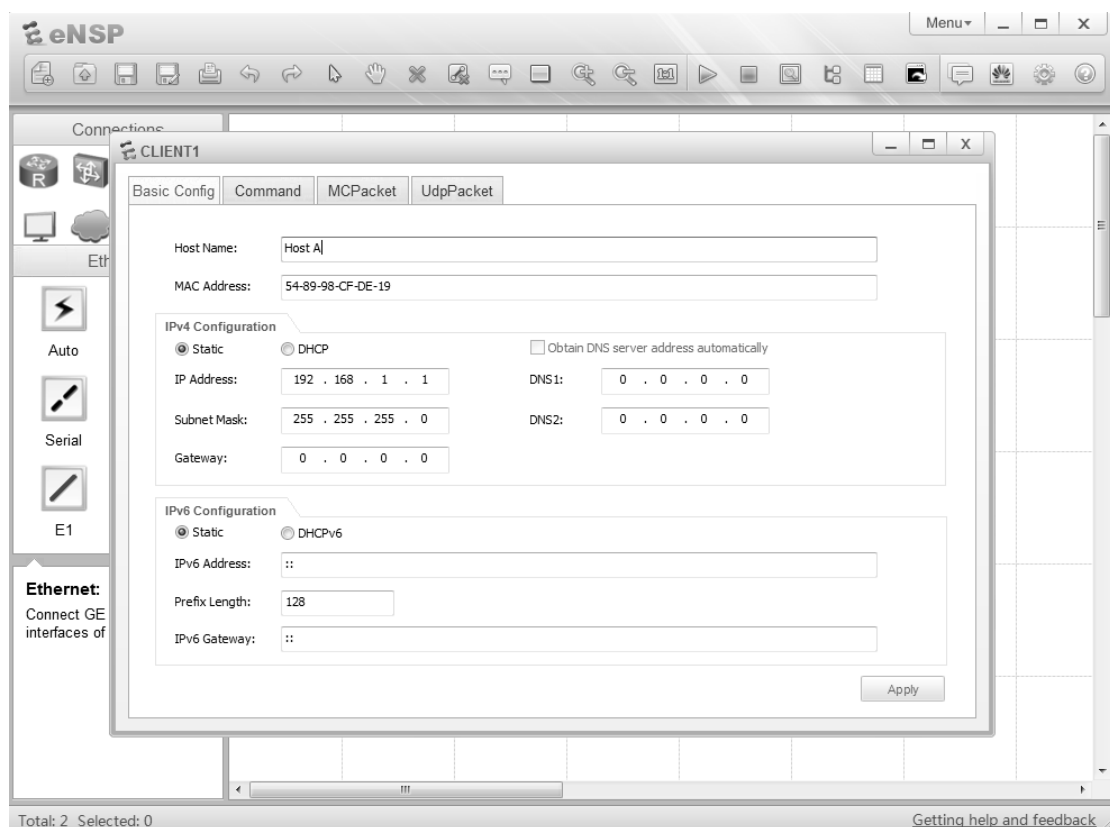
Select the end system and use the right click option to display a properties menu. The settings option should be selected in order to display the current system settings for the end system devices.



The settings option in the properties window reveals a set of four tabs for establishment of basic configuration, the device command line interface, multicast traffic generator configuration, and UDP packet generator configuration.

Step 5 **Configure the end system**

Ensure the Basic Config tab is selected and enter a host name in the Host Name field window. Ensure the IPv4 configuration is currently set to static and configure an IP address in the IP address window. It is recommended that the address (together with the subnet mask) be configured as shown in the below example. Once this has been configured, click the Apply button in the bottom left corner of the window before closing with the x in the top left corner of the CLIENT 1 window.



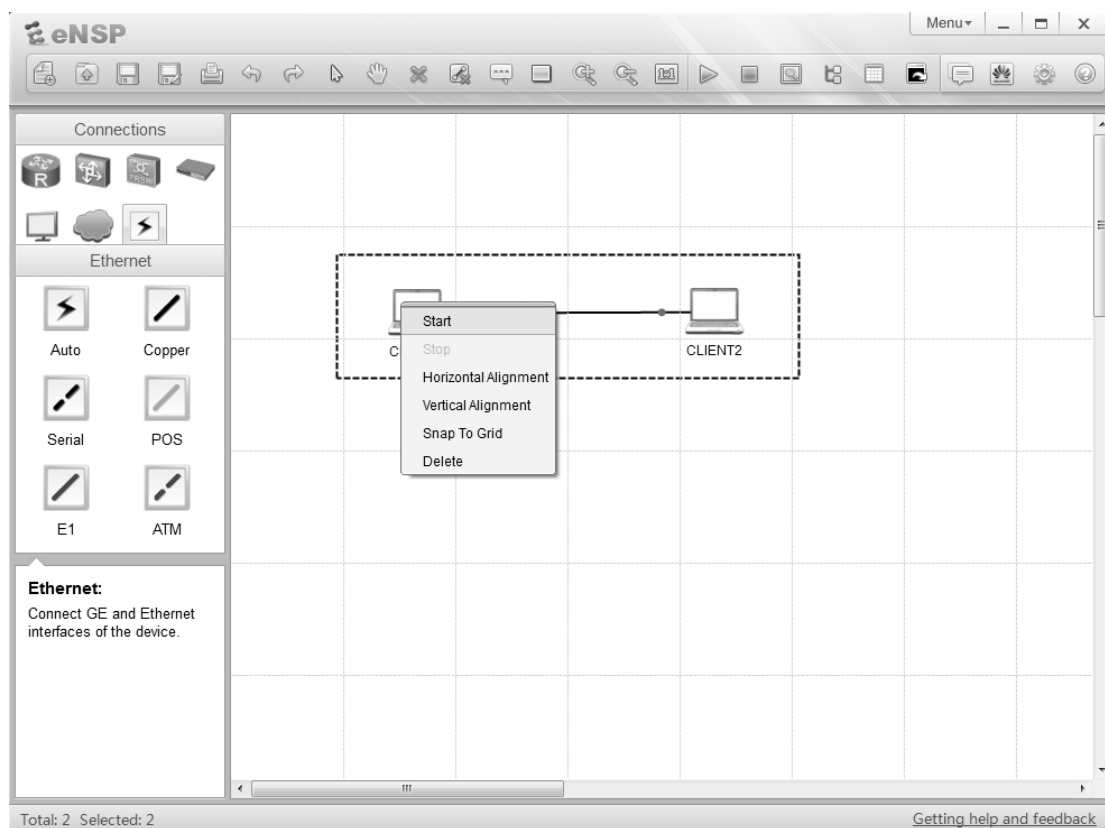
The same process is required for CLIENT2. It is recommended that initially the IP address 192.168.1.2 be configured, with a subnet mask of 255.255.255.0.

The basic configuration enables peer-to-peer communication to be supported

between the two end systems.

Step 6 **Initiate the end system devices**

The devices can be activated using one of two methods. The first involves using the right click option to open the properties menu and select start for the individual icons. The alternative involves dragging the cursor over the icons (as shown) to highlight multiple devices and using the right click settings option start multiple devices simultaneously.

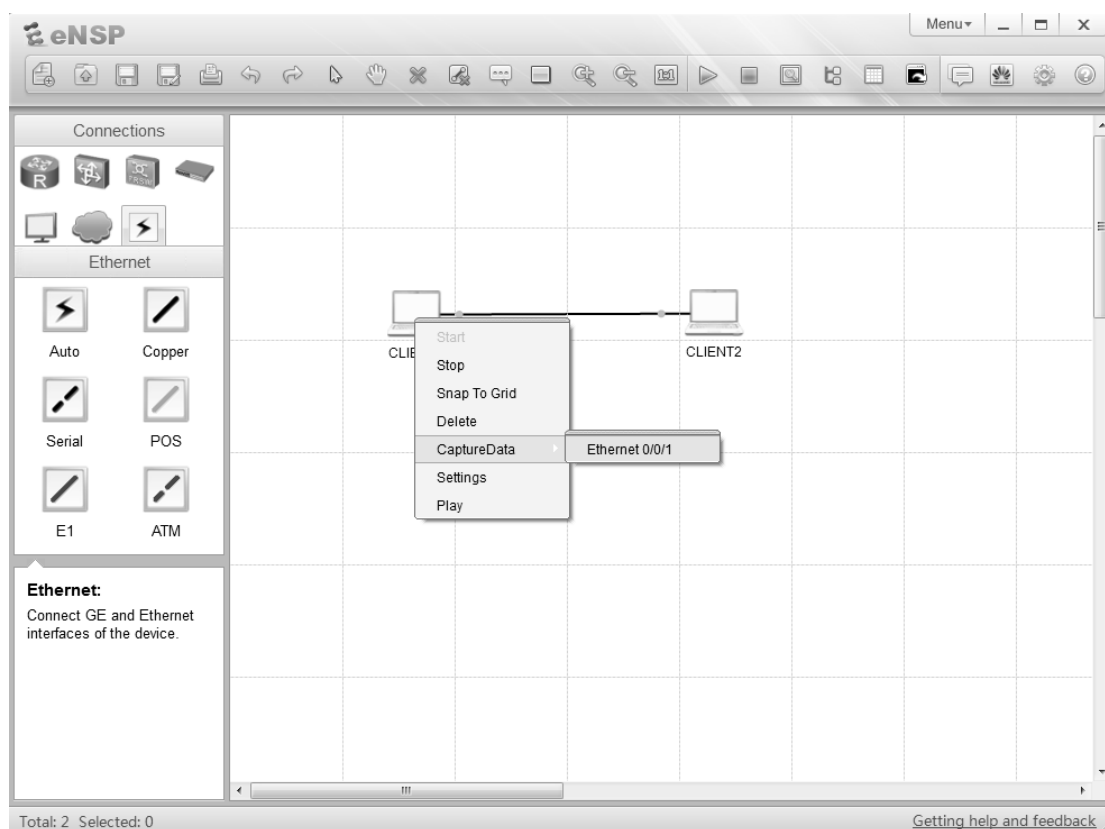


Once the devices are online and active, it is common to notice a change in the status of the connectors through a switch in the colour of the red dot on the medium to green, highlighting that the status of the connectors is now up.

Once the devices within the network topology are operational, it is possible to begin to monitor the flow of traffic that is carried over the medium and the interfaces via which the devices have established a physical peering.

Step 7 **Implement the capture of packets on an interface**

Select the device to for whose interface is to be monitored and use the right click option to display the settings menu. Highlight the capture data option to reveal a list of interfaces that belong to the device and are available for observation by the packet capture tool. Select the interface from the list that is to be monitored.



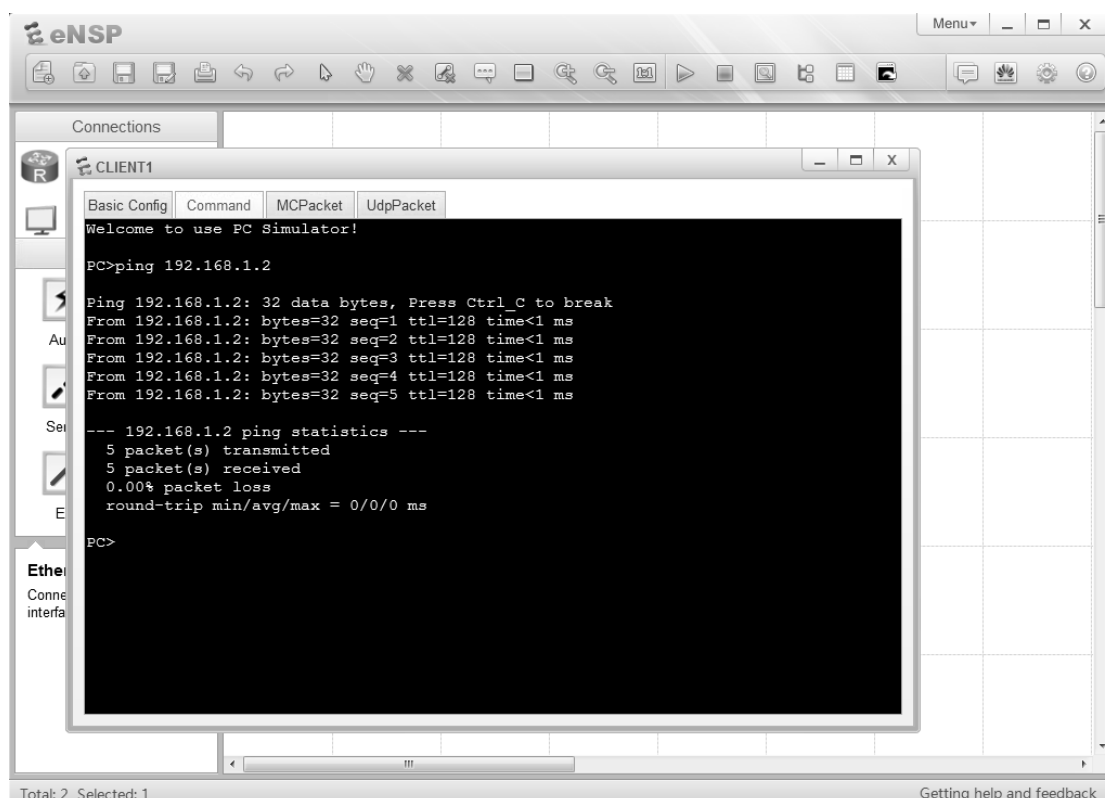
The selection of an interface will result in the activation of the Wireshark packet capture tool for the selected interface. If additional interfaces are to be monitored, separate instances of the same packet capture tool will be activated.

Depending on the devices being monitored, the packet capture tool may or may not begin to generate packet capture results for all traffic that passes through the selected interface. In the case of the peer-to-peer relationship, it will be necessary to generate some traffic.

Step 8 **Generate traffic on the interface**

Open the command window on the client by either double clicking the client icon and selecting the Command tab, or alternatively use the right click option to enter the properties menu and select settings from which point the Command tab can be selected.

The most basic means for generating traffic is through the ping command. This can be achieved by entering `ping <ip address>` where the IP address refers to the address of the peer.



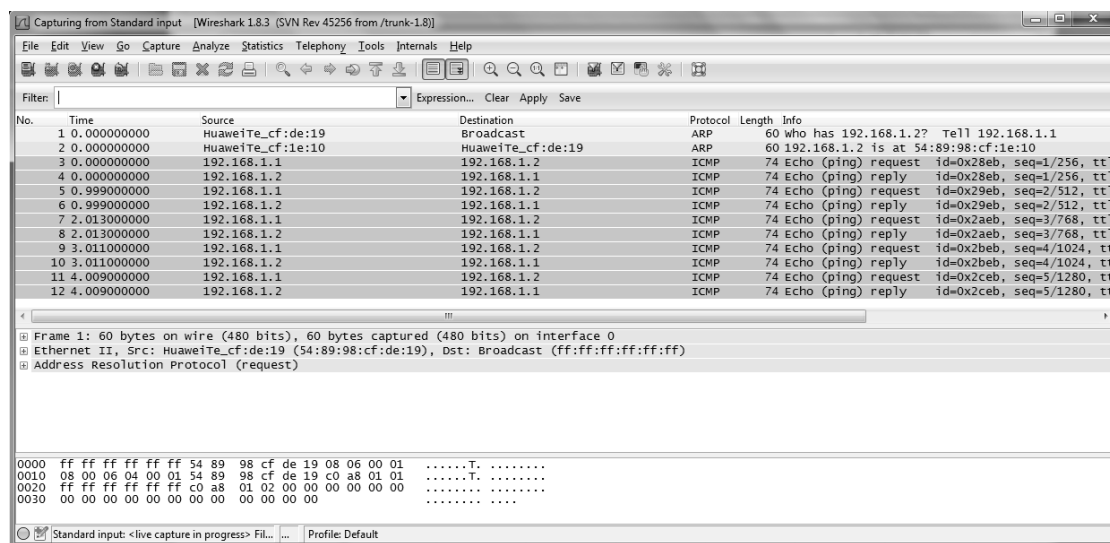
The generation of traffic will be confirmed by the resulting output in which case the

number of packets transmitted are shown to also be received.

Following the generation of traffic, the resulting traffic flow shall be captured by the packet capture tool and can be used for observation of the behavior of protocols within the IP network along with details of the various layers as referenced in the OSI reference model.

Step 9 **Observe the captured traffic flow**

An instance of the Wireshark packet capture tool should currently be active following the action to capture data on the client interface. Maximize the active window to observe the results of the packet capture process.



The Wireshark application contains many functions for management of the packet capture process. One of the more common functions includes the filter function to isolate the packet capture display to a select group of packets or protocols. This can be achieved using the filter field below the menu bar. The simplest filter method involves entering the protocol name (in lower case) and pressing Enter. In the given example packets for two protocols have been captured, entering either icmp, or arp into the filter window will result in only the protocol entered in the filter field being displayed in the output.

The packet capture tool consists of three panels, to show the list of packets, a breakdown of the content of each packet and finally display the equivalent data format of the packet. The breakdown is invaluable for understanding the format of protocol packets and displays the details for protocols as referenced at each layer of the OSI reference model.

Module 2 Basic Device Navigation and Configuration

Lab 2-1 Basic Device Navigation and Configuration

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Configure device system parameters including device name, the system time, and the system time zone.
- Configure the console port idle timeout duration.
- Configure the login information.
- Configure the login password.
- Save configuration files.
- Configure IP addresses for router interfaces.
- Test the connectivity between two directly connected routers.
- Restart a device using VRP.

Topology

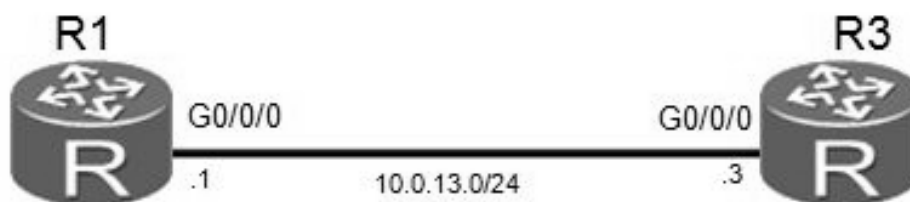


Figure 2.1 Lab topology for basic VRP navigation and operation.

Scenario

A company has purchased two AR G3 routers that require commissioning before they can be used in the enterprise network. Items to be commissioned include setting device names, the system time, and password management.

Tasks

Step 1 View the system information

Run the **display version** command to view the software version and hardware information for the system.

```
<Huawei>display version
Huawei Versatile Routing Platform Software
VRP (R) software, Version 5.160 (AR2200 V200R007C00SPC600)
Copyright (C) 2011-2013 HUAWEI TECH CO., LTD
Huawei AR2220E Router uptime is 0 week, 3 days, 21 hours, 43 minutes
BKP 0 version information:
.....output omitted.....
```

The command output includes the VRP operating system version, device model, and startup time.

Step 2 Change the system time parameter

The system automatically saves the time. If the time is incorrect, run the **clock timezone** and **clock datetime** commands in the user view to change the system time.

```
<Huawei>clock timezone Local add 08:00:00
<Huawei>clock datetime 12:00:00 2016-03-11
```

The keyword *Local* can be exchanged with the current regional timezone name, and **add** replaced with **minus** where the timezone is west of UTC+0.

Run the **display clock** command to check that the new system time has taken effect.

```
<Huawei>display clock
2016-03-11 12:00:10
Friday
Time Zone(Local) : UTC+08:00
```

Step 3 Help features & Auto-completion functions

The question mark (?) is a wildcard, and the Tab is used as a shortcut to enter commands.

```
<Huawei>display ?
Cellular                Cellular interface
aaa                     AAA
access-user             User access
accounting-scheme       Accounting scheme
acl                     <Group> acl command group
actual                  Current actual
adp-ipv4                Ipv4 information
adp-mpls                Adp-mpls module
alarm                   Alarm
antenna                 Current antenna that outputting radio
anti-attack             Specify anti-attack configurations
ap                      <Group> ap command group
ap-auth-mode            Display AP authentication mode
.....output omit.....
```

To display all the commands that start with a specific letter or string of letters, enter the desired letters and the question mark (?). The system displays all the commands that start with the letters entered. For example, if the string **dis?** is entered, the system displays all the commands that start with **dis**.

If a space exists between the character string and the question mark (?), the system will identify the commands corresponding to the string and display the parameters of the command. For example, if the string **dis ?** is entered and only the **display** command matches the **dis** string, the system displays the parameters of the **display** command. If multiple commands start with **dis**, the system displays an error.

The **Tab** key can also be pressed to complete a command. For example, if **dis** is entered followed by **Tab**, the system completes the **display** command. If multiple commands start with **dis**, the appropriate command can be selected.

If there are no other commands starting with the same letters, **dis** or **disp** can be

entered to indicate **display**, and **int** or **inter** to indicate **interface**.

Step 4 Access the system view

Run the **system-view** command to access the system view to configure interfaces and protocols.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]
```

Step 5 Change device names

To more easily identify devices, set device names during the device configuration. Change device names based on the lab topology, as shown below:

Change the name of the R1 router to **R1**.

```
[Huawei]sysname R1
[R1]
```

Change the name of the R3 router to **R3**.

```
[Huawei]sysname R3
[R3]
```

Step 6 Configure the login information

Configure the login information to indicate the login result.

```
[R1]header shell information "Welcome to the Huawei certification lab."
```

Run the preceding command to configure the login information. To check whether the login information has been changed, exit from the router command line interface, and log back in to view the login information.

```
[R1]quit
<R1>quit
```

```
Configuration console exit, please press any key to log on
Welcome to the Huawei certification lab.
```

<R1>

Step 7 **Configure console port parameters**

The console port by default does not have a login password. Users must configure a password for the console port before logging in to the device.

The password can be changed in the password authentication mode to **huawei** in plain text.

If there is no activity on the console port for the period of time specified by the timeout interval, the system will automatically log out the user. When this occurs, log in to the system again using the configured password.

The default timeout interval is set to 10 minutes. If a 10 minutes idle period is not a reasonable amount of time for the timeout interval, change the timeout interval to a more suitable duration, here this is set to 20 minutes.

```
[R1]user-interface console 0
[R1-ui-console0]authentication-mode password
[R1-ui-console0]set authentication password cipher
Warning: The "password" authentication mode is not secure, and it is strongly recommended to
use "aaa" authentication mode.
Enter Password(<8-128>):
Confirm password:
[R1-ui-console0] idle-timeout 20 0
```

Run the **display this** command to check the configuration results.

```
[R1-ui-console0]display this
[V200R007C00SPC600]
#
user-interface con 0
 authentication-mode password
 set authentication password cipher %$%$fIn'6>NZ6*~as(#J:WU%,#72Uy8cVlN^NXkT51E
^RX;>#75,%$%$
 idle-timeout 20 0
```

Log out of the system and log back in, using the password set. It should be noted that this password is required to be set when the router is first initialized.

```
[R1-ui-console0]return
<R1>quit
```

```
Configuration console exit, please press any key to log on

Login authentication

Password:

Welcome to Huawei certification lab

<R1>
```

Step 8 **Configure interface IP addresses and descriptions**

Configure an IP address for the GigabitEthernet 0/0/0 interface of R1. The subnet mask can be configured using a dotted decimal format (255.255.255.0), or based on the subnet mask prefix length.

```
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.0.13.1 24
[R1-GigabitEthernet0/0/0]description This interface connects to R3-G0/0/0
```

Run the **display this** command to check the configuration results at the current interface view.

```
[R1-GigabitEthernet0/0/0]display this
[V200R007C00SPC600]
#
interface GigabitEthernet0/0/0
description This interface connects to R3-G0/0/0
ip address 10.0.13.1 255.255.255.0
#
return
```

Run the **display interface** command to view the interface description.

```
[R1]display interface GigabitEthernet0/0/0
GigabitEthernet0/0/0 current state : UP
Line protocol current state : UP
Last line protocol up time : 2016-03-11 04:13:09
Description:This interface connects to R3-G0/0/0
Route Port,The Maximum Transmit Unit is 1500
Internet Address is 10.0.13.1/24
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 5489-9876-830b
Last physical up time : 2016-03-10 03:24:01
Last physical down time: 2016-03-10 03:25:29
Current system time: 2016-03-11 04:15:30
```

```

Port Mode: FORCE COPPER
Speed : 100, Loopback: NONE
Duplex: FULL, Negotiation: ENABLE
Mdi : AUTO, Clock : -
Last 300 seconds input rate 2296 bits/sec, 1 packets/sec
Last 300 seconds output rate 88 bits/sec, 0 packets/sec
Input peak rate 7392 bits/sec,Record time: 2016-03-10 04:08:41
Output peak rate 1120 bits/sec,Record time: 2016-03-10 03:27:56
Input: 3192 packets, 895019 bytes
  Unicast:          0,      Multicast:          1592
  Broadcast:        1600,    Jumbo:              0
  Discard:          0,      Total Error:          0
  CRC:              0,      Giants:            0
  Jabbers:          0,      Throttles:          0
  Runts:            0,      Symbols:          0
  Ignoreds:         0,      Frames:          0
Output: 181 packets, 63244 bytes
  Unicast:          0,      Multicast:          0
  Broadcast:        181,    Jumbo:              0
  Discard:          0,      Total Error:          0
  Collisions:       0,      ExcessiveCollisions: 0
  Late Collisions:  0,      Deferreds:          0
  Input bandwidth utilization threshold : 100.00%
  Output bandwidth utilization threshold: 100.00%
  Input bandwidth utilization : 0.01%
Output bandwidth utilization : 0%

```

The command output shows that the physical status and protocol status of the interface are **UP**, and the corresponding physical layer and data link layer are functional.

Once the status has been verified, configure the IP address and description for the interface of R3.

```

[R3]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]ip address 10.0.13.3 255.255.255.0
[R3-GigabitEthernet0/0/0]description This interface connects to R1-G0/0/0

```

After completing the configuration, run the **ping** command to test the connection between R1 and R3.

```

<R1>ping 10.0.13.3
  PING 10.0.13.3: 56 data bytes, press CTRL_C to break
    Reply from 10.0.13.3: bytes=56 Sequence=1 ttl=255 time=35 ms

```

```

Reply from 10.0.13.3: bytes=56 Sequence=2 ttl=255 time=32 ms
Reply from 10.0.13.3: bytes=56 Sequence=3 ttl=255 time=32 ms
Reply from 10.0.13.3: bytes=56 Sequence=4 ttl=255 time=32 ms
Reply from 10.0.13.3: bytes=56 Sequence=5 ttl=255 time=32 ms
--- 10.0.13.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
round-trip min/avg/max = 32/32/35 ms

```

Step 9 View the file list stored on the current device

Run the **dir** command in the user view to display the list of files in the current directory.

```

<R1>dir
Directory of flash:/
  Idx  Attr  Size(Byte)  Date          Time(LMT)    FileName
  0  -rw-   1,738,816  Mar 10 2016 11:50:24    web.zip
  1  -rw-  68,288,896  Mar 10 2016 14:17:5    ar2220E-v200r007c00spc600.cc
  2  -rw-           739   Mar 10 2016 16:01:17    vrpcfg.zip
1,927,476 KB total (1,856,548 KB free)

<R3>dir
Directory of flash:/
  Idx  Attr  Size(Byte)  Date          Time(LMT)    FileName
  0  -rw-   1,738,816  Mar 10 2016 11:50:58    web.zip
  1  -rw-  68,288,896  Mar 10 2016 14:19:0    ar2220E-v200r007c00spc600.cc
  2  -rw-           739   Mar 10 2016 16:03:04    vrpcfg.zip
1,927,476 KB total (1,855,076 KB free)

```

Step 10 Manage device configuration files

Attempt to display the saved-configuration file.

```

<R1>display saved-configuration
    There is no correct configuration file in FLASH

```

Since no save-configuration file exists, save the current configuration file.

```

<R1>save
    The current configuration will be written to the device.

```

```
Are you sure to continue? (y/n) [n]:y
It will take several minutes to save configuration file, please wait.....
Configuration file had been saved successfully
Note: The configuration file will take effect after being activated
```

Run the following command again to view the saved configuration information:

```
<R1>display saved-configuration
[V200R007C00SPC600]
#
sysname R1
header shell information "Welcome to Huawei certification lab"
#
board add 0/1 1SA
board add 0/2 1SA
.....output omit.....
```

Run the following command to view the current configuration information:

```
<R1>display current-configuration
[V200R007C00SPC600]
#
sysname R1
header shell information "Welcome to Huawei certification lab"
#
board add 0/1 1SA
board add 0/2 1SA
board add 0/3 2FE
.....output omit.....
```

A router can store multiple configuration files. Run the following command to view the configuration file to currently be used after the next startup:

```
<R3>display startup
MainBoard:
  Startup system software:          flash:/ar2220E-V200R007C00SPC600.cc
  Next startup system software:     flash:/ar2220E-V200R007C00SPC600.cc
  Backup system software for next startup: null
  Startup saved-configuration file: null
  Next startup saved-configuration file: flash:/vrpcfg.zip
  Startup license file:             null
  Next startup license file:        null
  Startup patch package:            null
```

```
Next startup patch package:          null
Startup voice-files:                 null
Next startup voice-files:            null
```

Delete configuration files from the flash memory.

```
<R1>reset saved-configuration
```

This will delete the configuration in the flash memory.

The device configurations will be erased to reconfigure.

Are you sure? (y/n) [n]:**y**

Clear the configuration in the device successfully.

```
<R3>reset saved-configuration
```

This will delete the configuration in the flash memory.

The device configurations will be erased to reconfigure.

Are you sure? (y/n) [n]:**y**

Clear the configuration in the device successfully.

Step 11 Device restart procedure

Use the **reboot** command to restart the router.

```
<R1>reboot
```

Info: The system is now comparing the configuration, please wait.

Warning: All the configuration will be saved to the next startup configuration. Continue ?

[y/n]:n

System will reboot! Continue ? [y/n]:y

Info: system is rebooting ,please wait...

```
<R3>reboot
```

Info: The system is now comparing the configuration, please wait.

Warning: All the configuration will be saved to the next startup configuration. Continue ?

[y/n]:n

System will reboot! Continue ? [y/n]:y

The system asks to save the current configuration. It is necessary to determine whether the current configuration should be saved based on the requirements for the lab. If unsure as to whether the current configuration should be saved, do not save.

Final Configuration

```
[R1]display current-configuration
[V200R007C00SPC600]
#
sysname R1
header shell information "Welcome to Huawei certification lab"
#
interface GigabitEthernet0/0/0
description This interface connects to R3-G0/0/0
ip address 10.0.13.1 255.255.255.0
#
user-interface con 0
authentication-mode password
set authentication password cipher %$%$4D0K*-E"t/I7[{HD~kgW,%dgkQQ!&|;XTDq9SFQJ.27M%dj,%$%
$
idle-timeout 20 0
#
return

[R3]display current-configuration
[V200R007C00SPC600]
#
sysname R3
#
interface GigabitEthernet0/0/0
description This interface connect to R1-G0/0/0
ip address 10.0.13.3 255.255.255.0
#
user-interface con 0
authentication-mode password
set authentication password cipher %$%$M8\HO3:72:ERQ8JLoHU8,%t+lE:$9=a7"8%yMoARB]$B%t.,%$%
$
user-interface vty 0 4
#
return
```

Module 3 STP and RSTP

Lab 3-1 Configuring STP

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Enable and disable STP
- Change the STP mode that is used by a switch
- Change the bridge priority to control root bridge election
- Change the port priority to control election of the root port and designated port
- Change the port cost to control election of the root port and designated port
- Configure an edge port

Topology

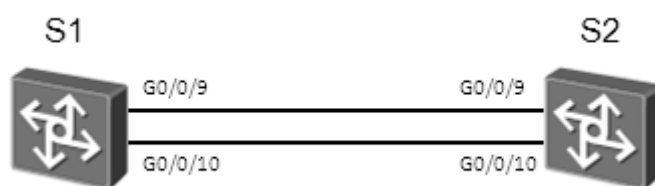


Figure 3.1 STP topology

Scenario

Assume that you are a network administrator of a company. The company network consists of two layers: core layer and access layer. The network uses a design that

supports network redundancy. STP will be used to prevent loops. The STP network should include setting the bridge priority to control STP root bridge election, and configuration of features to speed up STP route convergence.

Tasks

Step 1 **Configure STP and verify the STP configuration.**

Irrelevant interfaces must be disabled to ensure test result accuracy.

Shut down port interfaces Ethernet 0/0/1 on S3, Ethernet 0/0/13 and Ethernet 0/0/7 on S3; GigabitEthernet 0/0/1, GigabitEthernet 0/0/2, GigabitEthernet 0/0/3, GigabitEthernet 0/0/13, GigabitEthernet 0/0/14 on S1; GigabitEthernet 0/0/1, GigabitEthernet 0/0/2, GigabitEthernet 0/0/3, GigabitEthernet 0/0/6, GigabitEthernet 0/0/7 on S2; as well as Ethernet 0/0/1, Ethernet 0/0/14 and Ethernet 0/0/6 on S4 before starting STP configuration. Ensure that the devices start without any configuration files. If STP is disabled, run the **stp enable** command to enable STP.

```
<Quidway>system-view
Enter system view, return user view with Ctrl+Z.
```

```
[Quidway]sysname S1
[S1]interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1]shutdown
[S1-GigabitEthernet0/0/1]quit
[S1]interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2]shutdown
[S1-GigabitEthernet0/0/2]quit
[S1]interface GigabitEthernet 0/0/3
[S1-GigabitEthernet0/0/3]shutdown
[S1-GigabitEthernet0/0/3]quit
[S1]interface GigabitEthernet 0/0/13
[S1-GigabitEthernet0/0/13]shutdown
[S1-GigabitEthernet0/0/13]quit
[S1]interface GigabitEthernet 0/0/14
[S1-GigabitEthernet0/0/14]shutdown
[S1-GigabitEthernet0/0/14]quit
```

```
<Quidway>system-view
Enter system view, return user view with Ctrl+Z.
[Quidway]sysname S2
[S2]interface GigabitEthernet 0/0/1
[S2-GigabitEthernet0/0/1]shutdown
```

```
[S2-GigabitEthernet0/0/1]quit
[S2]interface GigabitEthernet 0/0/2
[S2-GigabitEthernet0/0/2]shutdown
[S2-GigabitEthernet0/0/2]quit
[S2]interface GigabitEthernet 0/0/3
[S2-GigabitEthernet0/0/3]shutdown
[S2-GigabitEthernet0/0/3]quit
[S2]interface GigabitEthernet 0/0/6
[S2-GigabitEthernet0/0/6]shutdown
[S2-GigabitEthernet0/0/6]quit
[S2]interface GigabitEthernet 0/0/7
[S2-GigabitEthernet0/0/7]shutdown
[S2-GigabitEthernet0/0/7]quit
```

<Quidway>system-view

Enter system view, return user view with Ctrl+Z.

```
[Quidway]sysname S3
[S3]interface Ethernet 0/0/1
[S3-Ethernet0/0/1]shutdown
[S3-Ethernet0/0/1]quit
[S3]interface Ethernet 0/0/13
[S3-Ethernet0/0/13]shutdown
[S3-Ethernet0/0/13]quit
[S3]interface Ethernet 0/0/7
[S3-Ethernet0/0/7]shutdown
```

<Quidway>system-view

Enter system view, return user view with Ctrl+Z.

```
[Quidway]sysname S4
[S4]inter Ethernet 0/0/1
[S4-Ethernet0/0/1]shutdown
[S4-Ethernet0/0/1]quit
[S4]inter Ethernet 0/0/14
[S4-Ethernet0/0/14]shutdown
[S4-Ethernet0/0/14]quit
[S4]interface Ethernet 0/0/6
[S4-Ethernet0/0/6]shutdown
```

In the lab, S1 and S2 are connected through two links, and STP is used. Enable STP on S1 and S2 and set S1 as the root.

```
[S1]stp mode stp
```

Info: This operation may take a few seconds. Please wait for a moment...done.

```
[S1]stp root primary
```

```
[S2]stp mode stp
```

Info: This operation may take a few seconds. Please wait for a moment...done.

```
[S2]stp root secondary
```

Run the **display stp brief** command to view brief information about STP.

```
<S1>display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/9	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/10	DESI	FORWARDING	NONE

```
<S2>display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/9	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/10	ALTE	DISCARDING	NONE

Run the **display stp interface** command to view the STP status of a port.

```
<S1>display stp interface GigabitEthernet 0/0/10
```

```
-----[CIST Global Info][Mode STP]-----
```

```
CIST Bridge      :0      .d0d0-4ba6-aab0
Config Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC   :0      .d0d0-4ba6-aab0 / 0 (This bridge is the root)
CIST RegRoot/IRPC :0      .d0d0-4ba6-aab0 / 0
CIST RootPortId  :0.0
BPDU-Protection  :Disabled
CIST Root Type   :Primary root
TC or TCN received :11
TC count per hello :0
STP Converge Mode :Normal
Share region-configuration :Enabled
Time since last TC :0 days 1h:43m:55s
Number of TC     :29
Last TC occurred  :GigabitEthernet0/0/9
----[Port10(GigabitEthernet0/0/10)] [FORWARDING]----
Port Protocol    :Enabled
Port Role        :Designated Port
Port Priority     :128
```

```

Port Cost(Dot1T )    :Config=auto / Active=20000
Designated Bridge/Port :0.d0d0-4ba6-aab0 / 128.10
Port Edged           :Config=default / Active=disabled
Point-to-point       :Config=auto / Active=true
Transit Limit        :6 packets/s
Protection Type      :None
Port STP Mode        :STP
Port Protocol Type   :Config=auto / Active=dot1s
BPDU Encapsulation   :Config=stp / Active=stp
PortTimes            :Hello 2s MaxAge 20s FwDly 15s RemHop 20
TC or TCN send       :52
TC or TCN received   :0
BPDU Sent            :3189
                    TCN: 0, Config: 3189, RST: 0, MST: 0
BPDU Received        :5
                    TCN: 0, Config: 5, RST: 0, MST: 0
Last forwarding time: 2016/11/21 14:55:11 UTC

```

```

<S2>display stp interface GigabitEthernet 0/0/10
-----[CIST Global Info][Mode STP]-----
CIST Bridge          :4096 .d0d0-4ba6-ac20
Config Times         :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times         :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC       :0 .d0d0-4ba6-aab0 / 20000
CIST RegRoot/IRPC    :4096 .d0d0-4ba6-ac20 / 0
CIST RootPortId      :128.9 (GigabitEthernet0/0/9)
BPDU-Protection      :Disabled
CIST Root Type       :Secondary root
TC or TCN received   :122
TC count per hello   :0
STP Converge Mode    :Normal
Share region-configuration :Enabled
Time since last TC   :0 days 1h:50m:0s
Number of TC         :17
Last TC occurred     :GigabitEthernet0/0/9
----[Port10(GigabitEthernet0/0/10)][DISCARDING]----
Port Protocol        :Enabled
Port Role            :Alternate Port
Port Priority         :128
Port Cost(Dot1T )    :Config=auto / Active=20000

```

```

Designated Bridge/Port   :0.d0d0-4ba6-aab0 / 128.10
Port Edged               :Config=default / Active=disabled
Point-to-point          :Config=auto / Active=true
Transit Limit           :6 packets/s
Protection Type          :None
Port STP Mode            :STP
Port Protocol Type       :Config=auto / Active=dot1s
BPDU Encapsulation      :Config=stp / Active=stp
PortTimes                :Hello 2s MaxAge 20s FwDly 15s RemHop 0
TC or TCN send          :0
TC or TCN received       :18
BPDU Sent                :2
                        TCN: 0, Config: 2, RST: 0, MST: 0
BPDU Received            :3317
                        TCN: 0, Config: 3317, RST: 0, MST: 0

```

Step 2 Control root bridge election.

Run the **display stp** command to view information about the root bridge.

```

<S1>display stp
-----[CIST Global Info][Mode STP]-----
CIST Bridge             :0      .d0d0-4ba6-aab0
Config Times            :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times            :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC          :0      .d0d0-4ba6-aab0 / 0 (This bridge is the root)
CIST RegRoot/IRPC       :0      .d0d0-4ba6-aab0 / 0
CIST RootPortId         :0.0
BPDU-Protection         :Disabled
CIST Root Type          :Primary root
TC or TCN received      :11
TC count per hello      :0
STP Converge Mode       :Normal
Share region-configuration :Enabled
Time since last TC      :0 days 2h:32m:25s
.....output omit.....

<S2>display stp
-----[CIST Global Info][Mode STP]-----
CIST Bridge             :4096   .d0d0-4ba6-ac20

```

```

Config Times      :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC   :0 .d0d0-4ba6-aab0 / 20000
CIST RegRoot/IRPC :4096 .d0d0-4ba6-ac20 / 0
CIST RootPortId  :128.9 (GigabitEthernet0/0/9)
BPDU-Protection  :Disabled
CIST Root Type   :Secondary root
TC or TCN received :122
TC count per hello :0
STP Converge Mode :Normal
Share region-configuration :Enabled
Time since last TC :0 days 2h:35m:57s
.....output omit.....

```

Configure S2 as the root bridge and S1 as the backup root bridge using priority values. The device with the same value for the **CIST Bridge** and **CIST Root/ERPC** is the root bridge. A smaller bridge priority value indicates a higher bridge priority. Change the priorities of S1 and S2 to 8192 and 4096 respectively so that S2 becomes the root bridge.

```

[S1]undo stp root
[S1]stp priority 8192

[S2]undo stp root
[S2]stp priority 4096

```

Run the **display stp** command to view information about the new root bridge.

```

<S1>display stp
-----[CIST Global Info][Mode STP]-----
CIST Bridge      :8192 .d0d0-4ba6-aab0
Config Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC   :4096 .d0d0-4ba6-ac20 / 20000
CIST RegRoot/IRPC :8192 .d0d0-4ba6-aab0 / 0
CIST RootPortId  :128.9 (GigabitEthernet0/0/9)
BPDU-Protection  :Disabled
TC or TCN received :47
TC count per hello :0
STP Converge Mode :Normal
Share region-configuration :Enabled
Time since last TC :0 days 0h:6m:55s
.....output omit.....

```

```

<S2>display stp
-----[CIST Global Info][Mode STP]-----
CIST Bridge      :4096 .d0d0-4ba6-ac20
Config Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC   :4096 .d0d0-4ba6-ac20 / 0 (This bridge is the root)
CIST RegRoot/IRPC :4096 .d0d0-4ba6-ac20 / 0
CIST RootPortId  :0.0
BPDU-Protection  :Disabled
TC or TCN received :135
TC count per hello :0
STP Converge Mode :Normal
Share region-configuration :Enabled
Time since last TC :0 days 0h:8m:4s
.....output omit.....

```

The highlighted lines in the preceding information indicate that S2 has become the new root bridge.

Shut down interfaces Gigabit Ethernet 0/0/9 and Gigabit Ethernet 0/0/10 on S2 to isolate S2.

```

[S2]interface GigabitEthernet 0/0/9
[S2-GigabitEthernet0/0/9]shutdown
[S2-GigabitEthernet0/0/9]quit
[S2]interface GigabitEthernet 0/0/10
[S2-GigabitEthernet0/0/10]shutdown

```

```

<S1>display stp
-----[CIST Global Info][Mode STP]-----
CIST Bridge      :8192 .d0d0-4ba6-aab0
Config Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC   :8192 .d0d0-4ba6-aab0 / 0 (This bridge is the root)
CIST RegRoot/IRPC :8192 .d0d0-4ba6-aab0 / 0
CIST RootPortId  :0.0
BPDU-Protection  :Disabled
TC or TCN received :174
TC count per hello :0
STP Converge Mode :Normal
Share region-configuration :Enabled
Time since last TC :0 days 0h:12m:51s

```

.....output omit.....

The highlighted lines in the preceding information indicate that S1 becomes the root bridge when S2 is faulty.

Re-enable the interfaces that have been disabled on S2.

```
[S2]interface GigabitEthernet 0/0/9
[S2-GigabitEthernet0/0/9]undo shutdown
[S2-GigabitEthernet0/0/9]quit
[S2]interface GigabitEthernet 0/0/10
[S2-GigabitEthernet0/0/10]undo shutdown
```

```
<S1>display stp
-----[CIST Global Info][Mode STP]-----
CIST Bridge      :8192 .d0d0-4ba6-aab0
Config Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC   :4096 .d0d0-4ba6-ac20 / 20000
CIST RegRoot/IRPC :8192 .d0d0-4ba6-aab0 / 0
CIST RootPortId  :128.9 (GigabitEthernet0/0/9)
BPDU-Protection  :Disabled
TC or TCN received :47
TC count per hello :0
STP Converge Mode :Normal
Share region-configuration :Enabled
Time since last TC :0 days 0h:6m:55s
.....output omit.....
```

```
<S2>display stp
-----[CIST Global Info][Mode STP]-----
CIST Bridge      :4096 .d0d0-4ba6-ac20
Config Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times     :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC   :4096 .d0d0-4ba6-ac20 / 0 (This bridge is the root)
CIST RegRoot/IRPC :4096 .d0d0-4ba6-ac20 / 0
CIST RootPortId  :0.0
BPDU-Protection  :Disabled
TC or TCN received :135
TC count per hello :0
STP Converge Mode :Normal
Share region-configuration :Enabled
Time since last TC :0 days 0h:8m:4s
```

.....output omit.....

The highlighted lines in the preceding information indicate that S2 has been restored and has become the root bridge once again.

Step 3 **Control root port election.**

Run the **display stp brief** command on S1 to view the roles of the interfaces.

```
<S1>display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/9	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/10	ALTE	DISCARDING	NONE

The preceding information shows that G0/0/9 is the root port and G0/0/10 is the alternate port. You can change port priorities so that port interface G0/0/10 will become the root port and G0/0/9 will become the alternate port.

Change priorities of G0/0/9 and G0/0/10 on S2.

The default port priority is 128. A larger port priority value indicates a lower priority. The priorities of G0/0/9 and G0/0/10 on S2 are set to 32 and 16; therefore, G0/0/10 on S1 becomes the root port.

```
[S2]interface GigabitEthernet 0/0/9
[S2-GigabitEthernet0/0/9]stp port priority 32
[S2-GigabitEthernet0/0/9]quit
[S2]interface GigabitEthernet 0/0/10
[S2-GigabitEthernet0/0/10]stp port priority 16
```

Note that the port priorities are changed on S2, not S1.

```
<S2>display stp interface GigabitEthernet 0/0/9
-----[CIST Global Info][Mode STP]-----
CIST Bridge           :4096 .d0d0-4ba6-ac20
Config Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC        :4096 .d0d0-4ba6-ac20 / 0 (This bridge is the root)
CIST RegRoot/IRPC     :4096 .d0d0-4ba6-ac20 / 0
CIST RootPortId       :0.0
BPDU-Protection       :Disabled
TC or TCN received    :147
TC count per hello    :0
STP Converge Mode     :Normal
```

```

Share region-configuration :Enabled
Time since last TC :0 days 0h:7m:35s
Number of TC :41
Last TC occurred :GigabitEthernet0/0/10
----[Port34(GigabitEthernet0/0/9)][FORWARDING]----
Port Protocol :Enabled
Port Role :Designated Port
Port Priority :32
Port Cost(Dot1T ) :Config=auto / Active=20000
Designated Bridge/Port :4096.d0d0-4ba6-ac20 / 32.34
Port Edged :Config=default / Active=disabled
Point-to-point :Config=auto / Active=true
Transit Limit :6 packets/s
Protection Type :None
Port STP Mode :STP
Port Protocol Type :Config=auto / Active=dot1s
BPDU Encapsulation :Config=stp / Active=stp
PortTimes :Hello 2s MaxAge 20s FwDly 15s RemHop 20
TC or TCN send :35
TC or TCN received :2
BPDU Sent :1013
    TCN: 0, Config: 1013, RST: 0, MST: 0
BPDU Received :2
    TCN: 2, Config: 0, RST: 0, MST: 0
Last forwarding time: 2016/11/22 10:00:00 UTC

```

```

<S2>display stp interface GigabitEthernet 0/0/10
-----[CIST Global Info][Mode STP]-----
CIST Bridge :4096 .d0d0-4ba6-ac20
Config Times :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC :4096 .d0d0-4ba6-ac20 / 0 (This bridge is the root)
CIST RegRootIRPC :4096 .d0d0-4ba6-ac20 / 0
CIST RootPortId :0.0
BPDU-Protection :Disabled
TC or TCN received :147
TC count per hello :0
STP Converge Mode :Normal
Share region-configuration :Enabled
Time since last TC :0 days 0h:8m:19s
Number of TC :41
Last TC occurred :GigabitEthernet0/0/10

```

```

----[Port35(GigabitEthernet0/0/10)][FORWARDING]----
Port Protocol      :Enabled
Port Role          :Designated Port
Port Priority       :16
Port Cost(Dot1T )  :Config=auto / Active=20000
Designated Bridge/Port :4096.d0d0-4ba6-ac20 / 16.35
Port Edged         :Config=default / Active=disabled
Point-to-point     :Config=auto / Active=true
Transit Limit      :6 packets/s
Protection Type    :None
Port STP Mode      :STP
Port Protocol Type :Config=auto / Active=dot1s
BPDU Encapsulation :Config=stp / Active=stp
PortTimes          :Hello 2s MaxAge 20s FwDly 15s RemHop 20
TC or TCN send     :35
TC or TCN received :1
BPDU Sent          :1032
                  TCN: 0, Config: 1032, RST: 0, MST: 0
BPDU Received      :2
                  TCN: 1, Config: 1, RST: 0, MST: 0
Last forwarding time: 2016/11/22 10:00:11 UTC

```

Run the **display stp brief** command on S1 to view the role of the interfaces.

```

<S1>display stp brief

```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/9	ALTE	DISCARDING	NONE
0	GigabitEthernet0/0/10	ROOT	FORWARDING	NONE

The highlighted lines in the preceding information indicate that G0/0/10 on S1 has become the root port and G0/0/9 has become the alternate port.

Shut down G0/0/10 on S1 and view the port roles.

```

[S1]interface GigabitEthernet 0/0/10
[S1-GigabitEthernet0/0/10]shutdown
<S1>display stp brief

```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/9	ROOT	FORWARDING	NONE

The highlighted line in the preceding information indicates that G0/0/9 has become the root port. Resume the default priorities of G0/0/9 and G0/0/10 on S2 and re-enable the shutdown interfaces on S1.

```
[S2]interface GigabitEthernet 0/0/9
[S2-GigabitEthernet0/0/9]undo stp port priority
[S2-GigabitEthernet0/0/9]quit
[S2]interface GigabitEthernet 0/0/10
[S2-GigabitEthernet0/0/10]undo stp port priority
```

```
[S1]interface GigabitEthernet 0/0/10
[S1-GigabitEthernet0/0/10]undo shutdown
```

Run the **display stp brief** and **display stp interface** command on S1 to view the roles of interfaces.

```
<S1>display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/9	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/10	ALTE	DISCARDING	NONE

```
[S1]display stp interface GigabitEthernet 0/0/9
```

```
----[CIST][Port9(GigabitEthernet0/0/9)][FORWARDING]----
```

```
Port Protocol      :Enabled
Port Role          :Root Port
Port Priority       :128
Port Cost(Dot1T )  :Config=auto / Active=20000
Designated Bridge/Port :4096.4c1f-cc45-aacc / 128.9
Port Edged         :Config=default / Active=disabled
Point-to-point     :Config=auto / Active=true
Transit Limit      :147 packets/hello-time
Protection Type    :None
Port STP Mode      :STP
Port Protocol Type  :Config=auto / Active=dot1s
BPDU Encapsulation :Config=stp / Active=stp
PortTimes          :Hello 2s MaxAge 20s FwDly 15s RemHop 0
TC or TCN send     :4
TC or TCN received :90
BPDU Sent          :5
                  TCN: 4, Config: 1, RST: 0, MST: 0
BPDU Received      :622
                  TCN: 0, Config: 622, RST: 0, MST: 0
```

```
[S1]display stp interface GigabitEthernet 0/0/10
```

```
----[CIST][Port10(GigabitEthernet0/0/10)][DISCARDING]----
```

```
Port Protocol      :Enabled
```

```

Port Role           :Alternate Port
Port Priority        :128
Port Cost(Dot1T )   :Config=auto / Active=20000
Designated Bridge/Port :4096.4c1f-cc45-aacc / 128.10
Port Edged           :Config=default / Active=disabled
Point-to-point       :Config=auto / Active=true
Transit Limit        :147 packets/hello-time
Protection Type      :None
Port STP Mode        :STP
Port Protocol Type   :Config=auto / Active=dot1s
BPDU Encapsulation   :Config=stp / Active=stp
PortTimes            :Hello 2s MaxAge 20s FwDly 15s RemHop 0
TC or TCN send       :3
TC or TCN received   :90
BPDU Sent            :4
                    TCN: 3, Config: 1, RST: 0, MST: 0
BPDU Received        :637
                    TCN: 0, Config: 637, RST: 0, MST: 0

```

The greyed line in the preceding information indicates that G0/0/9 and G0/0/10 cost is 20000 by default.

Change the cost of G0/0/9 to 200000 on S1.

```

[S1]interface GigabitEthernet 0/0/9
[S1-GigabitEthernet0/0/9]stp cost 200000

```

Run the **display stp brief** and **display stp interface** command on S1 to view the roles of interfaces.

```

<S1>display stp interface GigabitEthernet 0/0/9
----[CIST][Port9(GigabitEthernet0/0/9)][DISCARDING]----
Port Protocol       :Enabled
Port Role           :Alternate Port
Port Priority        :128
Port Cost(Dot1T )   :Config=200000 / Active=200000
Designated Bridge/Port :4096.4c1f-cc45-aacc / 128.9
Port Edged           :Config=default / Active=disabled
Point-to-point       :Config=auto / Active=true
Transit Limit        :147 packets/hello-time
Protection Type      :None
Port STP Mode        :STP
Port Protocol Type   :Config=auto / Active=dot1s
BPDU Encapsulation   :Config=stp / Active=stp

```

```

PortTimes           :Hello 2s MaxAge 20s FwDly 15s RemHop 0
TC or TCN send      :4
TC or TCN received  :108
BPDU Sent           :5
                    TCN: 4, Config: 1, RST: 0, MST: 0
BPDU Received       :818
                    TCN: 0, Config: 818, RST: 0, MST: 0

```

```
<S1>display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/9	ALTE	DISCARDING	NONE
0	GigabitEthernet0/0/10	ROOT	FORWARDING	NONE

The highlighted lines in the preceding information indicates that G0/0/10 has become the root port.

Final Configuration

```

<S1>display current-configuration
#
!Software Version V200R008C00SPC500
sysname S1
#
stp mode stp
stp instance 0 priority 8192
#
interface GigabitEthernet0/0/1
shutdown
#
interface GigabitEthernet0/0/2
shutdown
#
interface GigabitEthernet0/0/3
shutdown
#
interface GigabitEthernet0/0/9
stp instance 0 cost 200000
#
interface GigabitEthernet0/0/10
#
interface GigabitEthernet0/0/13
shutdown

```

```
#
interface GigabitEthernet0/0/14
 shutdown
#
user-interface con 0
user-interface vty 0 4
#
return

<S2>display current-configuration
#
!Software Version V200R008C00SPC500
 sysname S2
#
 stp mode stp
 stp instance 0 priority 4096
#
interface GigabitEthernet0/0/1
 shutdown
#
interface GigabitEthernet0/0/2
 shutdown
#
interface GigabitEthernet0/0/3
 shutdown
#
interface GigabitEthernet0/0/6
 shutdown
#
interface GigabitEthernet0/0/7
 shutdown
#
interface GigabitEthernet0/0/9
#
interface GigabitEthernet0/0/10
#
user-interface con 0
user-interface vty 0 4
#
return

<S3>display current-configuration
```

```
#
!Software Version V100R006C05
 sysname S3
#
interface Ethernet0/0/1
 shutdown
#
interface Ethernet0/0/13
 shutdown
#
interface Ethernet0/0/7
 shutdown
#
user-interface con 0
user-interface vty 0 4
#
return
```

```
<S4>display current-configuration
```

```
#
!Software Version V100R006C05
 sysname S4
#
interface Ethernet0/0/14
 shutdown
#
interface Ethernet0/0/1
 shutdown
#
interface Ethernet0/0/6
 shutdown
#
user-interface con 0
user-interface vty 0 4
#
return
```

Lab 3-2 Configuring RSTP

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Enable and disable RSTP .
- Configuration of an edge port.
- Configuration of RSTP BPDU protection.
- Configuration of RSTP loop protection

Topology

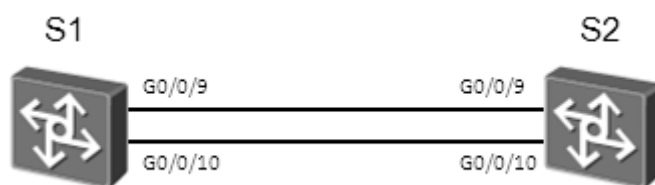


Figure 3.2 RSTP topology

Scenario

Assume that you are a network administrator of a company. The company network consists of two layers: core layer and access layer. The network uses a redundancy design. RSTP will be used to prevent loops. You can configure features to speed up RSTP route convergence at the edge network and configure RSTP protection function.

Tasks

Step 1 Preparing the environment

If you are starting this section with a non-configured device, begin here and then move to step 3. For those continuing from previous labs, begin at step 2.

Irrelevant interfaces must be disabled to ensure test result accuracy.

Shut down port interfaces Ethernet 0/0/1 on S3, Ethernet 0/0/13 and Ethernet 0/0/7 on S3; GigabitEthernet 0/0/1, GigabitEthernet 0/0/2, GigabitEthernet 0/0/3, GigabitEthernet 0/0/13, GigabitEthernet 0/0/14 on S1; GigabitEthernet 0/0/1, GigabitEthernet 0/0/2, GigabitEthernet 0/0/3, GigabitEthernet 0/0/6, GigabitEthernet 0/0/7 on S2; as well as Ethernet 0/0/1, Ethernet 0/0/14 and Ethernet 0/0/6 on S4 before starting STP configuration. Ensure that the devices start without any configuration files. If STP is disabled, run the **stp enable** command to enable STP.

```
<Quidway>system-view
Enter system view, return user view with Ctrl+Z.
```

```
[Quidway]sysname S1
[S1]interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1]shutdown
[S1-GigabitEthernet0/0/1]quit
[S1]interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2]shutdown
[S1-GigabitEthernet0/0/2]quit
[S1]interface GigabitEthernet 0/0/3
[S1-GigabitEthernet0/0/3]shutdown
[S1-GigabitEthernet0/0/3]quit
[S1]interface GigabitEthernet 0/0/13
[S1-GigabitEthernet0/0/13]shutdown
[S1-GigabitEthernet0/0/13]quit
[S1]interface GigabitEthernet 0/0/14
[S1-GigabitEthernet0/0/14]shutdown
[S1-GigabitEthernet0/0/14]quit
```

```
<Quidway>system-view
Enter system view, return user view with Ctrl+Z.
[Quidway]sysname S2
[S2]interface GigabitEthernet 0/0/1
```

```
[S2-GigabitEthernet0/0/1]shutdown
[S2-GigabitEthernet0/0/1]quit
[S2]interface GigabitEthernet 0/0/2
[S2-GigabitEthernet0/0/2]shutdown
[S2-GigabitEthernet0/0/2]quit
[S2]interface GigabitEthernet 0/0/3
[S2-GigabitEthernet0/0/3]shutdown
[S2-GigabitEthernet0/0/3]quit
[S2]interface GigabitEthernet 0/0/6
[S2-GigabitEthernet0/0/6]shutdown
[S2-GigabitEthernet0/0/6]quit
[S2]interface GigabitEthernet 0/0/7
[S2-GigabitEthernet0/0/7]shutdown
[S2-GigabitEthernet0/0/7]quit

<Quidway>system-view
Enter system view, return user view with Ctrl+Z.
[Quidway]sysname S3
[S3]interface Ethernet 0/0/1
[S3-Ethernet0/0/1]shutdown
[S3-Ethernet0/0/1]quit
[S3]interface Ethernet 0/0/13
[S3-Ethernet0/0/13]shutdown
[S3-Ethernet0/0/13]quit
[S3]interface Ethernet 0/0/7
[S3-Ethernet0/0/7]shutdown

<Quidway>system-view
Enter system view, return user view with Ctrl+Z.
[Quidway]sysname S4
[S4]inter Ethernet 0/0/1
[S4-Ethernet0/0/1]shutdown
[S4-Ethernet0/0/1]quit
[S4]inter Ethernet 0/0/14
[S4-Ethernet0/0/14]shutdown
[S4-Ethernet0/0/14]quit
[S4]interface Ethernet 0/0/6
[S4-Ethernet0/0/6]shutdown
```

Step 2 Clean up the previous configuration

Remove the configured STP priority from S1 and S2, and assigned cost on S1.

```
[S1]undo stp priority
[S1]inter GigabitEthernet 0/0/9
[S1-GigabitEthernet0/0/9]undo stp cost
```

```
[S2]undo stp priority
```

Step 3 Configure RSTP and verify the RSTP configuration.

Configure S1 and S2 to use RSTP as the spanning tree protocol.

```
[S1]stp mode rstp
```

```
[S2]stp mode rstp
```

Run the **display stp** command to view brief information about RSTP.

```
[S1]display stp
-----[CIST Global Info] [Mode RSTP]-----
CIST Bridge           :32768.d0d0-4ba6-aab0
Config Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC        :32768.d0d0-4ba6-aab0 / 0 (This bridge is the root)
CIST RegRoot/IRPC     :32768.d0d0-4ba6-aab0 / 0
CIST RootPortId       :0.0
BPDU-Protection       :Disabled
TC or TCN received    :362
TC count per hello    :0
STP Converge Mode     :Normal
Share region-configuration :Enabled
Time since last TC    :0 days 0h:0m:45s
.....output omit.....
```

```
[S2]display stp
-----[CIST Global Info] [Mode RSTP]-----
CIST Bridge           :32768.d0d0-4ba6-ac20
Config Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC        :32768.d0d0-4ba6-aab0 / 20000
```

```
CIST RegRoot/IRPC      :32768.d0d0-4ba6-ac20 / 0
CIST RootPortId        :128.34 (GigabitEthernet0/0/9)
BPDU-Protection         :Disabled
TC or TCN received     :186
TC count per hello      :0
STP Converge Mode       :Normal
Share region-configuration :Enabled
Time since last TC      :0 days 0h:3m:55s
.....output omit.....
```

Step 4 **Configure an edge port.**

Configure ports connected to the user terminals as edge ports. An edge port can transition to the forwarding state without participating in the RSTP calculation. In this example, interface GigabitEthernet 0/0/1 on S1 and S2 connect to a router and can be configured as edge ports.

```
[S1]interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1]undo shutdown
[S1-GigabitEthernet0/0/1]stp edged-port enable

[S2]interface GigabitEthernet 0/0/1
[S2-GigabitEthernet0/0/1]undo shutdown
[S2-GigabitEthernet0/0/1]stp edged-port enable
```

Step 5 **Configure BPDU protection.**

Edge ports are directly connected to user terminal and will not receive BPDUs. Attackers may send pseudo BPDUs to attack the switching device. If the edge ports receive the BPDUs, the switching device configures the edge ports as non-edge ports and triggers a new spanning tree calculation. Network flapping then occurs. BPDU protection can be used to protect switching devices against malicious attacks.

Configure BPDU protection on both S1 and S2.

```
[S1]stp bpdu-protection

[S2]stp bpdu-protection
```

Run the **display stp brief** command to view the port protection.

```
<S1>display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/1	DESI	FORWARDING	BPDU
0	GigabitEthernet0/0/9	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/10	DESI	FORWARDING	NONE

```
<S2>display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/1	DESI	FORWARDING	BPDU
0	GigabitEthernet0/0/9	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/10	ALTE	DISCARDING	NONE

After the configuration is complete, interface Gigabit Ethernet 0/0/1 on S1 and S2 shows as supporting BPDU protection.

Step 6 Configure Loop protection

On a network running RSTP, a switching device maintains the root port status and status of alternate ports by receiving BPDUs from an upstream switching device. If the switching device cannot receive BPDUs from the upstream device because of link congestion or unidirectional-link failure, the switching device re-selects a root port. The original root port becomes a designated port and the original discarding ports change to the Forwarding state. This switching may cause network loops, which can be mitigated by configuring loop protection.

Configure loop protection on both the root port and the alternate port.

```
[S2]display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/1	DESI	FORWARDING	BPDU
0	GigabitEthernet0/0/9	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/10	ALTE	DISCARDING	NONE

G0/0/9 and G0/0/10 on S2 are now the root port and alternate port. Configure loop protection on these two ports.

```
[S2]interface GigabitEthernet 0/0/9
[S2-GigabitEthernet0/0/9]stp loop-protection
[S2-GigabitEthernet0/0/9]quit
[S2]interface GigabitEthernet 0/0/10
[S2-GigabitEthernet0/0/10]stp loop-protection
```

Run the **display stp brief** command to view the port protection.

```
<S2>display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/1	DESI	FORWARDING	BPDU
0	GigabitEthernet0/0/9	ROOT	FORWARDING	LOOP
0	GigabitEthernet0/0/10	ALTE	DISCARDING	LOOP

Since S1 is root, all the ports are designated ports and therefore do not need to configure loop protection. After completing the configuration, you may wish to set S2 as the root, and configure loop protection on the root port and alternate port of S1 using the same process as with S2.

Final Configuration

```
<S1>display current-configuration
```

```
#
```

```
!Software Version V200R008C00SPC500
```

```
sysname S1
```

```
#
```

```
stp mode rstp
```

```
stp bpdu-protection
```

```
#
```

```
interface GigabitEthernet0/0/1
```

```
undo shutdown
```

```
stp edged-port enable
```

```
#
```

```
interface GigabitEthernet0/0/2
```

```
shutdown
```

```
#
```

```
interface GigabitEthernet0/0/3
```

```
shutdown
```

```
#
```

```
interface GigabitEthernet0/0/13
```

```
shutdown
```

```
#
```

```
interface GigabitEthernet0/0/14
```

```
shutdown
```

```
#
```

```
user-interface con 0
```

```
user-interface vty 0 4
```

```
#
```

```
return

<S2>display current-configuration
#
!Software Version V200R008C00SPC500
sysname S2
#
stp mode rstp
stp bpdu-protection
#
interface GigabitEthernet0/0/1
undo shutdown
stp edged-port enable
#
interface GigabitEthernet0/0/2
shutdown
#
interface GigabitEthernet0/0/3
shutdown
#
interface GigabitEthernet0/0/6
shutdown
#
interface GigabitEthernet0/0/7
shutdown
#
interface GigabitEthernet0/0/9
stp loop-protection
#
interface GigabitEthernet0/0/10
stp loop-protection
#
user-interface con 0
user-interface vty 0 4
#
return
```

```
<S3>display current-configuration
#
!Software Version V100R006C05
sysname S3
#
```

```
interface Ethernet0/0/1
 shutdown
#
interface Ethernet0/0/13
 shutdown
#
interface Ethernet0/0/7
 shutdown
#
user-interface con 0
user-interface vty 0 4
#
return

<S4>display current-configuration
#
!Software Version V100R006C05
 sysname S4
#
interface Ethernet0/0/14
 shutdown
#
interface Ethernet0/0/1
 shutdown
#

interface Ethernet0/0/6
 shutdown
#
user-interface con 0
user-interface vty 0 4
#
return
```

Module 4 Routing Configuration

Lab 4-1 Configuring Static Routes and Default Routes

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Configuration of a static route using an interface and an IP address as the next hop.
- Verification of static route operation.
- Implementation of the interconnection between a local and external network using a default route.
- Configuration of a backup static route on a router.

Topology

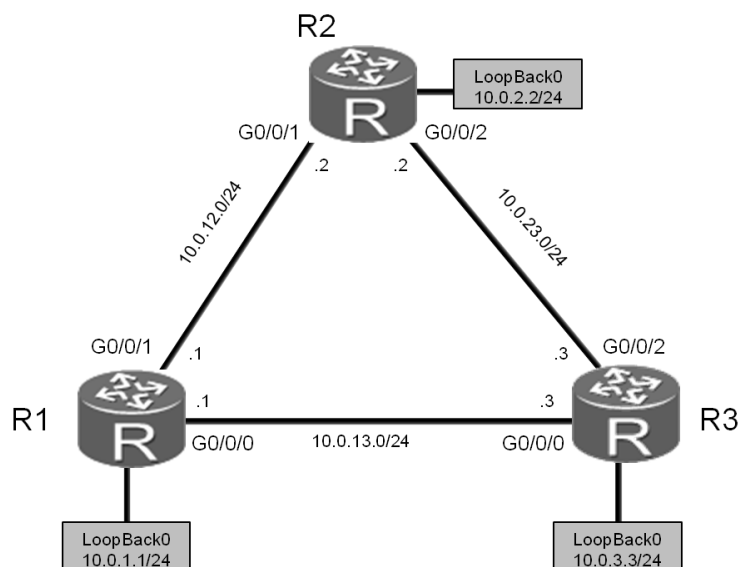


Figure 4.1 Lab topology for static and default routes

Scenario

Assume that you are a network administrator of a company that contains a single administrative domain and within the administrative domain, multiple networks have been defined, for which currently no method of routing exists.

Since the network scale is small, with only a few networks, static routes and default routes are to be used to implement interwork communication. The network addressing is to be applied as shown in Figure 4.1.

If a password is requested, and unless otherwise stated, please use the password: huawei

Tasks

Step 1 Perform basic system and IP address configuration.

Configure the device names and IP addresses for R1, R2, and R3.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.0.13.1 24
[R1-GigabitEthernet0/0/0]quit
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.0.12.1 24
[R1-GigabitEthernet0/0/1]quit
[R1]interface LoopBack 0
[R1-LoopBack0]ip address 10.0.1.1 24
```

Run the **display current-configuration** command to check the configuration.

```
<R1>display ip interface brief
Interface                IP Address/Mask      Physical  Protocol
.....output omitted.....
GigabitEthernet0/0/0     10.0.13.1/24         up        up
GigabitEthernet0/0/1     10.0.12.1/24         up        up
GigabitEthernet0/0/2     unassigned           up        down
LoopBack0                10.0.1.1/24          up        up(s)
```

.....output omitted.....

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet0/0/1]ip address 10.0.12.2 24
[R2-GigabitEthernet0/0/1]quit
[R2]interface GigabitEthernet0/0/2
[R2-GigabitEthernet0/0/2]ip add 10.0.23.2 24
[R2-GigabitEthernet0/0/2]quit
[R2]interface LoopBack0
[R2-LoopBack0]ip address 10.0.2.2 24
```

```
<R2>display ip interface brief
```

Interface	IP Address/Mask	Physical	Protocol
-----------	-----------------	----------	----------

.....output omitted.....

GigabitEthernet0/0/0	unassigned	up	down
GigabitEthernet0/0/1	10.0.12.2/24	up	up
GigabitEthernet0/0/2	10.0.23.2/24	up	up
LoopBack0	10.0.2.2/24	up	up(s)

.....output omitted.....

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]ip address 10.0.13.3 24
[R3-GigabitEthernet0/0/0]quit
[R3]interface GigabitEthernet0/0/2
[R3-GigabitEthernet0/0/2]ip address 10.0.23.3 24
[R3-GigabitEthernet0/0/2]quit
[R3]interface LoopBack 0
[R3-LoopBack0]ip address 10.0.3.3 24
```

```
<R3>display ip interface brief
```

Interface	IP Address/Mask	Physical	Protocol
-----------	-----------------	----------	----------

.....output omitted.....

GigabitEthernet0/0/0	10.0.13.3/24	up	up
GigabitEthernet0/0/1	unassigned	up	down
GigabitEthernet0/0/2	10.0.23.3/24	up	up

```
LoopBack0          10.0.3.3/24          up          up(s)
.....output omitted.....
```

Use the **ping** command to test network connectivity from R1.

```
<R1>ping 10.0.12.2
PING 10.0.12.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=30 ms
  Reply from 10.0.12.2: bytes=56 Sequence=2 ttl=255 time=30 ms
  Reply from 10.0.12.2: bytes=56 Sequence=3 ttl=255 time=30 ms
  Reply from 10.0.12.2: bytes=56 Sequence=4 ttl=255 time=30 ms
  Reply from 10.0.12.2: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 10.0.12.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 30/30/30 ms

<R1>ping 10.0.13.3
PING 10.0.13.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.13.3: bytes=56 Sequence=1 ttl=255 time=6 ms
  Reply from 10.0.13.3: bytes=56 Sequence=2 ttl=255 time=2 ms
  Reply from 10.0.13.3: bytes=56 Sequence=3 ttl=255 time=2 ms
  Reply from 10.0.13.3: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.0.13.3: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.0.13.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/6 ms
```

Use the ping command to test network connectivity from R2

```
<R2>ping 10.0.23.3
PING 10.0.23.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=31 ms
  Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=255 time=31 ms
  Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=255 time=41 ms
  Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=255 time=31 ms
  Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=255 time=41 ms
```

```
--- 10.0.23.3 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 31/35/41 ms
```

Step 2 Test connectivity

Use the ping command to test network connectivity from R2 to networks 10.0.13.0/24 and 10.0.3.0/24

```
<R2>ping 10.0.13.3
PING 10.0.13.3: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out
```

```
--- 10.0.13.3 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

```
<R2>ping 10.0.3.3
PING 10.0.3.3: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out
```

```
--- 10.0.3.3 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

If R2 wishes to communicate with the network segment 10.0.3.0, a route destined for this network segment must be configured on R2, and routes destined for the R2 interface must be configured on R3.

The preceding test result shows that R2 cannot communicate with 10.0.3.3 and 10.0.13.3.

Run the **display ip routing-table** command to view the routing table of R2. The routing table does not contain the routes of the two networks.

```
<R2>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----  
Routing Tables: Public
```

```
Destinations : 13      Routes : 13
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.2.0/24	Direct	0	0	D	10.0.2.2	LoopBack0
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GigabitEthernet0/0/2
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/2
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Step 3 Configure static routes on R2.

Configure a static route for destination networks 10.0.13.0/24 and 10.0.3.0/24, with the next hop set as the IP address 10.0.23.3 of R3, a preference value of 60 is the default and need not be set.

```
[R2]ip route-static 10.0.13.0 24 10.0.23.3
```

```
[R2]ip route-static 10.0.3.0 24 10.0.23.3
```

Note: In the **ip route-static** command, **24** indicates the subnet mask length, which can also be expressed using the decimal format 255.255.255.0.

```
<R2>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.3.0/24	Static	60	0	RD	10.0.23.3	GigabitEthernet0/0/2
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	Static	60	0	RD	10.0.23.3	GigabitEthernet0/0/2
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GigabitEthernet0/0/2
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/2

Step 4 Configure backup static routes.

The data exchanged between R2 and 10.0.13.3 and 10.0.3.3 is transmitted through the link between R2 and R3. R2 fails to communicate with 10.0.13.3 and 10.0.3.3 if the link between R2 and R3 is faulty.

According to the topology, R2 can communicate with R3 through R1 if the link between R2 and R3 fails. A backup static route can be configured to enable this redundancy. Backup static routes do not take effect in normal cases. If the link between R2 and R3 fails, backup static routes are used to transfer data.

Amend the preferences for the backup static routes to ensure that the routes are used only when the primary link fails. In this example, the preference of the backup static route is set to 80.

```
[R1]ip route-static 10.0.3.0 24 10.0.13.3
```

```
[R2]ip route-static 10.0.13.0 255.255.255.0 10.0.12.1 preference 80
```

```
[R2]ip route-static 10.0.3.0 24 10.0.12.1 preference 80
```

```
[R3]ip route-static 10.0.12.0 24 10.0.13.1
```

Step 5 Test the static routes.

View the current static route configuration in the routing table of R2.

```
<R2>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----
Routing Tables: Public
```

```
Destinations : 15      Routes : 15
```

```
Destination/Mask Proto Pre Cost Flags NextHop Interface
```

10.0.2.0/24	Direct	0	0	D	10.0.2.2	LoopBack0
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.0/24	Static	60	0	RD	10.0.23.3	GigabitEthernet0/0/2
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	Static	60	0	RD	10.0.23.3	GigabitEthernet0/0/2
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GigabitEthernet0/0/2
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/2
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The routing table contains two static routes that were configured in step 3. The value of the **Protocol** field is **Static**, indicating a static route. The value of the **Preference** field is **60**, indicating the default preference is used for the route.

Test network connectivity to ensure the route between R2 and R3 exists.

```
<R2>ping 10.0.13.3
PING 10.0.13.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.13.3: bytes=56 Sequence=1 ttl=255 time=34 ms
  Reply from 10.0.13.3: bytes=56 Sequence=2 ttl=255 time=34 ms
  Reply from 10.0.13.3: bytes=56 Sequence=3 ttl=255 time=34 ms
  Reply from 10.0.13.3: bytes=56 Sequence=4 ttl=255 time=34 ms
  Reply from 10.0.13.3: bytes=56 Sequence=5 ttl=255 time=34 ms

--- 10.0.13.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 34/34/34 ms
```

```
<R2>ping 10.0.3.3
PING 10.0.3.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.3.3: bytes=56 Sequence=1 ttl=255 time=41 ms
  Reply from 10.0.3.3: bytes=56 Sequence=2 ttl=255 time=41 ms
  Reply from 10.0.3.3: bytes=56 Sequence=3 ttl=255 time=41 ms
  Reply from 10.0.3.3: bytes=56 Sequence=4 ttl=255 time=41 ms
  Reply from 10.0.3.3: bytes=56 Sequence=5 ttl=255 time=41 ms
```

```
--- 10.0.3.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
round-trip min/avg/max = 41/41/41 ms
```

The command output shows that the route is functioning normally. The **tracert** command can also be run to view the path over which the data is transferred.

```
<R2>tracert 10.0.13.3
tracert to 10.0.13.3(10.0.13.3), max hops: 30 ,packet length: 40,
press CTRL_C to break
 1 10.0.23.3 40 ms 31 ms 30 ms

<R2>tracert 10.0.3.3
tracert to 10.0.3.3(10.0.3.3), max hops: 30 ,packet length: 40,
press CTRL_C to break
 1 10.0.23.3 40 ms 30 ms 30 ms
```

The command output verifies that R2 directly sends data to R3.

Step 6 Test the backup static routes.

Disable the path to 10.0.23.3 via GigabitEthernet0/0/2 on R2 and observe the changes in the IP routing tables.

```
[R2]interface GigabitEthernet0/0/2
[R2-GigabitEthernet0/0/2]shutdown
[R2-GigabitEthernet0/0/2]quit
```

Compare the routing tables with the previous routing tables before Gigabit Ethernet 0/0/2 was disabled.

```
<R2>display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 12      Routes : 12

Destination/Mask    Proto    Pre  Cost   Flags NextHop         Interface
10.0.2.0/24         Direct   0     0        D  10.0.2.2         LoopBack0
```

10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.0/24	Static	80	0	RD	10.0.12.1	GigabitEthernet0/0/1
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	Static	80	0	RD	10.0.12.1	GigabitEthernet0/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The next hops and preferences of the two routes as shown in the preceding routing table for R2 have changed.

Test connectivity between R2 and the destination addresses 10.0.13.3 and 10.0.3.3 on R2.

<R2>ping 10.0.3.3

PING 10.0.3.3: 56 data bytes, press CTRL_C to break

Reply from 10.0.3.3: bytes=56 Sequence=1 ttl=255 time=3 ms

Reply from 10.0.3.3: bytes=56 Sequence=2 ttl=255 time=2 ms

Reply from 10.0.3.3: bytes=56 Sequence=3 ttl=255 time=2 ms

Reply from 10.0.3.3: bytes=56 Sequence=4 ttl=255 time=2 ms

Reply from 10.0.3.3: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.0.3.3 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 2/2/3 ms

<R2>ping 10.0.13.3

PING 10.0.13.3: 56 data bytes, press CTRL_C to break

Reply from 10.0.13.3: bytes=56 Sequence=1 ttl=255 time=3 ms

Reply from 10.0.13.3: bytes=56 Sequence=2 ttl=255 time=2 ms

Reply from 10.0.13.3: bytes=56 Sequence=3 ttl=255 time=2 ms

Reply from 10.0.13.3: bytes=56 Sequence=4 ttl=255 time=2 ms

Reply from 10.0.13.3: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.0.13.3 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

```
0.00% packet loss
round-trip min/avg/max = 2/2/3 ms
```

The network is not disconnected when the link between R2 and R3 is shut down.

The **tracert** command can also be run to view through over which path the data is being forwarded.

```
<R2>tracert 10.0.13.3
  traceroute to 10.0.13.3(10.0.13.3), max hops: 30 ,packet length: 40,press CTRL_C to break
 1 10.0.12.1 40 ms 21 ms 21 ms
 2 10.0.13.3 30 ms 21 ms 21 ms

<R2>tracert 10.0.3.3
  traceroute to 10.0.3.3(10.0.3.3), max hops: 30 ,packet length: 40,press CTRL_C to break
 1 10.0.12.1 40 ms 21 ms 21 ms
 2 10.0.13.3 30 ms 21 ms 21 ms
```

The command output shows that the data sent by R2 reaches R3 via the 10.0.12.0 and 10.0.13.0 networks connected to R1.

Step 7 Using default routes to implement network connectivity.

Enable the interface that was disabled in step 6 on R2.

```
[R2]interface GigabitEthernet 0/0/2
[R2-GigabitEthernet0/0/2]undo shutdown
```

Verify connectivity to the network 10.0.23.0 from R1.

```
[R1]ping 10.0.23.3
  PING 10.0.23.3: 56 data bytes, press CTRL_C to break
    Request time out
    Request time out
    Request time out
    Request time out
    Request time out

  --- 10.0.23.3 ping statistics ---
    5 packet(s) transmitted
    0 packet(s) received
  100.00% packet loss
```

R3 cannot be reached because the route destined for 10.0.23.3 is not configured on R1.

```
<R1>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----  
Routing Tables: Public
```

Destinations : 14	Routes : 14						
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface	
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0	
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0	
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0	
10.0.3.0/24	Static	60	0	RD	10.0.13.3	GigabitEthernet0/0/0	
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet0/0/1	
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1	
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1	
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/0	
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0	
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0	
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0	
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0	
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0	
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0	

A default route can be configured on R1 to implement network connectivity via a next hop of 10.0.13.3.

```
[R1]ip route-static 0.0.0.0 0.0.0.0 10.0.13.3
```

After the configuration is complete, test connectivity between R1 and 10.0.23.3.

```
<R1>ping 10.0.23.3
```

```
PING 10.0.23.3: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=3 ms
```

```
Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=255 time=2 ms
```

```
Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=255 time=2 ms
```

```
Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=255 time=2 ms
```

```
Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=255 time=2 ms
```

```
--- 10.0.23.3 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/2/3 ms
```

The default route forwards traffic destined for 10.0.23.3 to the next hop of 10.0.13.3 on R3. R3 is directly connected to the 10.0.23.0 network.

Step 8 Configure a backup default route.

If the link between R1 and R3 fails, a backup default route can be used to communicate with 10.0.23.3 and 10.0.3.3 via the 10.0.12.0 network.

However, R1 is not directly connected to these networks and therefore a backup route (in both directions) must be configured to provide a forwarding path.

```
[R1]ip route-static 0.0.0.0 0.0.0.0 10.0.12.2 preference 80
```

```
[R3]ip route-static 10.0.12.0 24 10.0.23.2 preference 80
```

Step 9 Test the backup default route.

View the routes of R1 when the link between R1 and R3 is operational.

```
<R1>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----
Routing Tables: Public
```

```
Destinations : 15      Routes : 15
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Static	60	0	RD	10.0.13.3	GigabitEthernet0/0/0
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.0/24	Static	60	0	RD	10.0.13.3	GigabitEthernet0/0/0
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet0/0/1
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/0
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
```

Disable Gigabit Ethernet 0/0/0 on R1 and disable interface Gigabit Ethernet 0/0/0 on R3 to simulate a link failure, and then view the routes of R1. Compare the current routes with the routes before Gigabit Ethernet 0/0/0 was disabled.

```
[R1]interface GigabitEthernet0/0/0
[R1-GigabitEthernet0/0/0]shutdown
[R1-GigabitEthernet0/0/0]quit
```

```
[R3]interface GigabitEthernet0/0/0
[R3-GigabitEthernet0/0/0]shutdown
[R3-GigabitEthernet0/0/0]quit
```

```
<R1>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----
Routing Tables: Public
```

Destinations : 11	Routes : 11					
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Static	80	0	RD	10.0.12.2	GigabitEthernet0/0/1
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet0/0/1
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

According to the preceding routing table, the value of **80** in the Preference column indicates that the backup default route 0.0.0.0 is actively forwarding traffic to the next hop of 10.0.23.3.

Test network connectivity on R1.

```
<R1>ping 10.0.23.3
```

```
PING 10.0.23.3: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=254 time=76 ms
```

```
Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=254 time=250 ms
```

```

Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=254 time=76 ms
Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=254 time=76 ms
Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=254 time=76 ms
--- 10.0.23.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
    round-trip min/avg/max = 76/110/250 ms
<R1>tracert 10.0.23.3
tracert to 10.0.23.3(10.0.23.2), max hops: 30 ,packet length: 40,press CTRL_C to break
 1 10.0.12.2 30 ms 26 ms 26 ms
 2 10.0.23.3 60 ms 53 ms 56 ms

```

The IP packets are reaching R3 (10.0.23.3) via the next hop 10.0.12.2 of R2.

Final Configuration

```

<R1>dis current-configuration
[V200R007C00SPC600]
#
 sysname R1
#
interface GigabitEthernet0/0/0
 shutdown
 ip address 10.0.13.1 255.255.255.0
#
interface GigabitEthernet0/0/1
 ip address 10.0.12.1 255.255.255.0
#
interface LoopBack0
 ip address 10.0.1.1 255.255.255.0
#
ip route-static 0.0.0.0 0.0.0.0 10.0.13.3
ip route-static 0.0.0.0 0.0.0.0 10.0.12.2 preference 80
ip route-static 10.0.3.0 255.255.255.0 10.0.13.3
#
user-interface con 0
 authentication-mode password
 set authentication password cipher %$%$+L'YR&IZt'4,) >-*#lH",}%K-oJ_M9+'lOU~bD (\WTqB}%N,%
 $$$$
user-interface vty 0 4
#

```

```

return
<R2>display current-configuration
[V200R007C00SPC600]
#
 sysname R2
interface GigabitEthernet0/0/1
 ip address 10.0.12.2 255.255.255.0
#
interface GigabitEthernet0/0/2
 ip address 10.0.23.2 255.255.255.0
#
interface LoopBack0
 ip address 10.0.2.2 255.255.255.0
#
ip route-static 10.0.3.0 255.255.255.0 10.0.23.3
ip route-static 10.0.3.0 255.255.255.0 10.0.12.1 preference 80
ip route-static 10.0.13.0 255.255.255.0 10.0.23.3
ip route-static 10.0.13.0 255.255.255.0 10.0.12.1 preference 80
#
user-interface con 0
 authentication-mode password
 set authentication password cipher %$%$1=cd%b%/O%Id-8X:by1N,+s}'4wD6TvO<I||/pd#
#44C@+s#,%$%$
user-interface vty 0 4
#
return

<R3>display current-configuration
[V200R007C00SPC600]
#
 sysname R3
#
interface GigabitEthernet0/0/0
 shutdown
 ip address 10.0.13.3 255.255.255.0
#
interface GigabitEthernet0/0/2
 ip address 10.0.23.3 255.255.255.0
#
interface LoopBack0
 ip address 10.0.3.3 255.255.255.0
#

```

```
ip route-static 10.0.12.0 255.255.255.0 10.0.13.1
ip route-static 10.0.12.0 255.255.255.0 10.0.23.2 preference 80
#
user-interface con 0
 authentication-mode password
  set authentication password cipher %$$$ksXDMg7Ry6yUU:63:DQ),#/sQg"@*S\U#.s.bHW
xQ,y%#/v,%$$$
user-interface vty 0 4
#
return
```

Lab 4-2 Configuring RIPv1 and RIPv2

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Establish routing loop prevention mechanisms for RIP
- Configuration of RIPv1.
- Enable RIP for a specified network and interface.
- Use of the **display** and **debugging** commands to view RIP operation.
- Procedure for testing connectivity of the RIP network.
- Configuration of RIPv2.

Topology

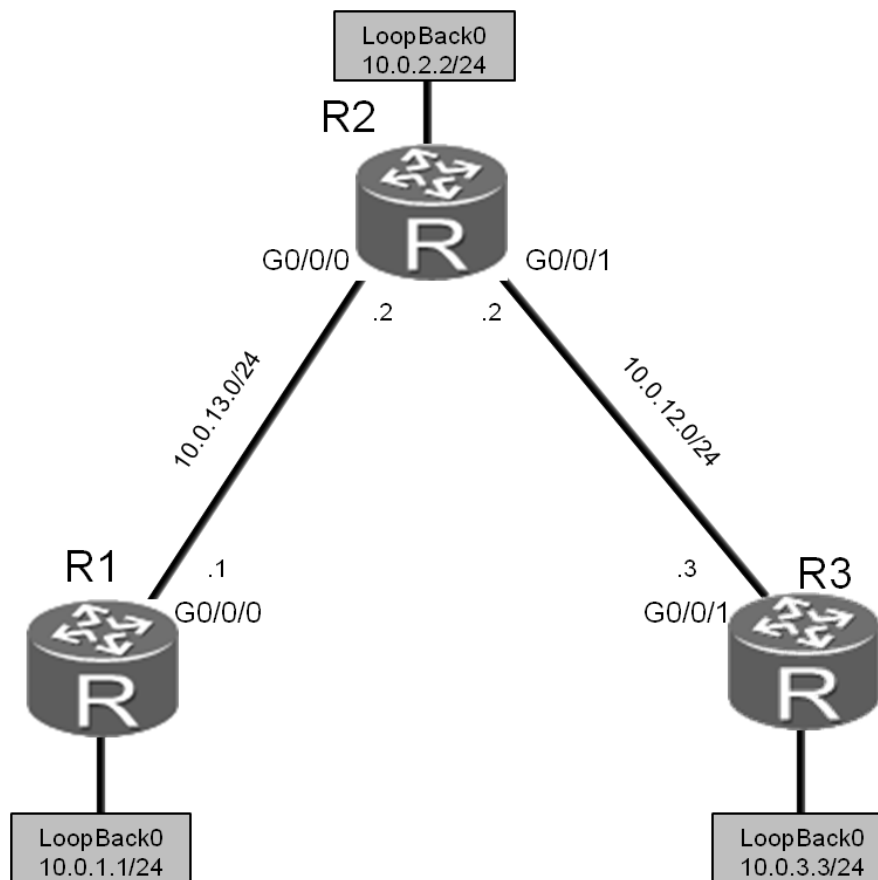


Figure 4.2 Lab topology for RIPv1 and RIPv2

Scenario

Assume that you are a network administrator in charge of managing a small administrative domain consisting of three routers and five networks. Due to the limited requirement, RIP is to be used to support routing. RIPv1 is initially configured, but you realize that RIPv2 has many advantages. After some consideration, you transition the domain to support RIPv2.

Tasks

Step 1 **Preparing the environment.**

If you are starting this section with a non-configured device begin here and then move to step 3. For those continuing from previous labs, begin at step 2.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.0.13.1 24
[R1-GigabitEthernet0/0/0]quit
[R1]interface LoopBack 0
[R1-LoopBack0]ip address 10.0.1.1 24
```

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet0/0/1]ip address 10.0.12.2 24
[R2-GigabitEthernet0/0/1]quit
[R2]interface LoopBack 0
[R2-LoopBack0]ip address 10.0.2.2 24
```

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]interface LoopBack 0
[R3-LoopBack0]ip address 10.0.3.3 24
```

Step 2 **Clean up the previous configuration**

Clean up previous static route configuration and disable all unused interfaces

```
[R1]interface GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1]shutdown
[R1-GigabitEthernet0/0/1]quit
```

```
[R1]interface GigabitEthernet0/0/0
[R1-GigabitEthernet0/0/0]undo shutdown
[R1-GigabitEthernet0/0/0]quit
[R1]undo ip route-static 0.0.0.0 0.0.0.0
[R1]undo ip route-static 10.0.3.0 255.255.255.0
[R1]undo ip route-static 10.0.12.0 255.255.255.0
```

```
[R2]interface GigabitEthernet 0/0/2
[R2-GigabitEthernet0/0/2]shutdown
[R2-GigabitEthernet0/0/2]quit
[R2]undo ip route-static 10.0.3.0 255.255.255.0
[R2]undo ip route-static 10.0.13.0 255.255.255.0
```

```
[R3]interface GigabitEthernet 0/0/2
[R3-GigabitEthernet0/0/2]shutdown
[R3-GigabitEthernet0/0/2]quit
[R3]undo ip route-static 10.0.12.0 255.255.255.0
```

Step 3 **Additional address configuration**

Configure the following additional interfaces for R2 and R3.

```
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]ip address 10.0.13.2 24

[R3]interface GigabitEthernet0/0/1
[R3-GigabitEthernet0/0/1]ip address 10.0.12.3 24
```

Verify that R1 and R2 can communicate with one another over the 10.0.13.0 network.

```
<R1>ping 10.0.13.2
PING 10.0.13.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.13.2: bytes=56 Sequence=1 ttl=255 time=30 ms
  Reply from 10.0.13.2: bytes=56 Sequence=2 ttl=255 time=30 ms
  Reply from 10.0.13.2: bytes=56 Sequence=3 ttl=255 time=30 ms
  Reply from 10.0.13.2: bytes=56 Sequence=4 ttl=255 time=30 ms
  Reply from 10.0.13.2: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 10.0.13.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 30/30/30 ms
```

Verify that R2 can successfully reach R3 over the 10.0.12.0 network.

```
<R2>ping 10.0.12.3
PING 10.0.12.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.3: bytes=56 Sequence=1 ttl=255 time=31 ms
  Reply from 10.0.12.3: bytes=56 Sequence=2 ttl=255 time=31 ms
  Reply from 10.0.12.3: bytes=56 Sequence=3 ttl=255 time=41 ms
  Reply from 10.0.12.3: bytes=56 Sequence=4 ttl=255 time=31 ms
  Reply from 10.0.12.3: bytes=56 Sequence=5 ttl=255 time=41 ms

--- 10.0.12.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 31/35/41 ms
```

Step 4 **Configure RIPv1.**

Enable RIP on R1, and then advertise the 10.0.0.0 network segment.

```
[R1]rip 1
[R1-rip-1]network 10.0.0.0
```

Enable RIP on R2, and then advertise the 10.0.0.0 network segment.

```
[R2]rip 1
[R2-rip-1]network 10.0.0.0
```

Enable RIP on R3, and then advertise the 10.0.0.0 network segment.

```
[R3]rip 1
[R3-rip-1]network 10.0.0.0
```

Step 5 Verify RIPv1 routes.

View the routing tables of R1, R2, and R3. Make sure that these routers have learned the RIP routes that are highlighted in gray in the following command output.

```
<R1>display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 13 Routes : 13

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.0/24	RIP	100	1	D	10.0.13.2	GigabitEthernet0/0/0
10.0.3.0/24	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
10.0.12.0/24	RIP	100	1	D	10.0.13.2	GigabitEthernet0/0/0
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/0
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
<R2>display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 15 Routes : 15

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.0/24	RIP	100	1	D	10.0.13.1	GigabitEthernet0/0/0

10.0.2.0/24	Direct	0	0	D	10.0.2.2	LoopBack0
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.0/24	RIP	100	1	D	10.0.12.3	GigabitEthernet0/0/1
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	Direct	0	0	D	10.0.13.2	GigabitEthernet0/0/0
10.0.13.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

<R3>display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 13 Routes : 13

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.0/24	RIP	100	2	D	10.0.12.2	GigabitEthernet0/0/1
10.0.2.0/24	RIP	100	1	D	10.0.12.2	GigabitEthernet0/0/1
10.0.3.0/24	Direct	0	0	D	10.0.3.3	LoopBack0
10.0.3.3/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	Direct	0	0	D	10.0.12.3	GigabitEthernet0/0/1
10.0.12.3/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	RIP	100	1	D	10.0.12.2	GigabitEthernet0/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Test connectivity from R1 to IP address 10.0.23.3. R1 and R3 can communicate with one another.

```
[R1]ping 10.0.12.3
PING 10.0.12.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.3: bytes=56 Sequence=1 ttl=254 time=70 ms
  Reply from 10.0.12.3: bytes=56 Sequence=2 ttl=254 time=65 ms
  Reply from 10.0.12.3: bytes=56 Sequence=3 ttl=254 time=65 ms
  Reply from 10.0.12.3: bytes=56 Sequence=4 ttl=254 time=65 ms
  Reply from 10.0.12.3: bytes=56 Sequence=5 ttl=254 time=65 ms

--- 10.0.12.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 65/66/70 ms
```

The **debugging** command can be used to view RIP periodic updates.

Run the **debugging** command to enable the RIP debugging function. The **debugging** command can be used only in the user view. To identify the currently enabled debugging information, use the **display debugging** command. Run the **terminal debugging** command to display the debugging information.

The information about RIP interactions between routers is displayed.

```
<R1>debugging rip 1
<R1>display debugging
RIP Process id: 1
  Debugs ON: SEND, RECEIVE, PACKET, TIMER, EVENT, BRIEF,
             JOB, ROUTE-PROCESSING, ERROR,
             REPLAY-PROTECT, GR
<R1>terminal debugging
Info: Current terminal debugging is on.
<R1>
Mar 29 2016 09:45:07.860.1+00:00 R1 RIP/7/DBG: 6: 12734: RIP 1: Receiving v1 response on
GigabitEthernet0/0/0 from 10.0.13.2 with 3 RTEs
<R1>
Mar 29 2016 09:45:07.860.2+00:00 R1 RIP/7/DBG: 6: 12785: RIP 1: Receive response from 10.0.13.2
on GigabitEthernet0/0/0
<R1>
Mar 29 2016 09:45:07.860.3+00:00 R1 RIP/7/DBG: 6: 12796: Packet: Version 1, Cmd response, Length
64
<R1>
Mar 29 2016 09:45:07.860.4+00:00 R1 RIP/7/DBG: 6: 12845: Dest 10.0.2.0, Cost 1
<R1>
Mar 29 2016 09:45:07.860.5+00:00 R1 RIP/7/DBG: 6: 12845: Dest 10.0.3.0, Cost 2
```

```
<R1>
```

```
Mar 29 2016 09:45:07.860.6+00:00 R1 RIP/7/DBG: 6: 12845: Dest 10.0.12.0, Cost 1
```

```
<R1>
```

```
Mar 29 2016 09:45:09.370.1+00:00 R1 RIP/7/DBG: 25: 5071: RIP 1: Periodic timer expired for  
interface GigabitEthernet0/0/1
```

Run the **undo debugging rip <process-id>** or **undo debugging all** command to disable the debugging functions.

```
<R1>undo debugging rip 1
```

Individual parameters can be specified to control the debugging information viewed. For example, the **debugging rip 1 event** command allows for only periodical update events sent or received by routers to be viewed. The question mark (?) can be added to the command to query other parameters.

```
<R1>debugging rip 1 event
```

```
<R1>
```

```
Mar 29 2016 10:00:04.880.1+00:00 R1 RIP/7/DBG: 25: 5719: RIP 1: Periodic timer expired for  
interface GigabitEthernet0/0/0 (10.0.13.1) and its added to periodic update queue
```

```
<R1>
```

```
Mar 29 2016 10:00:04.890.1+00:00 R1 RIP/7/DBG: 25: 6048: RIP 1: Interface GigabitEthernet0/0/0  
(10.0.13.1) is deleted from the periodic update queue
```

```
<R1>undo debugging all
```

```
Info: All possible debugging has been turned off
```

Warning: If too many debugging functions are enabled, a large number of router resources will be utilized that may result in system service failure. Therefore, the use of commands (such as **debug all**) for enabling debugging functions in batches should be performed with caution.

Step 6 **Configure RIPv2.**

After the preceding configuration, you need to configure only **version 2** in the RIP sub view.

```
[R1]rip 1
```

```
[R1-rip-1]version 2
```

```
[R2]rip 1
```

```
[R2-rip-1]version 2
```

```
[R3]rip 1
[R3-rip-1]version 2
```

Step 7 Verify RIPv2 routes.

View the routing tables of R1, R2, and R3.

Run the **display ip routing-table** command to view the routing tables of R1, R2, and R3. Compare the routes that are highlighted with RIPv1 routes.

```
<R1>display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 13 Routes : 13

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.0/24	RIP	100	1	D	10.0.13.2	GigabitEthernet0/0/0
10.0.3.0/24	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
10.0.12.0/24	RIP	100	1	D	10.0.13.2	GigabitEthernet0/0/0
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/0
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
<R2>display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 15 Routes : 15

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
------------------	-------	-----	------	-------	---------	-----------

10.0.1.0/24	RIP	100	1	D	10.0.13.1	GigabitEthernet0/0/0
10.0.2.0/24	Direct	0	0	D	10.0.2.2	LoopBack0
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.0/24	RIP	100	1	D	10.0.12.3	GigabitEthernet0/0/1
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	Direct	0	0	D	10.0.13.2	GigabitEthernet0/0/0
10.0.13.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R3]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 13 Routes : 13

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.0/24	RIP	100	2	D	10.0.12.2	GigabitEthernet0/0/1
10.0.2.0/24	RIP	100	1	D	10.0.12.2	GigabitEthernet0/0/1
10.0.3.0/24	Direct	0	0	D	10.0.3.3	LoopBack0
10.0.3.3/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	Direct	0	0	D	10.0.12.3	GigabitEthernet0/0/1
10.0.12.3/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	RIP	100	1	D	10.0.12.2	GigabitEthernet0/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Test connectivity from R1 to the IP destination 10.0.12.3 on interface Gigabit Ethernet 0/0/1 of R3.

```
<R1>ping 10.0.12.3
PING 10.0.12.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.3: bytes=56 Sequence=1 ttl=254 time=74 ms
  Reply from 10.0.12.3: bytes=56 Sequence=2 ttl=254 time=75 ms
  Reply from 10.0.12.3: bytes=56 Sequence=3 ttl=254 time=75 ms
  Reply from 10.0.12.3: bytes=56 Sequence=4 ttl=254 time=75 ms
  Reply from 10.0.12.3: bytes=56 Sequence=5 ttl=254 time=75 ms

--- 10.0.12.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 74/74/75 ms
```

The **debugging** command can be used to view the RIPv2 periodic updates.

```
<R1>terminal debugging
Info: Current terminal debugging is on.
<R1>debugging rip 1 event
<R1>
Mar 29 2016 10:41:04.490.1+00:00 R1 RIP/7/DBG: 25: 5719: RIP 1: Periodic timer expired for
interface GigabitEthernet0/0/0 (10.0.13.1) and its added to periodic update queue
<R1>
Mar 29 2016 10:41:04.500.1+00:00 R1 RIP/7/DBG: 25: 6048: RIP 1: Interface GigabitEthernet0/0/0
(10.0.13.1) is deleted from the periodic update queue

<R1>undo debugging rip 1

<R1>debugging rip 1 packet
<R1>
Mar 29 2016 10:43:07.770.1+00:00 R1 RIP/7/DBG: 6: 12776: RIP 1: Sending response on interface
GigabitEthernet0/0/0 from 10.0.13.1 to 224.0.0.9
<R1>
Mar 29 2016 10:43:07.770.2+00:00 R1 RIP/7/DBG: 6: 12796: Packet: Version 2, Cmd response, Length
24
<R1>
Mar 29 2016 10:43:07.770.3+00:00 R1 RIP/7/DBG: 6: 12864: Dest 10.0.1.0/24, Nexthop 0.0.0.0,
Cost 1, Tag 0

<R1>undo debugging rip 1
```

Additional Exercises: Analyzing and Verifying

When using RIPv1, a router sends network IDs and other route update information to its neighbor routers without sending subnet masks. How do neighbor routers process the route update information and generate the corresponding subnet masks?

How are RIPv1 and RIPv2 compatible with each other?

Final Configuration

```
<R1>display current-configuration
[V200R007C00SPC600]
#
 sysname R1
#
interface GigabitEthernet0/0/0
 ip address 10.0.13.1 255.255.255.0
#
interface GigabitEthernet0/0/1
 shutdown
 ip address 10.0.12.1 255.255.255.0
#
interface LoopBack0
 ip address 10.0.1.1 255.255.255.0
#
rip 1
 version 2
 network 10.0.0.0
#
user-interface con 0
 authentication-mode password
 set authentication password cipher %$%$+L'YR&IZt'4,) >-*#lH",}%K-oJ_M9+'lOU~bD (\WTqB}%N,%
$%$
user-interface vty 0 4
#
return
```

```

<R2>display current-configuration
[V200R007C00SPC600]
#
 sysname R2
#
interface GigabitEthernet0/0/0
 ip address 10.0.13.2 255.255.255.0
#
interface GigabitEthernet0/0/1
 ip address 10.0.12.2 255.255.255.0
#
interface GigabitEthernet0/0/2
 shutdown
 ip address 10.0.23.2 255.255.255.0
#
interface LoopBack0
 ip address 10.0.2.2 255.255.255.0
#
rip 1
 version 2
 network 10.0.0.0
#
user-interface con 0
 authentication-mode password
 set authentication password cipher %$%$1=cd%b%/O%Id-8X:by1N,+s}'4wD6TvO<I|/pd# #44C@+s#,%
$%$
user-interface vty 0 4
#
return

```

```

<R3>display current-configuration
[V200R007C00SPC600]
#
 sysname R3
#
interface GigabitEthernet0/0/0
 shutdown
 ip address 10.0.13.3 255.255.255.0
#
interface GigabitEthernet0/0/1

```

```
ip address 10.0.12.3 255.255.255.0
#
interface GigabitEthernet0/0/2
shutdown
ip address 10.0.23.3 255.255.255.0
#
interface LoopBack0
ip address 10.0.3.3 255.255.255.0
#
rip 1
version 2
network 10.0.0.0
#
user-interface con 0
authentication-mode password
set authentication password cipher %$%$ksXDMg7Ry6yUU:63:DQ),#/sQg"@*S\U#.s.bHW xQ,y%#/v,%
$%$
user-interface vty 0 4
#
return
```

Lab 4-3 RIPv2 Route Aggregation and Authentication

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Aggregation of routes in RIPv2
- Implementation of authentication between RIPv2 peers
- Troubleshoot RIPv2 peer authentication failures.

Topology

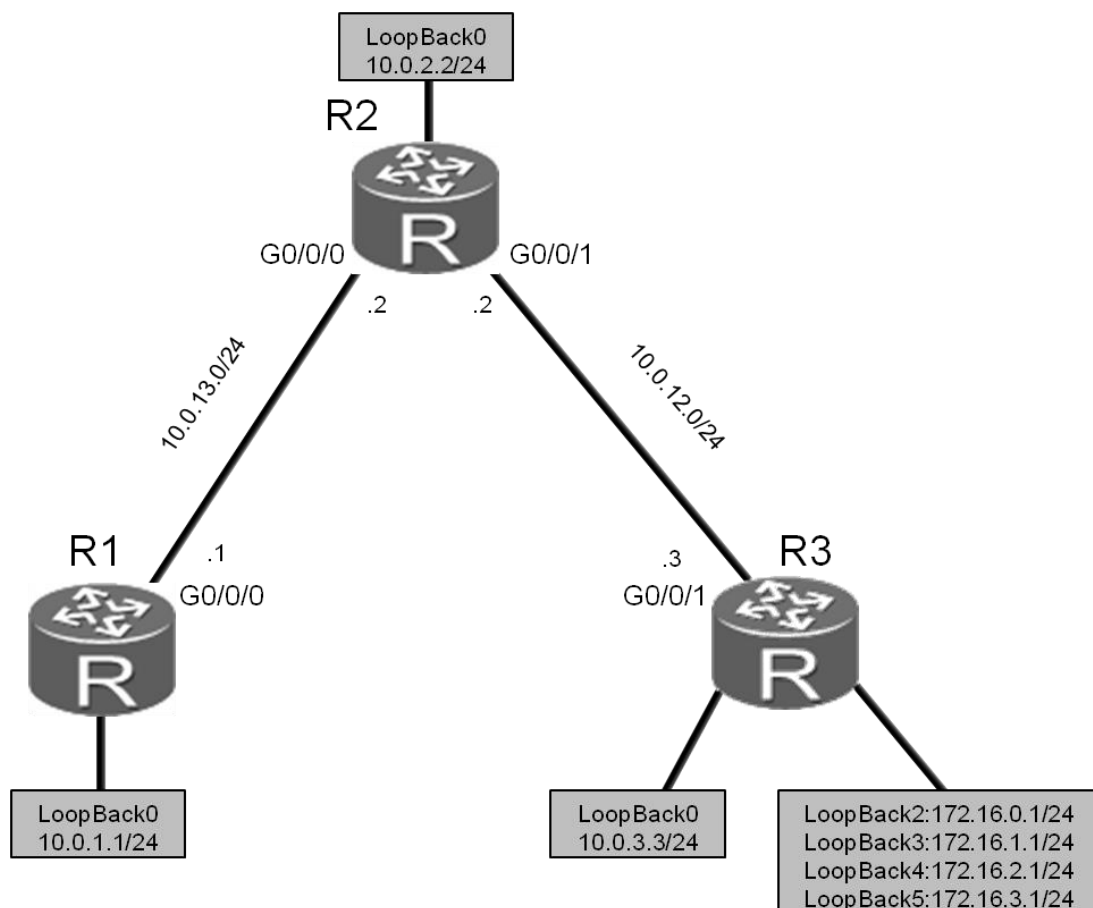


Figure 4.3 RIPv2 route aggregation and authentication topology

Scenario

As the network administrator of a small company you are responsible for the support of a RIPv2 based enterprise network. In order to better manage and optimize the routing table, route aggregation is required.

Additionally, concerns over the insertion of rogue devices into the network that may affect routing tables means that RIP authentication is required to protect the network.

Tasks

Step 1 Preparing the environment

If you are starting this section with a non-configured device begin here and then move to step 2. For those continuing from previous labs, begin at step 2.

Configure the base system information and addressing for R1, R2 and R3 on the network.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.0.13.1 24
[R1-GigabitEthernet0/0/0]quit
[R1]interface LoopBack 0
[R1-LoopBack0]ip address 10.0.1.1 24
```

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]ip address 10.0.13.2 24
[R2-GigabitEthernet0/0/0]quit
[R2]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet0/0/1]ip address 10.0.12.2 24
[R2-GigabitEthernet0/0/1]quit
[R2]interface LoopBack 0
```

```
[R2-LoopBack0]ip address 10.0.2.2 24
```

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1]ip address 10.0.12.3 24
[R3-GigabitEthernet0/0/1]quit
[R3]interface LoopBack 0
[R3-LoopBack0]ip address 10.0.3.3 24
```

After the IP addresses have been configured for the interfaces, test the network connectivity.

```
<R1>ping 10.0.13.2
PING 10.0.13.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.13.2: bytes=56 Sequence=1 ttl=255 time=30 ms
  Reply from 10.0.13.2: bytes=56 Sequence=2 ttl=255 time=30 ms
  Reply from 10.0.13.2: bytes=56 Sequence=3 ttl=255 time=30 ms
  Reply from 10.0.13.2: bytes=56 Sequence=4 ttl=255 time=30 ms
  Reply from 10.0.13.2: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 10.0.13.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 30/30/30 ms
```

```
<R2>ping 10.0.12.3
PING 10.0.12.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.3: bytes=56 Sequence=1 ttl=255 time=31 ms
  Reply from 10.0.12.3: bytes=56 Sequence=2 ttl=255 time=31 ms
  Reply from 10.0.12.3: bytes=56 Sequence=3 ttl=255 time=41 ms
  Reply from 10.0.12.3: bytes=56 Sequence=4 ttl=255 time=31 ms
  Reply from 10.0.12.3: bytes=56 Sequence=5 ttl=255 time=41 ms

--- 10.0.12.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 31/35/41 ms
```

Configure RIPv2 on R1, R2, and R3.

```
[R1]rip 1
[R1-rip-1]version 2
[R1-rip-1]network 10.0.0.0
```

```
[R2]rip 1
[R2-rip-1]version 2
[R2-rip-1]network 10.0.0.0
```

```
[R3]rip 1
[R3-rip-1]version 2
[R3-rip-1]network 10.0.0.0
```

Step 2 Configuration of additional loopback addresses

Establish additional loopback interfaces to represent multiple networks on R3.

```
[R3]interface LoopBack 2
[R3-LoopBack2]ip address 172.16.0.1 24
[R3-LoopBack2]quit
[R3]interface LoopBack 3
[R3-LoopBack3]ip address 172.16.1.1 24
[R3-LoopBack3]quit
[R3]interface LoopBack 4
[R3-LoopBack4]ip address 172.16.2.1 24
[R3-LoopBack4]quit
[R3]interface LoopBack 5
[R3-LoopBack5]ip address 172.16.3.1 24
```

Step 3 Advertize the loopback addresses in RIP.

The networks for the configured loopback interfaces need to be advertized.

Advertise the 172.16.0.0 network range on R3.

```
[R3]rip 1
[R3-rip-1]network 172.16.0.0
```

View the routing table of R1 to verify the new networks are being advertized.

```
<R1>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----  
Routing Tables: Public
```

```
Destinations : 17      Routes : 17
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.0/24	RIP	100	1	D	10.0.13.2	GigabitEthernet0/0/0
10.0.3.0/24	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
10.0.12.0/24	RIP	100	1	D	10.0.13.2	GigabitEthernet0/0/0
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/0
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.0.0/24	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
172.16.1.0/24	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
172.16.2.0/24	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
172.16.3.0/24	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The information in grey shows that R1 has learned specific routes but not aggregated routes.

Test network connectivity from R1 to the 172.16.0.0 network range.

```
<R1>ping 172.16.0.1
```

```
PING 172.16.0.1: 56 data bytes, press CTRL_C to break
```

```
Reply from 172.16.0.1: bytes=56 Sequence=1 ttl=254 time=80 ms  
Reply from 172.16.0.1: bytes=56 Sequence=2 ttl=254 time=79 ms  
Reply from 172.16.0.1: bytes=56 Sequence=3 ttl=254 time=79 ms  
Reply from 172.16.0.1: bytes=56 Sequence=4 ttl=254 time=79 ms  
Reply from 172.16.0.1: bytes=56 Sequence=5 ttl=254 time=79 ms
```

```
--- 172.16.0.1 ping statistics ---
```

```
5 packet(s) transmitted  
5 packet(s) received  
0.00% packet loss
```

round-trip min/avg/max = 79/79/80 ms

Step 4 **Configure RIP manual route aggregation on R2.**

Run the **rip summary-address** command on S1/0/0 of R2 to configure RIP route aggregation. The four routes (172.16.0.0/24, 172.16.1.0/24, 172.16.2.0/24, and 172.16.3.0/24) are to be aggregated into one route, 172.16.0.0/16.

```
[R2]interface GigabitEthernet0/0/0
[R2-GigabitEthernet0/0/0]rip summary-address 172.16.0.0 255.255.0.0
```

View the routing table of R1 that should now include an aggregated route.

```
<R1>display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 14 Routes : 14

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.0/24	RIP	100	1	D	10.0.13.2	GigabitEthernet0/0/0
10.0.3.0/24	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
10.0.12.0/24	RIP	100	1	D	10.0.13.2	GigabitEthernet0/0/0
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/0
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.0.0/16	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The highlighted information shows the aggregated route. There is now no specific route is listed in the routing table.

Verify that the routes are still supported for the 172.16.0.0 network range.

```
<R1>ping 172.16.0.1
PING 172.16.0.1: 56 data bytes, press CTRL_C to break
  Reply from 172.16.0.1: bytes=56 Sequence=1 ttl=254 time=60 ms
  Reply from 172.16.0.1: bytes=56 Sequence=2 ttl=254 time=59 ms
  Reply from 172.16.0.1: bytes=56 Sequence=3 ttl=254 time=80 ms
  Reply from 172.16.0.1: bytes=56 Sequence=4 ttl=254 time=60 ms
  Reply from 172.16.0.1: bytes=56 Sequence=5 ttl=254 time=60 ms

--- 172.16.0.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
round-trip min/avg/max = 59/63/80 ms
```

The preceding information shows that route aggregation reduces the size of the routing table without affecting communication to aggregated networks.

Step 5 **Configure RIP authentication.**

Configure plain text authentication between R1 and R2 and MD5 based authentication between R2 and R3. The authentication password in all cases should be **huawei**.

```
[R1]interface GigabitEthernet0/0/0
[R1-GigabitEthernet0/0/0]rip authentication-mode simple huawei

[R2]interface GigabitEthernet0/0/0
[R2-GigabitEthernet0/0/0]rip authentication-mode simple huawei
[R2-GigabitEthernet0/0/0]quit
[R2]interface GigabitEthernet0/0/1
[R2-GigabitEthernet0/0/1]rip authentication-mode md5 usual huawei

[R3]interface GigabitEthernet0/0/1
[R3-GigabitEthernet0/0/1]rip authentication-mode md5 usual huawei
```

After the configuration is complete, verify that the routes are not affected.

```
<R1>display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 14 Routes : 14

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.0/24	RIP	100	1	D	10.0.13.2	GigabitEthernet0/0/0
10.0.3.0/24	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
10.0.12.0/24	RIP	100	1	D	10.0.13.2	GigabitEthernet0/0/0
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/0
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.0.0/16	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
<R2>display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 19 Routes : 19

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.0/24	RIP	100	1	D	10.0.13.1	GigabitEthernet0/0/0
10.0.2.0/24	Direct	0	0	D	10.0.2.2	LoopBack0
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.0/24	RIP	100	1	D	10.0.12.3	GigabitEthernet0/0/1
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1

10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	Direct	0	0	D	10.0.13.2	GigabitEthernet0/0/0
10.0.13.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.0.0/24	RIP	100	1	D	10.0.12.3	GigabitEthernet0/0/1
172.16.1.0/24	RIP	100	1	D	10.0.12.3	GigabitEthernet0/0/1
172.16.2.0/24	RIP	100	1	D	10.0.12.3	GigabitEthernet0/0/1
172.16.3.0/24	RIP	100	1	D	10.0.12.3	GigabitEthernet0/0/1
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

<R3>display ip routing-table

Route Flags: R - relay, D - download to fib

```

-----
Routing Tables: Public
      Destinations : 25          Routes : 25

Destination/Mask    Proto    Pre  Cost  Flags  NextHop    Interface

10.0.1.0/24         RIP      100  2      D      10.0.12.2   GigabitEthernet0/0/1
10.0.2.0/24         RIP      100  1      D      10.0.12.2   GigabitEthernet0/0/1
10.0.3.0/24         Direct   0     0      D      10.0.3.3     LoopBack0
10.0.3.3/32         Direct   0     0      D      127.0.0.1    LoopBack0
10.0.3.255/32       Direct   0     0      D      127.0.0.1    LoopBack0
10.0.12.0/24        Direct   0     0      D      10.0.12.3    GigabitEthernet0/0/1
10.0.12.3/32        Direct   0     0      D      127.0.0.1    GigabitEthernet0/0/1
10.0.12.255/32      Direct   0     0      D      127.0.0.1    GigabitEthernet0/0/1
10.0.13.0/24        RIP      100  1      D      10.0.12.2   GigabitEthernet0/0/1
127.0.0.0/8         Direct   0     0      D      127.0.0.1    InLoopBack0
127.0.0.1/32        Direct   0     0      D      127.0.0.1    InLoopBack0
127.255.255.255/32  Direct   0     0      D      127.0.0.1    InLoopBack0
172.16.0.0/24       Direct   0     0      D      172.16.0.1   LoopBack2
172.16.0.1/32       Direct   0     0      D      127.0.0.1    LoopBack2
172.16.0.255/32     Direct   0     0      D      127.0.0.1    LoopBack2
172.16.1.0/24       Direct   0     0      D      172.16.1.1   LoopBack3
172.16.1.1/32       Direct   0     0      D      127.0.0.1    LoopBack3
172.16.1.255/32     Direct   0     0      D      127.0.0.1    LoopBack3
172.16.2.0/24       Direct   0     0      D      172.16.2.1   LoopBack4
172.16.2.1/32       Direct   0     0      D      127.0.0.1    LoopBack4
172.16.2.255/32     Direct   0     0      D      127.0.0.1    LoopBack4
172.16.3.0/24       Direct   0     0      D      172.16.3.1   LoopBack5

```

```

172.16.3.1/32   Direct  0    0      D    127.0.0.1   LoopBack5
172.16.3.255/32 Direct  0    0      D    127.0.0.1   LoopBack5
255.255.255.255/32 Direct  0    0      D    127.0.0.1   InLoopBack0

```

Step 6 **Generate and rectify RIPv2 authentication faults.**

Change the authentication password on G0/0/0 of R2 to **huawei2**.

```

[R2]interface GigabitEthernet0/0/0
[R2-GigabitEthernet0/0/0]rip authentication-mode simple huawei2

```

View the routing table of R1.

```

<R1>display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 10          Routes : 10

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface
-----
10.0.1.0/24         Direct  0    0          D    10.0.1.1         LoopBack0
10.0.1.1/32         Direct  0    0          D    127.0.0.1         LoopBack0
10.0.1.255/32       Direct  0    0          D    127.0.0.1         LoopBack0
10.0.13.0/24        Direct  0    0          D    10.0.13.1         GigabitEthernet0/0/0
10.0.13.1/32        Direct  0    0          D    127.0.0.1         GigabitEthernet0/0/0
10.0.13.255/32      Direct  0    0          D    127.0.0.1         GigabitEthernet0/0/0
127.0.0.0/8         Direct  0    0          D    127.0.0.1         InLoopBack0
127.0.0.1/32        Direct  0    0          D    127.0.0.1         InLoopBack0
127.255.255.255/32  Direct  0    0          D    127.0.0.1         InLoopBack0
255.255.255.255/32  Direct  0    0          D    127.0.0.1         InLoopBack0

```

Since R1 and R2 use different RIP authentication passwords, R1 will not receive any advertised RIP routes from R2.

Restore the authentication password on G0/0/0 of R2 to **huawei**.

```

[R2]interface GigabitEthernet0/0/0
[R2- GigabitEthernet0/0/0]rip authentication-mode simple huawei

```

Change the authentication mode on G0/0/1 of R2 to plain text authentication.

```

[R2]interface GigabitEthernet0/0/1
[R2-GigabitEthernet0/0/1]rip authentication-mode simple huawei

```

Run the following command to delete the routes learned by R3 from R2 before you change the authentication password.

View the routing table of R3.

```
<R3>display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 22 Routes : 22

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.3.0/24	Direct	0	0	D	10.0.3.3	LoopBack0
10.0.3.3/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	Direct	0	0	D	10.0.12.3	GigabitEthernet0/0/1
10.0.12.3/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.0.0/24	Direct	0	0	D	172.16.0.1	LoopBack2
172.16.0.1/32	Direct	0	0	D	127.0.0.1	LoopBack2
172.16.0.255/32	Direct	0	0	D	127.0.0.1	LoopBack2
172.16.1.0/24	Direct	0	0	D	172.16.1.1	LoopBack3
172.16.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack3
172.16.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack3
172.16.2.0/24	Direct	0	0	D	172.16.2.1	LoopBack4
172.16.2.1/32	Direct	0	0	D	127.0.0.1	LoopBack4
172.16.2.255/32	Direct	0	0	D	127.0.0.1	LoopBack4
172.16.3.0/24	Direct	0	0	D	172.16.3.1	LoopBack5
172.16.3.1/32	Direct	0	0	D	127.0.0.1	LoopBack5
172.16.3.255/32	Direct	0	0	D	127.0.0.1	LoopBack5
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Since R2 and R3 use different RIP authentication modes, R3 cannot receive any advertised RIP routes from R2.

Restore the authentication mode on G0/0/1 of R2 to MD5.

```
[R2]interface GigabitEthernet0/0/1
```

```
[R2-GigabitEthernet0/0/1]rip authentication-mode md5 usual huawei
```

Verify that routes in routing tables of R1, R2, and R3 have been restored. Note that RIP updates routes periodically, so may take a moment to be restored.

```
<R1>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----  
Routing Tables: Public
```

```
Destinations : 14      Routes : 14
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.0/24	RIP	100	1	D	10.0.13.2	GigabitEthernet0/0/0
10.0.3.0/24	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
10.0.12.0/24	RIP	100	1	D	10.0.13.2	GigabitEthernet0/0/0
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/0
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.0.0/16	RIP	100	2	D	10.0.13.2	GigabitEthernet0/0/0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
[R2]display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----  
Routing Tables: Public
```

```
Destinations : 19      Routes : 19
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.0/24	RIP	100	1	D	10.0.13.1	GigabitEthernet0/0/0
10.0.2.0/24	Direct	0	0	D	10.0.2.2	LoopBack0
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.0/24	RIP	100	1	D	10.0.12.3	GigabitEthernet0/0/1
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1

10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	Direct	0	0	D	10.0.13.2	GigabitEthernet0/0/0
10.0.13.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.0.0/24	RIP	100	1	D	10.0.12.3	GigabitEthernet0/0/1
172.16.1.0/24	RIP	100	1	D	10.0.12.3	GigabitEthernet0/0/1
172.16.2.0/24	RIP	100	1	D	10.0.12.3	GigabitEthernet0/0/1
172.16.3.0/24	RIP	100	1	D	10.0.12.3	GigabitEthernet0/0/1
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

<R3>display ip routing-table

Route Flags: R - relay, D - download to fib

```

-----
Routing Tables: Public
      Destinations : 25          Routes : 25

Destination/Mask    Proto    Pre  Cost  Flags  NextHop    Interface

10.0.1.0/24         RIP      100  2      D      10.0.12.2  GigabitEthernet0/0/1
10.0.2.0/24         RIP      100  1      D      10.0.12.2  GigabitEthernet0/0/1
10.0.3.0/24         Direct   0     0      D      10.0.3.3   LoopBack0
10.0.3.3/32         Direct   0     0      D      127.0.0.1  LoopBack0
10.0.3.255/32       Direct   0     0      D      127.0.0.1  LoopBack0
10.0.12.0/24        Direct   0     0      D      10.0.12.3  GigabitEthernet0/0/1
10.0.12.3/32        Direct   0     0      D      127.0.0.1  GigabitEthernet0/0/1
10.0.12.255/32      Direct   0     0      D      127.0.0.1  GigabitEthernet0/0/1
10.0.13.0/24        RIP      100  1      D      10.0.12.2  GigabitEthernet0/0/1
127.0.0.0/8         Direct   0     0      D      127.0.0.1  InLoopBack0
127.0.0.1/32        Direct   0     0      D      127.0.0.1  InLoopBack0
127.255.255.255/32  Direct   0     0      D      127.0.0.1  InLoopBack0
172.16.0.0/24       Direct   0     0      D      172.16.0.1 LoopBack2
172.16.0.1/32       Direct   0     0      D      127.0.0.1  LoopBack2
172.16.0.255/32     Direct   0     0      D      127.0.0.1  LoopBack2
172.16.1.0/24       Direct   0     0      D      172.16.1.1 LoopBack3
172.16.1.1/32       Direct   0     0      D      127.0.0.1  LoopBack3
172.16.1.255/32     Direct   0     0      D      127.0.0.1  LoopBack3
172.16.2.0/24       Direct   0     0      D      172.16.2.1 LoopBack4
172.16.2.1/32       Direct   0     0      D      127.0.0.1  LoopBack4
172.16.2.255/32     Direct   0     0      D      127.0.0.1  LoopBack4
172.16.3.0/24       Direct   0     0      D      172.16.3.1 LoopBack5

```

172.16.3.1/32	Direct	0	0	D	127.0.0.1	LoopBack5
172.16.3.255/32	Direct	0	0	D	127.0.0.1	LoopBack5
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Final Configuration

```
<R1>display current-configuration
[V200R007C00SPC600]
#
sysname R1
#
interface GigabitEthernet0/0/0
ip address 10.0.13.1 255.255.255.0
rip authentication-mode simple cipher %$$$S2AJ2_mJ)Hf++RSng6^NN|Xl%$$$
#
interface LoopBack0
ip address 10.0.1.1 255.255.255.0
#
rip 1
version 2
network 10.0.0.0
#
user-interface con 0
authentication-mode password
set authentication password cipher %$$$+L'YR&IZt'4,) >-*#lH",}%K-oJ_M9+'lOU~bD (\WTqB}%N,%
$$$
user-interface vty 0 4
#
return

<R2>display current-configuration
[V200R007C00SPC600]
#
sysname R2
#
interface GigabitEthernet0/0/0
ip address 10.0.13.2 255.255.255.0
rip authentication-mode simple cipher %$$$+Ob&JcQxU6mUJ(ZXLZY#OEXz%$$$
rip summary-address 172.16.0.0 255.255.0.0
#
interface GigabitEthernet0/0/1
ip address 10.0.12.2 255.255.255.0
```

```

rip authentication-mode md5 usual cipher %$$$C] '$.`NWGZ}|gLV%:XF>OG}|%$$$
#
interface LoopBack0
ip address 10.0.2.2 255.255.255.0
#
rip 1
version 2
network 10.0.0.0
#
user-interface con 0
authentication-mode password
set authentication password cipher %$$$1=cd%b%/O%Id-8X:by1N,+s}'4wD6TvO<I|/pd# #44C@+s#,%
$$$
user-interface vty 0 4
#
return

```

```

<R3>display current-configuration
[V200R007C00SPC600]
#
sysname R3
#
interface GigabitEthernet0/0/1
ip address 10.0.12.3 255.255.255.0
rip authentication-mode md5 usual cipher %$$$_5VL+wN6FNe]rVKbh[E(O=E>%$$$
#
interface LoopBack0
ip address 10.0.3.3 255.255.255.0
#
interface LoopBack2
ip address 172.16.0.1 255.255.255.0
#
interface LoopBack3
ip address 172.16.1.1 255.255.255.0
#
interface LoopBack4
ip address 172.16.2.1 255.255.255.0
#
interface LoopBack5
ip address 172.16.3.1 255.255.255.0
#

```

```
rip 1
version 2
network 10.0.0.0
network 172.16.0.0
#
user-interface con 0
authentication-mode password
set authentication password cipher %$%$ksXDMg7Ry6yUU:63:DQ),#/sQg"@*S\U#.s.bHW xQ,y%#/v,%
$%$
user-interface vty 0 4
#
return
```

Lab 4-4 OSPF Single-Area Configuration

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Configuration of the Router-ID for OSPF.
- Establish OSPF on a specified interface or network.
- View OSPF operations using display commands.
- Advertisement of default routes in OSPF.
- Change of the OSPF hello interval and dead interval.
- Familiarization with DR or BDR election on multi-access networks.
- Change of the OSPF route priority to manipulate DR election.

Topology

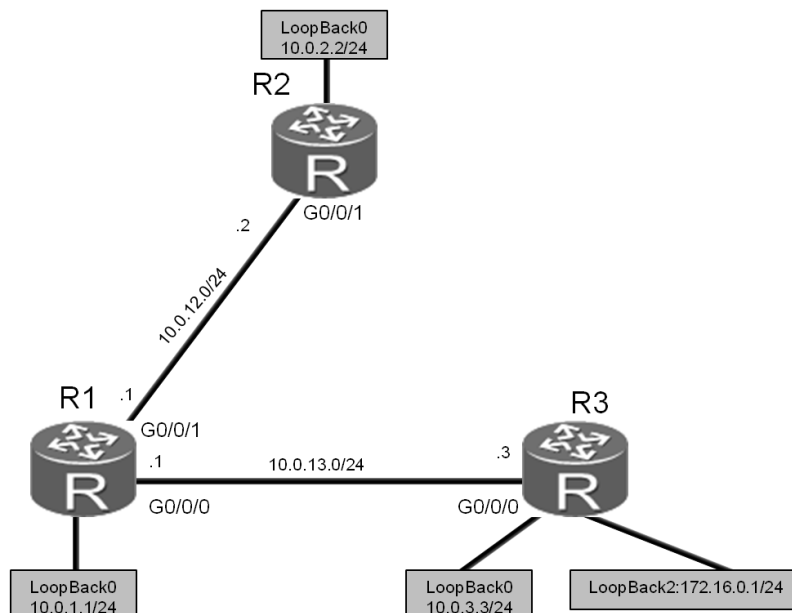


Figure 4.4 OSPF single area topology

Scenario

As the network administrator of an establishing small enterprise, it is required that a network be implemented using OSPF. Then network is to support a single area and with consideration for future expansion it is requested that this area be set as area 0. OSPF is required to advertise default routes and also elect both a DR and BDR for network resiliency.

Tasks

Step 1 Prepare the environment

If you are starting this section with a non-configured device, begin here and then move to step 3. For those continuing from previous labs, begin at step 2.

Establish the basic system configuration and addressing for the lab.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet 0/0/1]ip address 10.0.12.1 24
[R1-GigabitEthernet 0/0/1]quit
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.0.13.1 24
[R1-GigabitEthernet0/0/0]quit
[R1]interface LoopBack 0
[R1-LoopBack0]ip address 10.0.1.1 24
```

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet 0/0/1]ip address 10.0.12.2 24
[R2-GigabitEthernet 0/0/1]quit
[R2]interface LoopBack 0
[R2-LoopBack0]ip address 10.0.2.2 24
```

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]ip address 10.0.13.3 24
[R3-GigabitEthernet0/0/0]quit
[R3]interface LoopBack 0
[R3-LoopBack0]ip address 10.0.3.3 24
[R3-LoopBack0]quit
[R3]interface LoopBack 2
[R3-LoopBack2]ip address 172.16.0.1 24
```

Step 2 Clean up the previous configuration.

Enable the interfaces necessary for this lab and disable those not needed.

```
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]undo shutdown
[R1-GigabitEthernet0/0/1]quit

[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]undo rip summary-address 172.16.0.0 255.255.0.0
[R2-GigabitEthernet0/0/0]shutdown

[R3]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]undo shutdown
[R3-GigabitEthernet0/0/0]quit
[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1]shutdown
[R3-GigabitEthernet0/0/1]quit
[R3]undo interface LoopBack 3
Info: This operation may take a few seconds. Please wait for a moment...succeeded.
[R3]undo interface LoopBack 4
Info: This operation may take a few seconds. Please wait for a moment...succeeded.
[R3]undo interface LoopBack 5
Info: This operation may take a few seconds. Please wait for a moment...succeeded.
```

Remove the configured RIP authentication and RIP 1 process.

```
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]undo rip authentication-mode
[R1-GigabitEthernet0/0/0]quit
[R1]undo rip 1
Warning: The RIP process will be deleted. Continue?[Y/N]y
```

```
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]undo rip authentication-mode
[R2-GigabitEthernet0/0/0]quit
[R2]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet0/0/1]undo rip authentication-mode
[R2-GigabitEthernet0/0/1]quit
[R2]undo rip 1
Warning: The RIP process will be deleted. Continue?[Y/N]y
```

```
[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1]undo rip authentication-mode
[R3-GigabitEthernet0/0/1]quit
[R3]undo rip 1
Warning: The RIP process will be deleted. Continue?[Y/N]y
```

Step 3 **Configure OSPF.**

Assign the value 10.0.1.1 (as used on logical interface loopback 0 for simplicity) as the router ID. Use OSPF process 1 (the default process), and specify network segments 10.0.1.0/24, 10.0.12.0/24, and 10.0.13.0/24 as part of OSPF area 0.

```
[R1]ospf 1 router-id 10.0.1.1
[R1-ospf-1]area 0
[R1-ospf-1-area-0.0.0.0]network 10.0.1.0 0.0.0.255
[R1-ospf-1-area-0.0.0.0]network 10.0.13.0 0.0.0.255
[R1-ospf-1-area-0.0.0.0]network 10.0.12.0 0.0.0.255
```

Different process ID's will generate multiple link state databases, therefore ensure that all routers use the same OSPF process ID. The wildcard mask must be specified as part of the **network** command.

Manually assign the value 10.0.2.2 as the router ID. Use OSPF process 1, and advertise network segments 10.0.12.0/24 and 10.0.2.0/24 into OSPF area 0.

```
[R2]ospf 1 router-id 10.0.2.2
[R2-ospf-1]area 0
```

```
[R2-ospf-1-area-0.0.0.0]network 10.0.2.0 0.0.0.255
[R2-ospf-1-area-0.0.0.0]network 10.0.12.0 0.0.0.255
```

...output omitted...

```
Mar 30 2016 09:41:39+00:00 R2 %%01OSPF/4/NBR_CHANGE_E(1)[5]:Neighbor changes event: neighbor status
changed. (ProcessId=1, NeighborAddress=10.0.12.1, NeighborEvent=LoadingDone,
NeighborPreviousState=Loading, NeighborCurrentState=Full)
```

Adjacency is attained when "NeighborCurrentState=Full" . For R3, Manually assign the value 10.0.3.3 as the router ID. Use OSPF process 1, and advertise network segments 10.0.3.0/24 and 10.0.13.0/24 into OSPF area 0.

```
[R3]ospf 1 router-id 10.0.3.3
[R3-ospf-1]area 0
[R3-ospf-1-area-0.0.0.0]network 10.0.3.0 0.0.0.255
[R3-ospf-1-area-0.0.0.0]network 10.0.13.0 0.0.0.255
```

...output omitted...

```
Mar 30 2016 16:05:34+00:00 R3 %%01OSPF/4/NBR_CHANGE_E(1)[5]:Neighbor changes event: neighbor status
changed. (ProcessId=1, NeighborAddress=10.0.13.1, NeighborEvent=LoadingDone,
NeighborPreviousState=Loading, NeighborCurrentState=Full)
```

Step 4 Verify the OSPF configuration.

After OSPF route convergence is complete, view routing tables of R1, R2, and R3.

```
<R1>display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 15 Routes : 15

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.2/32	OSPF	10	1	D	10.0.12.2	GigabitEthernet0/0/1
10.0.3.3/32	OSPF	10	1	D	10.0.13.3	GigabitEthernet0/0/0
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet0/0/1
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1

```

10.0.12.255/32 Direct 0 0 D 127.0.0.1 GigabitEthernet0/0/1
10.0.13.0/24 Direct 0 0 D 10.0.13.1 GigabitEthernet0/0/0
10.0.13.1/32 Direct 0 0 D 127.0.0.1 GigabitEthernet0/0/0
10.0.13.255/32 Direct 0 0 D 127.0.0.1 GigabitEthernet0/0/0
127.0.0.0/8 Direct 0 0 D 127.0.0.1 InLoopBack0
127.0.0.1/32 Direct 0 0 D 127.0.0.1 InLoopBack0
127.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0

```

<R2>display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 13 Routes : 13

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	OSPF	10	1	D	10.0.12.1	GigabitEthernet0/0/1
10.0.2.0/24	Direct	0	0	D	10.0.2.2	LoopBack0
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.3/32	OSPF	10	2	D	10.0.12.1	GigabitEthernet0/0/1
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	OSPF	10	2	D	10.0.12.1	GigabitEthernet0/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

<R3>display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 16 Routes : 16

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	OSPF	10	1	D	10.0.13.1	GigabitEthernet0/0/0
10.0.2.2/32	OSPF	10	2	D	10.0.13.1	GigabitEthernet0/0/0
10.0.3.0/24	Direct	0	0	D	10.0.3.3	LoopBack0

10.0.3.3/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	OSPF	10	2	D	10.0.13.1	GigabitEthernet0/0/0
10.0.13.0/24	Direct	0	0	D	10.0.13.3	GigabitEthernet0/0/0
10.0.13.3/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.0.0/24	Direct	0	0	D	172.16.0.1	LoopBack2
172.16.0.1/32	Direct	0	0	D	127.0.0.1	LoopBack2
172.16.0.255/32	Direct	0	0	D	127.0.0.1	LoopBack2
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Test network connectivity between R2 and R1 at 10.0.1.1 and between R2 and R3 at 10.0.3.3.

<R2>ping 10.0.1.1

PING 10.0.1.1: 56 data bytes, press CTRL_C to break

Reply from 10.0.1.1: bytes=56 Sequence=1 ttl=255 time=37 ms

Reply from 10.0.1.1: bytes=56 Sequence=2 ttl=255 time=42 ms

Reply from 10.0.1.1: bytes=56 Sequence=3 ttl=255 time=42 ms

Reply from 10.0.1.1: bytes=56 Sequence=4 ttl=255 time=45 ms

Reply from 10.0.1.1: bytes=56 Sequence=5 ttl=255 time=42 ms

--- 10.0.1.1 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 37/41/45 ms

<R2>ping 10.0.3.3

PING 10.0.3.3: 56 data bytes, press CTRL_C to break

Reply from 10.0.3.3: bytes=56 Sequence=1 ttl=254 time=37 ms

Reply from 10.0.3.3: bytes=56 Sequence=2 ttl=254 time=42 ms

Reply from 10.0.3.3: bytes=56 Sequence=3 ttl=254 time=42 ms

Reply from 10.0.3.3: bytes=56 Sequence=4 ttl=254 time=42 ms

Reply from 10.0.3.3: bytes=56 Sequence=5 ttl=254 time=42 ms

--- 10.0.3.3 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

```
0.00% packet loss
round-trip min/avg/max = 37/41/42 ms
```

Run the **display ospf peer** command to view the OSPF neighbor status.

```
<R1>display ospf peer
```

```
OSPF Process 1 with Router ID 10.0.1.1
Neighbors
```

```
Area 0.0.0.0 interface 10.0.12.1(GigabitEthernet0/0/1)'s neighbors
```

```
Router ID: 10.0.2.2      Address: 10.0.12.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.0.12.1 BDR: 10.0.12.2 MTU: 0
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00:47:59
Authentication Sequence: [ 0 ]
```

```
Neighbors
```

```
Area 0.0.0.0 interface 10.0.13.1(GigabitEthernet0/0/0)'s neighbors
```

```
Router ID: 10.0.3.3      Address: 10.0.13.3
State: Full Mode:Nbr is Master Priority: 1
DR: 10.0.13.1 BDR: 10.0.13.3 MTU: 0
Dead timer due in 34 sec
Retrans timer interval: 5
Neighbor is up for 00:41:44
Authentication Sequence: [ 0 ]
```

The **display ospf peer** command displays detailed information about any peering neighbors. In the example given, the link 10.0.13.1 of R1 shows to be the DR. The DR election is non pre-emptive, meaning that the link of R3 will not take over the role of DR from R1 unless the OSPF process is reset.

The **display ospf peer brief** command can also be used to display a condensed version of the OSPF peer information.

```
<R1>display ospf peer brief
```

```
OSPF Process 1 with Router ID 10.0.1.1
Peer Statistic Information
```

Area Id	Interface	Neighbor id	State
0.0.0.0	GigabitEthernet0/0/0	10.0.3.3	Full
0.0.0.0	GigabitEthernet0/0/1	10.0.2.2	Full

<R2>display ospf peer brief

OSPF Process 1 with Router ID 10.0.2.2
Peer Statistic Information

Area Id	Interface	Neighbor id	State
0.0.0.0	GigabitEthernet0/0/1	10.0.1.1	Full

<R3>display ospf peer brief

OSPF Process 1 with Router ID 10.0.3.3
Peer Statistic Information

Area Id	Interface	Neighbor id	State
0.0.0.0	GigabitEthernet0/0/0	10.0.1.1	Full

Step 5 Change the OSPF hello interval and dead interval.

Run the **display ospf interface GigabitEthernet 0/0/0** command on R1 to view the default OSPF hello interval and dead interval.

<R1>display ospf interface GigabitEthernet 0/0/0

OSPF Process 1 with Router ID 10.0.1.1
Interfaces

```
Interface: 10.0.13.1 (GigabitEthernet0/0/0)
Cost: 1      State: DR      Type: Broadcast    MTU: 1500
Priority: 1
Designated Router: 10.0.13.1
Backup Designated Router: 10.0.13.3
Timers: Hello 10 , Dead 40 , Poll 120 , Retransmit 5 , Transmit Delay 1
```

Run the **ospf timer** command to change the OSPF hello interval and dead interval on GE0/0/0 of R1 to 15s and 60s respectively.

```
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ospf timer hello 15
[R1-GigabitEthernet0/0/0]ospf timer dead 60
Mar 30 2016 16:58:39+00:00 R1 %%01OSPF/3/NBR_DOWN_REASON(1)[1]:Neighbor state leaves full or
changed to Down. (ProcessId=1, NeighborRouterId=10.0.3.3, NeighborAreaId=0,
NeighborInterface=GigabitEthernet0/0/0,NeighborDownImmediate reason=Neighbor Down Due to
Inactivity, NeighborDownPrimeReason=Interface Parameter Mismatch,
NeighborChangeTime=2016-03-30 16:58:39)
```

```
<R1>display ospf interface GigabitEthernet 0/0/0
```

```
OSPF Process 1 with Router ID 10.0.1.1
  Interfaces
```

```
Interface: 10.0.13.1 (GigabitEthernet0/0/0)
Cost: 1      State: DR      Type: Broadcast    MTU: 1500
Priority: 1
Designated Router: 10.0.13.1
Backup Designated Router: 10.0.13.3
Timers: Hello 15 , Dead 60 , Poll 120 , Retransmit 5 , Transmit Delay 1
```

Check the OSPF neighbor status on R1.

```
<R1>display ospf peer brief
```

```
OSPF Process 1 with Router ID 10.0.1.1
  Peer Statistic Information
```

```
-----
Area Id      Interface      Neighbor id    State
0.0.0.0      GigabitEthernet0/0/1    10.0.2.2      Full
-----
```

The preceding information shows that R1 has only one neighbor, R2. Since the OSPF hello intervals and dead intervals on R1 and R3 are different, R1 and R3 will fail to establish an OSPF neighbor relationship.

Run the **ospf timer** command to change the OSPF hello interval and dead interval on GE0/0/0 of R3 to 15s and 60s respectively.

```
[R3]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]ospf timer hello 15
[R3-GigabitEthernet0/0/0]ospf timer dead 60
...output omitted...
```

```
Mar 30 2016 17:03:33+00:00 R3 %%01OSPF/4/NBR_CHANGE_E(1)[4]:Neighbor changes event: neighbor
status changed. (ProcessId=1, NeighborAddress=10.0.13.1, NeighborEvent=LoadingDone,
NeighborPreviousState=Loading, NeighborCurrentState=Full)
```

```
<R3>display ospf interface GigabitEthernet 0/0/0
```

```
OSPF Process 1 with Router ID 10.0.3.3
Interfaces
```

```
Interface: 10.0.13.3 (GigabitEthernet0/0/0)
Cost: 1      State: DR      Type: Broadcast      MTU: 1500
Priority: 1
Designated Router: 10.0.13.3
Backup Designated Router: 10.0.13.1
Timers: Hello 15 , Dead 60 , Poll 120 , Retransmit 5 , Transmit Delay 1
```

Check the OSPF neighbor status on R1 again.

```
<R1>display ospf peer brief
```

```
OSPF Process 1 with Router ID 10.0.1.1
Peer Statistic Information
```

```
-----
Area Id      Interface                Neighbor id    State
0.0.0.0      GigabitEthernet0/0/0      10.0.3.3      Full
0.0.0.0      GigabitEthernet0/0/1      10.0.2.2      Full
-----
```

Step 6 **Advertise default routes in OSPF.**

Configure OSPF to advertise default routes on R3.

```
[R3]ip route-static 0.0.0.0 0.0.0.0 LoopBack 2
[R3]ospf 1
[R3-ospf-1]default-route-advertise
```

View routing tables of R1 and R2. You can see that R1 and R2 have learned the default routes advertised by R3.

```
<R1>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----
Routing Tables: Public
```

Destinations : 16 Routes : 16

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	O_ASE	150	1	D	10.0.13.3	GigabitEthernet0/0/0
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.2/32	OSPF	10	1	D	10.0.12.2	GigabitEthernet0/0/1
10.0.3.3/32	OSPF	10	1	D	10.0.13.3	GigabitEthernet0/0/0
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet0/0/1
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/0
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

<R2>display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 14 Routes : 14

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	O_ASE	150	1	D	10.0.12.1	GigabitEthernet0/0/1
10.0.1.1/32	OSPF1	0	1	D	10.0.12.1	GigabitEthernet0/0/1
10.0.2.0/24	Direct	0	0	D	10.0.2.2	LoopBack0
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.3/32	OSPF	10	2	D	10.0.12.1	GigabitEthernet0/0/1
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.0/24	OSPF	10	2	D	10.0.12.1	GigabitEthernet0/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
127.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
```

```
<R3>display ip routing-table
```

```
Route Flags: R - relay, D - download to fib
```

```
-----
Routing Tables: Public
```

```
Destinations : 17 Routes : 17
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Static	60	0	D	172.16.0.1	LoopBack2
10.0.1.1/32	OSPF	10	1	D	10.0.13.1	GigabitEthernet0/0/0
10.0.2.2/32	OSPF	10	2	D	10.0.13.1	GigabitEthernet0/0/0
10.0.3.0/24	Direct	0	0	D	10.0.3.3	LoopBack0
10.0.3.3/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	OSPF	10	2	D	10.0.13.1	GigabitEthernet0/0/0
10.0.13.0/24	Direct	0	0	D	10.0.13.3	GigabitEthernet0/0/0
10.0.13.3/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.0.0/24	Direct	0	0	D	172.16.0.1	LoopBack2
172.16.0.1/32	Direct	0	0	D	127.0.0.1	LoopBack2
172.16.0.255/32	Direct	0	0	D	127.0.0.1	LoopBack2
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Run the **ping** command to test connectivity between R2 and Loopback2 at 172.16.0.1.

```
<R2>ping 172.16.0.1
```

```
PING 172.16.0.1: 56 data bytes, press CTRL_C to break
```

```
Reply from 172.16.0.1: bytes=56 Sequence=1 ttl=254 time=47 ms
Reply from 172.16.0.1: bytes=56 Sequence=2 ttl=254 time=37 ms
Reply from 172.16.0.1: bytes=56 Sequence=3 ttl=254 time=37 ms
Reply from 172.16.0.1: bytes=56 Sequence=4 ttl=254 time=37 ms
Reply from 172.16.0.1: bytes=56 Sequence=5 ttl=254 time=37 ms
```

```
--- 172.16.0.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 37/39/47 ms
```

Step 7 **Control OSPF DR or BDR election.**

Run the **display ospf peer** command to view the DR and BDR of R1 and R3.

```
<R1>display ospf peer 10.0.3.3

      OSPF Process 1 with Router ID 10.0.1.1
        Neighbors

Area 0.0.0.0 interface 10.0.13.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.0.3.3      Address: 10.0.13.3
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 10.0.13.3  BDR: 10.0.13.1  MTU: 0
  Dead timer due in 49 sec
  Retrans timer interval: 5
  Neighbor is up for 00:17:40
  Authentication Sequence: [ 0 ]
```

The preceding information shows that R3 is the DR and R1 is the BDR. This is because R3's router ID 10.0.3.3 is greater than R1's router ID 10.0.1.1. R1 and R3 use the default priority of 1, so their router IDs are used for DR or BDR election.

Run the **ospf dr-priority** command to change DR priorities of R1 and R3.

```
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ospf dr-priority 200

[R3]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]ospf dr-priority 100
```

A DR or BDR is elected in non-preemption mode, by default. After router priorities are changed, a DR is not re-elected, so you must reset the OSPF neighbor relationship between R1 and R3.

Shut down and re-enable Gigabit Ethernet 0/0/0 interfaces on R1 and R3 to reset the OSPF neighbor relationship between R1 and R3.

```
[R3]interface GigabitEthernet0/0/0
```

```
[R3-GigabitEthernet0/0/0]shutdown

[R1]interface GigabitEthernet0/0/0
[R1-GigabitEthernet0/0/0]shutdown

[R1-GigabitEthernet0/0/0]undo shutdown

[R3-GigabitEthernet0/0/0]undo shutdown
```

Run the **display ospf peer** command to view the DR and BDR of R1 and R3.

```
[R1]display ospf peer 10.0.3.3

      OSPF Process 1 with Router ID 10.0.1.1
        Neighbors

Area 0.0.0.0 interface 10.0.13.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.0.3.3      Address: 10.0.13.3
  State: Full  Mode:Nbr is Master  Priority: 100
  DR: 10.0.13.1  BDR: 10.0.13.3  MTU: 0
  Dead timer due in 52  sec
  Retrans timer interval: 5
  Neighbor is up for 00:00:25
  Authentication Sequence: [ 0 ]
```

According to the preceding information, R1's priority is higher than R3's priority, so R1 becomes DR and R3 becomes the BDR.

Final Configuration

```
<R1>display current-configuration
[V200R007C00SPC600]
#
 sysname R1
#
interface GigabitEthernet0/0/0
 ip address 10.0.13.1 255.255.255.0
 ospf dr-priority 200
 ospf timer hello 15
#
interface GigabitEthernet0/0/1
 ip address 10.0.12.1 255.255.255.0
```

```
#
interface LoopBack0
 ip address 10.0.1.1 255.255.255.0
#
ospf 1 router-id 10.0.1.1
 area 0.0.0.0
  network 10.0.1.0 0.0.0.255
  network 10.0.12.0 0.0.0.255
  network 10.0.13.0 0.0.0.255
#
user-interface con 0
 authentication-mode password
 set authentication password cipher %$%$+L'YR&IZt'4,)>-*#lH",}%K-oJ_M9+'lOU~bD (\WTqB}%N,%
 $%$
user-interface vty 0 4
#
return
```

<R2>display current-configuration

[V200R007C00SPC600]

```
#
 sysname R2
#
interface GigabitEthernet0/0/1
 ip address 10.0.12.2 255.255.255.0
#
interface LoopBack0
 ip address 10.0.2.2 255.255.255.0
#
ospf 1 router-id 10.0.2.2
 area 0.0.0.0
  network 10.0.2.0 0.0.0.255
  network 10.0.12.0 0.0.0.255
#
user-interface con 0
 authentication-mode password
 set authentication password cipher %$%$1=cd%b%/O%Id-8X:by1N,+s}'4wD6TvO<I|/pd#
#44C@+s#,%$%$
user-interface vty 0 4
#
return
```

```

<R3>display current-configuration
[V200R007C00SPC600]
#
sysname R3
#
interface GigabitEthernet0/0/0
ip address 10.0.13.3 255.255.255.0
ospf dr-priority 100
ospf timer hello 15
#
interface LoopBack0
ip address 10.0.3.3 255.255.255.0
#
interface LoopBack2
ip address 172.16.0.1 255.255.255.0
#
ospf 1 router-id 10.0.3.3
default-route-advertise
area 0.0.0.0
network 10.0.3.0 0.0.0.255
network 10.0.13.0 0.0.0.255
#
ip route-static 0.0.0.0 0.0.0.0 LoopBack2
#
user-interface con 0
authentication-mode password
set authentication password cipher %$%$ksXDMg7Ry6yUU:63:DQ),#/sQg"@*S\U#.s.bHW xQ,y%#/v,%
$%$
user-interface vty 0 4
#
return

```

Module 5 FTP and DHCP

Lab 5-1 Configuring FTP Services

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Establishment of the FTP service.
- Configuration of FTP server parameters.
- Successful file transfer from an FTP server.

Topology

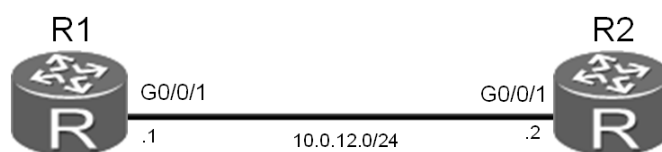


Figure 5.1 FTP topology

Scenario

As a network administrator of a company, you have been tasked with implementing FTP services on the network. You need to implement the FTP service on a router assigned to be an FTP server. The router should allow clients to successfully establish a TCP session to the FTP application and transfer files.

Tasks

Step 1 **Preparing the environment.**

If you are starting this section with a non-configured device, begin here and then move to step 2. For those continuing from previous labs, begin at step 2.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.0.12.1 24
```

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet0/0/1]ip address 10.0.12.2 24
```

Verify that R1 can reach R2, and vice versa..

```
[R1]ping 10.0.12.2
PING 10.0.12.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=10 ms
  Reply from 10.0.12.2: bytes=56 Sequence=2 ttl=255 time=1 ms
  Reply from 10.0.12.2: bytes=56 Sequence=3 ttl=255 time=1 ms
  Reply from 10.0.12.2: bytes=56 Sequence=4 ttl=255 time=10 ms
  Reply from 10.0.12.2: bytes=56 Sequence=5 ttl=255 time=1 ms

--- 10.0.12.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/4/10 ms
```

Step 2 **Enable the FTP service on the router.**

The FTP service is disabled by default on the router. It must be enabled before FTP can be used. Configure an FTP server using R1 with R2 as the client. The same steps

can be reversed to enable R2 to also act as an FTP server.

```
[R1]ftp server enable
Info: Succeeded in starting the FTP server
[R1]set default ftp-directory flash:/
```

Configure user authorization for FTP users to access the server. Unauthorized users will not be able to access the FTP server, reducing security risks.

```
[R1]aaa
[R1-aaa]local-user huawei password cipher huawei123
Info: Add a new user.

[R1-aaa]local-user huawei service-type ftp
Info: The cipher password has been changed to an irreversible-cipher password.
Warning: The user access modes include Telnet, FTP or HTTP, and so security risks exist.
Info: After you change the rights (including the password, access type, FTP directory, and level) of a local user, the rights of users already online do not change. The change takes effect to users who go online after the change.

[R1-aaa]local-user huawei privilege level 15
Info: After you change the rights (including the password, access type, FTP directory, and level) of a local user, the rights of users already online do not change. The change takes effect to users who go online after the change.

[R1-aaa]local-user huawei ftp-directory flash:
Info: After you change the rights (including the password, access type, FTP directory, and level) of a local user, the rights of users already online do not change. The change takes effect to users who go online after the change.
```

```
[R1]display ftp-server

FTP server is running

Max user number          5
User count                0
Timeout value(in minute) 30
Listening port           21
Acl number                0
FTP server's source address 0.0.0.0
```

The FTP server is running on R1 and listens on TCP port 21 by default.

Step 3 Establish an FTP client connection

Establish a connection to the FTP Server from R2.

```

<R2>ftp 10.0.12.1
Trying 10.0.12.1 ...
Press CTRL+K to abort
Connected to 10.0.12.1.
220 FTP service ready.
User(10.0.12.1:(none)):huawei
331 Password required for huawei.
Enter password:
230 User logged in.

```

```
[R2-ftp]
```

Following entry of the correct user name and password, the FTP server can be successfully logged into.

Run the **dir** command before downloading a file or after uploading a file to view the detailed information of the file.

```

[R2-ftp]dir
200 Port command okay.
150 Opening ASCII mode data connection for *.
drwxrwxrwx  1 noone  nogroup      0 May 03 18:03 .
-rwxrwxrwx  1 noone  nogroup 114552448 Jan 19  2012 AR2220E-V200R006C10SPC300.cc
-rwxrwxrwx  1 noone  nogroup  159858 May 03 17:59 mon_file.txt
-rwxrwxrwx  1 noone  nogroup  304700 Mar 03 11:11 sacrule.dat
-rwxrwxrwx  1 noone  nogroup    783 Mar 03 11:12 default_local.cer
-rwxrwxrwx  1 noone  nogroup     0 Dec 20  2015 brdxpon_snmp_cfg.efs
-rwxrwxrwx  1 noone  nogroup   777 May 03 18:03 vrpcfg.zip
drwxrwxrwx  1 noone  nogroup     0 Mar 10 11:14 update
drwxrwxrwx  1 noone  nogroup     0 May 03 18:03 localuser
drwxrwxrwx  1 noone  nogroup     0 Mar 17 10:45 dhcp
-rwxrwxrwx  1 noone  nogroup   460 May 03 18:03 private-data.txt
-rwxrwxrwx  1 noone  nogroup 126352896 Mar 10 11:09 AR2220E-V200R007C00SPC600.cc
drwxrwxrwx  1 noone  nogroup     0 Mar 10 11:15 shelldir
-rwxrwxrwx  1 noone  nogroup  11606 May 03 18:00 mon_lpu_file.txt
drwxrwxrwx  1 noone  nogroup     0 Mar 18 14:45 huawei
-rwxrwxrwx  1 noone  nogroup   120 Mar 18 15:02 text.txt226 Transfer complete.
FTP: 836 byte(s) received in 0.976 second(s) 856.55byte(s)/sec.

```

Set the transfer mode for the files to be transferred.

```
[R2-ftp]binary
```

200 Type set to I.

Retrieve a file from the FTP server. Note: If the `vrpcfg.zip` file is not present in the `sd1:` directory of R1, use the **save** command on R1 to create it.

```
[R2-ftp]get vrpcfg.zip vrpnew.zip
200 Port command okay.
150 Opening BINARY mode data connection for vrpcfg.zip.
226 Transfer complete.
FTP: 120 byte(s) received in 0.678 second(s) 176.99byte(s)/sec.
```

After downloading the file from FTP server, use the **bye** command to close the connection

```
[R2-ftp]bye
221 Server closing.
```

```
<R2>dir
Directory of flash:/
```

Idx	Attr	Size(Byte)	Date	Time(LMT)	FileName
0	-rw-	114,552,448	Jan 19 2012	15:32:52	AR2220E-V200R006C10SPC300.cc
1	-rw-	270,176	Apr 30 2016	03:17:08	mon_file.txt
2	-rw-	304,700	Mar 03 2016	11:11:44	sacrulc.dat
3	-rw-	783	Mar 03 2016	11:12:22	default_local.cer
4	-rw-	0	Dec 20 2015	00:06:14	brdxpon_snmp_cfg.efs
5	-rw-	775	Apr 29 2016	17:51:48	vrpcfg.zip
6	drw-	-	Mar 10 2016	11:28:46	update
7	drw-	-	Apr 23 2016	17:33:38	localuser
8	drw-	-	Mar 21 2016	20:59:46	dhcp
9	-rw-	394	Apr 29 2016	17:51:50	private-data.txt
10	-rw-	126,352,896	Mar 10 2016	11:14:40	AR2220E-V200R007C00SPC600.cc
11	drw-	-	Mar 10 2016	11:29:20	shelldir
12	-rw-	23,950	Apr 27 2016	16:06:06	mon_lpu_file.txt
13	-rw-	120	Mar 24 2016	11:45:44	huawei.zip
14	-rw-	777	May 10 2016	14:23:43	vrpnew.zip

A file can be uploaded to the FTP server by using the command **put**, for which a new file name can also be assigned.

```
[R2-ftp]put vrpnew.zip vrpnew2.zip
200 Port command okay.
150 Opening BINARY mode data connection for vrpnew2.zip.
226 Transfer complete.
FTP: 120 byte(s) sent in 0.443 second(s) 270.88byte(s)/sec.
```

After uploading the file, check for the presence of the file on FTP server.

```
<R1>dir
```

```
Directory of flash:/
```

Idx	Attr	Size(Byte)	Date	Time(LMT)	FileName
0	-rw-	286,620	Mar 14 2016	09:22:20	sacrule.dat
1	-rw-	512,000	Mar 28 2016	14:39:16	mon_file.txt
2	-rw-	1,738,816	Mar 17 2016	12:05:36	web.zip
3	-rw-	48,128	Mar 10 2016	14:16:56	ar2220E_v200r001sph001.pat
4	-rw-	120	Mar 28 2016	10:09:50	iascfg.zip
5	-rw-	699	Mar 28 2016	17:52:38	vrpcfg.zip
6	-rw-	93,871,872	Mar 14 2016	09:13:26	ar2220-V200R007C00SPC600.cc
7	-rw-	512,000	Mar 28 2016	14:40:20	mon_lpu_file.txt
8	-rw-	699	Mar 02 2016	15:44:16	vrpnew2.zip

Remove the created vrpnew.zip and vrpnew2.zip files on R1 and R2.

```
<R1>delete flash:/vrpnew2.zip
Delete flash:/vrpnew2.zip? (y/n) [n]:y
Info: Deleting file flash:/vrpnew2.zip...succeed.
```

```
<R2>delete flash:/vrpnew.zip
Delete flash:/vrpnew.zip? (y/n) [n]:y
Info: Deleting file flash:/vrpnew.zip...succeed.
```

Note: Please take extreme care when deleting the configuration files so to ensure that the entire flash:/ directory of R1 and R2 is not erased.

Final Configuration

```
<R1>display current-configuration
[V200R007C00SPC600]
#
sysname R1
ftp server enable
```

```

set default ftp-directory flash:
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$$$=i~>Xp&aY+*2cEVcS-A23Uwe%$$$
local-user admin service-type http
local-user huawei password cipher %$$$f+~&ZkCn]NUX7m.t;tF9R48s%$$$
local-user huawei privilege level 15
local-user huawei ftp-directory flash:/
local-user huawei service-type ftp
#
interface GigabitEthernet0/0/1
ip address 10.0.12.1 255.255.255.0
#
user-interface con 0
authentication-mode password
set authentication password cipher %$$$+L'YR&IZt'4,) >-*#lH",}%K-oJ_M9+'lOU~bD (\WTqB}%N,%
$$$
user-interface vty 0 4
#
return

<R2>display current-configuration
[V200R007C00SPC600]
#
sysname R2
ftp server enable
set default ftp-directory flash:
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$$$=i~>Xp&aY+*2cEVcS-A23Uwe%$$$
local-user admin service-type http
local-user huawei password cipher %$$$<;qM3D/O;ZLqy/"&6wEESdg$%$$$

```

```
local-user huawei privilege level 15
local-user huawei ftp-directory flash:/
local-user huawei service-type ftp
#
interface GigabitEthernet0/0/1
 ip address 10.0.12.2 255.255.255.0
#
user-interface con 0
 authentication-mode password
 set authentication password cipher %$%$1=cd%b%/O%Id-8X:by1N,+s}'4wD6TvO<I|/pd# #44C@+s#,%
$%$
user-interface vty 0 4
#
return
```

Lab 5-2 Implementing DHCP

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Configuration of a global DHCP pool.
- Configuration of an interface based DHCP pool.
- Enable DHCP discovery and IP allocation for switch interfaces.
- Method of global address pool configuration.
- Method of interface address pool configuration.

Topology

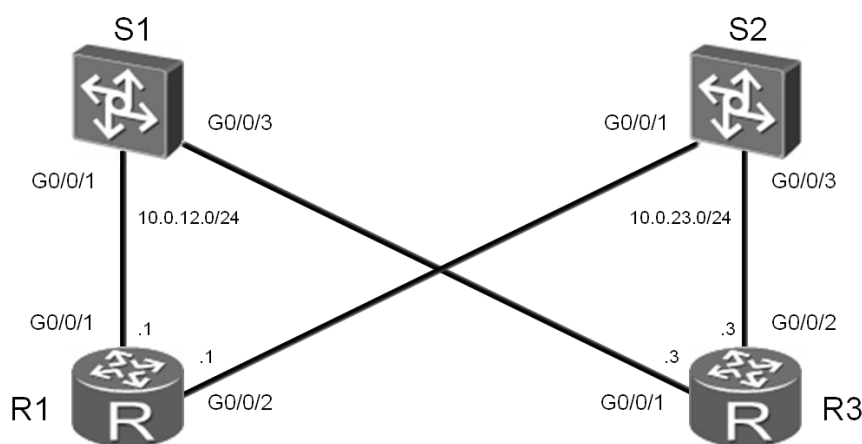


Figure 5.2 DHCP topology

Scenario

As the administrator of an enterprise you have been tasked with implementing DHCP application services within the network. The gateway router in the company network is to be configured as a DHCP server. IP addressing from an address pool are to be offered by the gateway(s) (R1 and R3) to respective access layer devices.

Tasks

Step 1 **Preparing the environment**

If you are starting this section with a non-configured device, begin here and then move to step 3. For those continuing from previous labs, begin at step 2.

Establish the addressing for the lab and temporarily shut down the interfaces Gigabit Ethernet 0/0/2 of R1 and Gigabit Ethernet 0/0/1 of R3.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.0.12.1 24
[R1-GigabitEthernet0/0/1]quit
```

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1]ip address 10.0.12.3 24
[R3-GigabitEthernet0/0/1]shutdown
[R3-GigabitEthernet0/0/1]quit
[R3]interface GigabitEthernet 0/0/2
[R3-GigabitEthernet0/0/2]ip address 10.0.23.3 24
```

```
<Quidway>system-view
Enter system view, return user view with Ctrl+Z.
[Quidway]sysname S1
```

```
<Quidway>system-view
Enter system view, return user view with Ctrl+Z.
[Quidway]sysname S2
```

Step 2 **Cleaning up the previous configuration**

Re-enable to Gigabit Ethernet 0/0/2 interface on R3.

```
[R3]interface GigabitEthernet 0/0/2
[R3-GigabitEthernet0/0/2]undo shutdown
```

Step 3 **Additional configuration**

Disable the port interfaces between S1 and S2 as well as other interfaces to prevent interference from other devices.

```
[S1]interface GigabitEthernet 0/0/9
[S1-GigabitEthernet0/0/9]shutdown
[S1-GigabitEthernet0/0/9]quit
[S1]interface GigabitEthernet 0/0/10
[S1-GigabitEthernet0/0/10]shutdown
[S1-GigabitEthernet0/0/10]quit
[S1]interface GigabitEthernet 0/0/13
[S1-GigabitEthernet0/0/13]shutdown
[S1-GigabitEthernet0/0/13]quit
[S1]interface GigabitEthernet 0/0/14
[S1-GigabitEthernet0/0/14]shutdown
```

```
[S2]interface GigabitEthernet 0/0/9
[S2-GigabitEthernet0/0/9]shutdown
[S2-GigabitEthernet0/0/9]quit
[S2]interface GigabitEthernet 0/0/10
[S2-GigabitEthernet0/0/10]shutdown
[S2-GigabitEthernet0/0/10]quit
[S2]interface GigabitEthernet 0/0/7
[S2-GigabitEthernet0/0/7]shutdown
[S2-GigabitEthernet0/0/23]quit
[S2]interface GigabitEthernet 0/0/6
[S2-GigabitEthernet0/0/6]shutdown
```

```
[R1]interface GigabitEthernet 0/0/2
```

```
[R1-GigabitEthernet0/0/2]ip address 10.0.23.1 24
[R1-GigabitEthernet0/0/2]shutdown
```

Verify that Gigabit Ethernet interfaces 0/0/9, 0/0/10, 0/0/13 and 0/0/14, have been shut down on S1 and that Gigabit Ethernet interfaces 0/0/9, 0/0/10, 0/0/6 and 0/0/7 have been shut down on S2.

```
<S1>display interface brief
```

...output omitted...

Interface	PHY	Protocol	InUti	OutUti	inErrors	outErrors
GigabitEthernet0/0/1	up	up	0.01%	0.01%	0	0
GigabitEthernet0/0/2	up	up	0.01%	0.01%	0	0
GigabitEthernet0/0/3	down	down	0%	0%	0	0
GigabitEthernet0/0/4	up	up	0%	0.01%	0	0
GigabitEthernet0/0/5	up	up	0%	0.01%	0	0
GigabitEthernet0/0/6	down	down	0%	0%	0	0
GigabitEthernet0/0/7	down	down	0%	0%	0	0
GigabitEthernet0/0/8	down	down	0%	0%	0	0
GigabitEthernet0/0/9	*down	down	0%	0%	0	0
GigabitEthernet0/0/10	*down	down	0%	0%	0	0
GigabitEthernet0/0/11	down	down	0%	0%	0	0
GigabitEthernet0/0/12	down	down	0%	0%	0	0
GigabitEthernet0/0/13	*down	down	0%	0%	0	0
GigabitEthernet0/0/14	*down	down	0%	0%	0	0

...output omitted...

```
<S2>display interface brief
```

...output omit...

GigabitEthernet0/0/1	up	up	0%	4.06%	0	0
GigabitEthernet0/0/2	up	up	0%	4.06%	0	0
GigabitEthernet0/0/3	up	up	0%	4.06%	0	0
GigabitEthernet0/0/4	up	up	0%	20.40%	0	0
GigabitEthernet0/0/5	up	up	0%	20.40%	0	0
GigabitEthernet0/0/6	*down	down	0%	2.04%	0	0
GigabitEthernet0/0/7	*down	down	2.03%	2.03%	0	0

GigabitEthernet0/0/8	down	down	0%	0%	0	0
GigabitEthernet0/0/9	*down	down	1.91%	1.91%	0	0
GigabitEthernet0/0/10	*down	down	3.95%	0.12%	0	0
GigabitEthernet0/0/11	up	up	0%	4.06%	0	0
GigabitEthernet0/0/12	up	up	0%	4.06%	0	0

...output omit...

Verify that only interface Gigabit Ethernet 0/0/2 is disabled on R1 and that only interface Gigabit Ethernet 0/0/1 is disabled on R3.

```
<R1>display ip interface brief
...output omitted...
GigabitEthernet0/0/1          10.0.12.1/24          up          up
GigabitEthernet0/0/2          10.0.23.1/24          *down       down
...output omitted...
```

```
<R3>display ip interface brief
...output omitted...
GigabitEthernet0/0/1          10.0.12.3/24          *down       down
GigabitEthernet0/0/2          10.0.23.3/24          up          up
...output omitted...
```

Step 4 Enable the DHCP function.

The DHCP service is not enabled by default, enable the DHCP service on the router(s).

```
[R1]dhcp enable
```

```
[R3]dhcp enable
```

Step 5 Create a global IP address pool

Create an address pool named **pool1** for R1 and **pool2** for R3. Configure attributes for **pool1** and **pool2**, including address range, egress gateway, and IP address lease period.

```
[R1]ip pool pool1
Info: It's successful to create an IP address pool.
[R1-ip-pool-pool1]network 10.0.12.0 mask 24
```

```
[R1-ip-pool-pool1]gateway-list 10.0.12.1
[R1-ip-pool-pool1]lease day 1 hour 12
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]dhcp select global

[R3]ip pool pool2
Info: It's successful to create an IP address pool.
[R3-ip-pool-pool2]network 10.0.23.0 mask 24
[R3-ip-pool-pool2]gateway-list 10.0.23.3
[R3-ip-pool-pool2]lease day 1 hour 12
[R3]interface GigabitEthernet 0/0/2
[R3-GigabitEthernet0/0/2]dhcp select global
```

Run the **display ip pool name <name>** command on the router to view the assigned IP address pool configuration parameters.

```
<R1>display ip pool name pool1
```

```
Pool-name       : pool1
Pool-No         : 0
Lease           : 1 Days 12 Hours 0 Minutes
Domain-name     : -
DNS-server0     : -
NBNS-server0    : -
Netbios-type    : -
Position        : Local          Status          : Unlocked
Gateway-0       : 10.0.12.1
Network         : 10.0.12.0
Mask            : 255.255.255.0
VPN instance    : --
```

-----							Start
End	Total	Used	Idle(Expired)	Conflict	Disable		

10.0.12.1	10.0.12.254	253	0	253 (0)	0	0	

Configure the default management interface for S1 to request an IP address from the DHCP server (R1). Perform the same steps on S2 for R3.

```
[S1]dhcp enable
```

```
[S1]interface Vlanif 1
[S1-Vlanif1]ip address dhcp-alloc
```

```
<S1>display ip interface brief
```

...output omitted...

Interface	IP Address/Mask	Physical	Protocol
MEth0/0/1	unassigned	down	down
NULL0	unassigned	up	up(s)
Vlanif1	10.0.12.254/24	up	up

Verify that this address was taken from the DHCP pool named pool1 on R1,
and for S2, from the DHCP pool named pool2 on R3.

```
<R1>display ip pool name pool1
```

```
Pool-name      : pool1
Pool-No        : 0
Lease          : 1 Days 12 Hours 0 Minutes
Domain-name    : -
DNS-server0    : -
NBNS-server0   : -
Netbios-type   : -
Position       : Local          Status      : Unlocked
Gateway-0      : 10.0.12.1
Network        : 10.0.12.0
Mask           : 255.255.255.0
VPN instance   : --
```

Start	End	Total	Used	Idle(Expired)	Conflict	Disable
10.0.12.1	10.0.12.254	253	1	252(0)	0	0

```
<R3>display ip pool name pool2
```

```
Pool-name      : pool2
Pool-No        : 0
Lease          : 1 Days 12 Hours 0 Minutes
```

```

Domain-name      : -
DNS-server0     : -
NBNS-server0    : -
Netbios-type    : -
Position        : Local          Status          : Unlocked
Gateway-0       : 10.0.23.3
Network         : 10.0.23.0
Mask            : 255.255.255.0
VPN instance    : --

```

Start	End	Total	Used	Idle (Expired)	Conflict	Disable
10.0.23.1	10.0.23.254	253	1	252 (0)	0	0

Ensure that global pool configuration has been completed for both R1 and R3 before continuing!

Step 6 Create an interface based IP address pool

Disable the interface Gigabit Ethernet 0/0/1 R1. For R3 disable interface Gigabit Ethernet 0/0/2.

```

[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]shutdown

```

```

[R3]interface GigabitEthernet 0/0/2
[R3-GigabitEthernet0/0/2]shutdown

```

Configure an interface address pool to allow the clients connected via Gigabit Ethernet 0/0/2 of R1 to obtain IP addresses. Perform the same operation for Gigabit Ethernet 0/0/1 of R3. Do not enable these interfaces, as we do not yet wish to activate the DHCP service on the network.

```

[R1]interface GigabitEthernet 0/0/2
[R1-GigabitEthernet0/0/2]dhcp select interface

```

```

[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1]dhcp select interface

```

Isolate addresses from the pool GigabitEthernet0/0/2 for R1, and the pool GigabitEthernet0/0/1 for R3, for DNS services. Additionally, set the IP address lease

period for the interface address pool.

```
[R1-GigabitEthernet0/0/2]dhcp server dns-list 10.0.23.254
[R1-GigabitEthernet0/0/2]dhcp server excluded-ip-address 10.0.23.254
[R1-GigabitEthernet0/0/2]dhcp server lease day 1 hour 12

[R3-GigabitEthernet0/0/1]dhcp server dns-list 10.0.12.254
[R3-GigabitEthernet0/0/1]dhcp server excluded-ip-address 10.0.12.254
[R3-GigabitEthernet0/0/1]dhcp server lease day 1 hour 12
```

Run the **display ip pool interface** command on the router to view the configured parameters of the interface address pool. For R3 the interface is Gigabit Ethernet 0/0/1.

```
<R1>display ip pool interface GigabitEthernet0/0/2
Pool-name       : GigabitEthernet0/0/2
Pool-No        : 1
Lease           : 1 Days 12 Hours 0 Minutes
Domain-name     : -
DNS-server0    : 10.0.23.254
NBNS-server0   : -
Netbios-type    : -
Position       : Interface      Status      : Unlocked
Gateway-0      : 10.0.23.1
Network        : 10.0.23.0
Mask           : 255.255.255.0
VPN instance    : --

-----
      Start      End      Total  Used  Idle(Expired)  Conflict  Disable
-----
      10.0.23.1  10.0.23.254   253    0    252(0)         0         1
-----
```

Flush the existing Vlanif1 address from S2 to allow for dynamic allocation of a new IP address from the interface GigabitEthernet0/0/2 pool.

```
[S2]interface Vlanif 1
[S2-Vlanif1]shutdown
[S2-Vlanif1]undo shutdown
```

Enable interface Gigabit Ethernet 0/0/2 to allow the DHCP server to become active on the network and to begin sending DHCP discover messages.

```
[R1]interface GigabitEthernet0/0/2
[R1-GigabitEthernet0/0/2]undo shutdown
```

```
<R1>display ip pool interface GigabitEthernet0/0/2
```

```
Pool-name       : GigabitEthernet0/0/2
Pool-No        : 1
Lease           : 1 Days 12 Hours 0 Minutes
Domain-name     : -
DNS-server0    : 10.0.23.254
NBNS-server0   : -
Netbios-type    : -
Position       : Interface      Status      : Unlocked
Gateway-0      : 10.0.23.1
Network        : 10.0.23.0
Mask           : 255.255.255.0
VPN instance    : --
```

Start	End	Total	Used	Idle(Expired)	Conflict	Disable
10.0.23.1	10.0.23.254	253	1	251 (0)	0	1

```
<S2>display ip interface brief
```

...output omitted...

Interface	IP Address/Mask	Physical	Protocol
MEth0/0/1	unassigned	down	down
NULL0	unassigned	up	up(s)
Vlanif1	10.0.23.253/24	up	up

The interface Vlanif1 shows to have been allocated an address from the GigabitEthernet0/0/2 address pool of R1.

Flush the existing Vlanif1 address from S1 to allow for dynamic allocation of a new IP address from the interface GigabitEther0/0/1 pool.

```
[S1]interface Vlanif 1
[S1-Vlanif1]shutdown
[S1-Vlanif1]undo shutdown
```

Enable interface Gigabit Ethernet 0/0/1 to allow the DHCP server to become active

on the network and to begin sending DHCP discover messages.

```
[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1]undo shutdown
```

Verify that the new IP address has been allocated from the interface pool.

```
<R3>display ip pool interface GigabitEthernet0/0/1
Pool-name       : GigabitEthernet0/0/1
Pool-No        : 1
Lease           : 1 Days 12 Hours 0 Minutes
Domain-name     : -
DNS-server0    : 10.0.12.254
NBNS-server0   : -
Netbios-type    : -
Position       : Interface      Status      : Unlocked
Gateway-0      : 10.0.12.3
Network        : 10.0.12.0
Mask           : 255.255.255.0
VPN instance    : --

-----
      Start      End      Total  Used  Idle(Expired)  Conflict  Disable
-----
      10.0.12.1  10.0.12.254  253    1    251 (0)        0         1
-----
```

```
<S1>display ip interface brief
...output omitted...
Interface                IP Address/Mask      Physical  Protocol
MEth0/0/1                 unassigned           down      down
NULL0                    unassigned           up        up(s)
Vlanif1                   10.0.12.253/24      up        up
```

It should also be noted that a default static route pointing to the DHCP server is automatically generated by the switch, as seen in the final configuration below.

Final Configuration

```
[R1]display current-configuration
[V200R007C00SPC600]
#
```

```

sysname R1
#
dhcp enable
#
ip pool pool1
gateway-list 10.0.12.1
network 10.0.12.0 mask 255.255.255.0
lease day 1 hour 12 minute 0
#
interface GigabitEthernet0/0/1
shutdown
ip address 10.0.12.1 255.255.255.0
dhcp select global
#
interface GigabitEthernet0/0/2
ip address 10.0.23.1 255.255.255.0
dhcp select interface
dhcp server excluded-ip-address 10.0.23.254
dhcp server lease day 1 hour 12 minute 0
dhcp server dns-list 10.0.23.254
#
user-interface con 0
authentication-mode password
set authentication password cipher %$%$+L'YR&IZt'4,) >-*#lH",}%K-oJ_M9+'lOU~bD(\WTqB}%N,%$%
$
user-interface vty 0 4
#
return

[R3]dis current-configuration
[V200R007C00SPC600]
#
sysname R3
#
dhcp enable
#
ip pool pool2
gateway-list 10.0.23.3
network 10.0.23.0 mask 255.255.255.0
lease day 1 hour 12 minute 0
#
interface GigabitEthernet0/0/1
ip address 10.0.12.3 255.255.255.0

```

```

dhcp select interface
dhcp server excluded-ip-address 10.0.12.254
dhcp server lease day 1 hour 12 minute 0
dhcp server dns-list 10.0.12.254
#
interface GigabitEthernet0/0/2
shutdown
ip address 10.0.23.3 255.255.255.0
dhcp select global
#
user-interface con 0
authentication-mode password
set authentication password cipher %$%$ksXDMg7Ry6yUU:63:DQ),#/sQg"@*S\U#.s.bHWxQ,y%#/v,%$%
$
user-interface vty 0 4
#
return

<S1>dis current-configuration
#
!Software Version V200R008C00SPC500
sysname S1
#
dhcp enable
#
interface Vlanif1
ip address dhcp-alloc
#
ip route-static 0.0.0.0 0.0.0.0 10.0.12.3
#
user-interface con 0
user-interface vty 0 4
#
return

<S2>display current-configuration
#
!Software Version V200R008C00SPC500
sysname S2
#
dhcp enable
#

```

```
interface Vlanif1
  ip address dhcp-alloc
#
  ip route-static 0.0.0.0 0.0.0.0 10.0.23.1
#
user-interface con 0
user-interface vty 0 4
#
return
```

The privilege of HCNA/HCNP/HCIE:

With any Huawei Career Certification, you have the privilege on <http://learning.huawei.com/en> to enjoy:

- 1、e-Learning Courses: Logon <http://learning.huawei.com/en> and enter [Huawei Training/e-Learning](#)
 - If you have the HCNA/HCNP certificate: You can access Huawei Career Certification and Basic Technology e-Learning courses.
 - If you have the HCIE certificate: You can access all the e-Learning courses which marked for HCIE Certification Users.
 - Methods to get the HCIE e-Learning *privilege*: Please associate HCIE certificate information with your Huawei account, and email the account to Learning@huawei.com to apply for HCIE e-Learning privilege.
- 2、Training Material Download
 - Content: Huawei product training material and Huawei career certification training material.
 - Method: Logon <http://learning.huawei.com/en> and enter [Huawei Training/Classroom Training](#), then you can download training material in the specific training introduction page.
- 3、Priority to participate in [Huawei Online Open Class \(LVC\)](#)
 - The Huawei career certification training and product training covering all ICT technical domains like R&S, UC&C, Security, Storage and so on, which are conducted by Huawei professional instructors.
- 4、Learning Tools:
 - [eNSP](#): Simulate single Router&Switch device and large network.
 - [WLAN Planner](#): Network planning tools for WLAN AP products.
- In addition, Huawei has built up [Huawei Technical Forum](#) which allows candidates to discuss technical issues with Huawei experts, share exam experiences with others or be acquainted with Huawei Products.
- Statement:

This material is for personal use only, and can not be used by any individual or organization for any commercial purposes.